

Schrödinger’s Cat on a Theory of Everything, Patchwork-Style

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so, you scientists are no thieves, no villains?
your little hands are groping in the pitch dark…
I’m the one who hung those ideas in the air.
help yourselves, my dears, don’t turn up your noses!

To curious and industrious luminaries and pupils alike.

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A Simple Solution to the Twin Paradox

Rejecting faster-than-light speeds (and with them negative time) runs counter to the logic of the theory of relativity. That rejection led to the idea that, after making a loop through outer space and returning to Earth, the travelling twin would be younger than his stay-at-home brother. A paradox still regarded as unresolved to this day.

Recognising the reality of negative time, on the other hand, resolves this paradox quite simply. But what exactly is time that runs backwards, from the future into the past?

A person registers photons only when they strike the retina. Any emission of photons takes place before it is seen, since photons need time to reach the observer.

Thus photons, and indeed all objects approaching the observer, move “from earlier to later”, from the past into the future, in ordinary positive time, with its absolute values increasing.

Moving away from the observer, accordingly, means going backwards, from the future into the past, and going deeper into it the farther one gets from the receiver.

In other words, when the traveller changes direction in space, he also changes direction in time.

On a round trip, then, the difference between the traveller’s clock and his stay-at-home twin’s, which first appears and grows on the outward leg, diminishes on the way back and disappears altogether when the twins meet.

The logical error that led to the paradox, as we can see, was to ignore the vector of motion and take only the magnitude of the velocity into account.

Unfortunately, negative time running backwards (from the future into the past) is still a long way from being accepted by all researchers.

That is because, at ordinary speeds below the speed of light, the moving object's time is "compressed" but remains the familiar, positive time.

At the speed of light, an object sets out and reaches its destination simultaneously. Its time is "zeroed out". The traveller seems to "multiply", occupying several places at once and thus acquiring the properties of a wave. As for faster-than-light objects, they are always in negative time, since they are always ahead of the light wave they were "on the same starting line" with at the outset. They cannot in principle be observed visually and make their presence known only indirectly.

Relations of Particular Importance

In the cosmos, when all is said and done, there are only two fundamental concepts: event and observer.

It is the relations between them, then, that we ought to examine.

This is a matter of the first importance.

At the Edge of the World

As is well known, scientists today put the limits of the **visible universe** at about 14 billion light-years.

This means that the light from the farthest points we can observe, the light we **register now**, began its journey to Earth 14 billion years ago.

What we see today are the “portions” of that light generated 14 billion years ago that have made it all the way from that distant past to us, in the present, at the speed of light.

In other words, **its motion through spacetime** is decidedly **positive** in direction: from the past into the future.

Motion in the opposite direction (away from the observer, out into space, towards the limits of the observable universe) accordingly goes **from the future into the past**, in the negative direction along spacetime.

The Clock, Eternal and Flawless

A universal clock, “the Almighty’s Clock”, ought to show not only the positive time we are used to but also negative time, running from the future into the past.

A reminder: a light-year (a unit for measuring cosmic distances) is the distance light travels in a year (about 9.5 trillion kilometres).

A year, accordingly, is the time it takes light to cover that number of kilometres (a distance of one light-year).

That is, cosmic time is measured by the distance light covers during it. And distance, accordingly, by the time light needs to cover it.

When the two are so inseparably linked, is it not natural to expect that time, like space, should be a **vector** with two directions, extending both ways from a chosen zero point on the coordinate axis?

That is exactly how things are in reality: photons from any point source (and point sources are what astrophysicists deal with) head off in **all** directions. In the **forward** light cone the rays travel towards the observer (towards Earth), while in the **reverse** light cone they travel in the opposite direction.

A human being, as a biological creature, can see only light that falls on the retina, light from the **forward** cone.

We do not see light from the reverse cone at all, except by **reflection**. In that case what we see is once again light in the **forward** cone, but from a different, “secondary” light source, even farther from Earth than the first, in time as well as space.

If we take a flash of light as zero on the coordinate axis, spacetime in its forward cone will be positive (which is how we register it). The reverse cone, by contrast, gives us grounds to speak of a negative spacetime continuum inaccessible to observation.

Earth’s coordinate is then positive and corresponds to the distance between Earth and the flash in question.

To generalise, the motion of photons (and of all other objects) **towards the observer**, towards Earth, from any point in the visible universe must **always** be taken as positive. Motion **away from** Earth, away from the observer, is the negative **direction of motion** along the spacetime coordinate axis (motion with a “negative increment” of spacetime, let’s say).

In choosing an origin, of course, we can take absolutely any “event” (as astrophysicists call points in spacetime) as the zero point.

Earth itself is one possibility. To keep the frame of reference in this case, however, **any** point away from Earth (away from the observer) gets a minus sign in front of the absolute value of its distance.

Here I would like to stress once more that, in studying the cosmos, human beings can see **only** sources of light, owing to the way we are built. Whether the light is emitted or reflected does not matter. What matters is this: an object **is registered** only when the photons from the flash it emitted or reflected

have travelled the entire distance separating them from Earth. No more and no less.

So although the cosmos is an entire ocean of light from every sort of source, Earth at any given moment is washed by **one single**, quite specific wave. It is made up of light from a flash that occurred, say, 8 minutes ago at a distance of eight light-minutes from Earth, 5 hours ago at a distance of 5 light-hours, a year ago at a distance of one light-year, 40 years ago at a distance of 40 light-years, a million (a billion) years ago at a distance of a million (a billion) light-years, and so on, all the way to the edge of the visible universe.

The “crests” of this wave are intervals of space and time that the speed of light binds “fast” together. They are, so to speak, equivalent: “x” light-years = “x” years. Time is equivalent to space.

All the other flashes, such as those that occurred 7 years ago at a distance of 8 light-years, or the other way round, 8 years ago at a distance of 7 light-years, are all “irrelevant”, so to speak.

Here is why. For convenience, let us put Earth at zero on the coordinate axis. Then the resulting position in space and time of the photons from the first of those “wrong” flashes, relative to Earth, can be written as $\mathbf{-8 + 7 = -1}$. From the point minus eight years, the photons have spent seven years drawing closer to Earth. That positive increment was not enough to reach the observer (the zero point).

The coordinates of the second flash are $\mathbf{-7 + 7 + (-1) = -1}$. That is because throughout the final, “extra” year of their journey the photons were **moving away** from Earth again, away from the zero point, on the negative-number scale.

In both cases the particles of light end up, relative to Earth, in the unobservable reverse light cone, **in negative time and**

space. And the light in the reverse cone, along with the objects in it, is, as we know, invisible to us.

Let us pick any stretch of spacetime separating the observer (Earth) from another point (an event) in outer space, say a stretch of “x” light-years, and have a swift rider on a spirited horse dash from one end to the other. Let his speed be one and a half times the speed of light.

We can put the observer at zero, which puts the starting point at “minus x”. At first, the motion is towards Earth. But after “x” years, when light would just have reached the retina of the eye registering it, point zero, our cosmic wanderer will have travelled so fast that he will be somewhere else entirely:

−x + x + (−0.5x) = −0.5x

Once again, the coordinate is negative!

And the result? The poor fellow sailed straight over Earth, like a sheet of plywood over Paris. Nobody even noticed him!

And, characteristically, as he travels away from Earth he will also find himself in the negative cone of an “event” lying “x” years from Earth.

Even when travelling **towards** the light, such an object invariably finds itself in the reverse cone of any **oncoming** flash-event! Picture it: the flash has coordinate zero. Earth is some distance away from it in the positive direction along the axis, “plus x”, let’s say. At some point the rider takes off towards the light. The countdown is on! If he were travelling at the speed of light, he would reach zero in “x” years, the epicentre of the light wave. But no, he’s racing along one and a half times as fast as light! So after “x” years he will be at “minus 0.5x”, in the reverse cone of the flash:

+x − x + (−0.5x) = −0.5x

Negative time again! And negative space, of course. As they say, “in the wrong place at the wrong time”. The result: **invisible**! (Note: the actual figures naturally differ from those given here, since light rays and objects with mass follow different trajectories, but the principle remains the same.)

Since “x” can be any natural number at all, the conclusion suggests itself: it is absolutely impossible for us to see anything capable of moving faster than light, because an object with that speed is **permanently** in the reverse light cone. That holds wherever it starts within the visible universe and whichever way it heads.

A faster-than-light object can never be overtaken by light or “collide with it”, and so can never be a source of reflected rays. Consequently it cannot be detected by an observer. Never. Under no circumstances.

All material bodies, on the other hand, are barred even from reaching the speed of light, let alone exceeding it. Sooner or later they are bound to come into view as sources of reflected light.

In Another Dimension

After reading my piece about “the Almighty’s Clock”, a certain gentleman declared that there was nothing in it about why we cannot see a photon from a faster-than-light object.

At first I was surprised, but I had little trouble imagining what had prompted my opponent’s staggering declaration. Apparently, he had missed a detail of the thought experiment and taken the photons to have started out in precisely the same place as the postulated “invisible fellow”. If they had reflected off him, the photons could then have made the object in question observable.

But that is not what happens.

In fact, the faster-than-light object is only on the same “starting line” as the photons, at the same distance from the subluminal observer. By the rule of CPT symmetry, it is at the mirror-symmetric counterpart of the material source of light (whether that source emits or reflects it), and by no means at the same point.

So what I said above still holds. At any given moment we see only photons from events that took place n years (hours, minutes) ago at a distance of the same n light-years (light-hours, light-minutes). Those are the photons that reach the observer’s eye.

This is precisely what our dimension, our spacetime, is. The speed of light “marks it out”, and it is always positive. The observer registers neither photons that have not had enough time to arrive nor those that have had more time than they needed. The objects occupying the corresponding points of spacetime at the moment in question are therefore invisible too.

And those are precisely the points where objects made of antimatter are permanently located by virtue of their off-the-scale speed. They are always in the reverse light cone, in negative spacetime, in another dimension, beyond the sight of the subluminal observer, a product of the material world. “Dark matter”, gentlemen.

And anyone who wants to find their way around the complex constructions of modern physics must grasp this simple truth.

Kozma Prutkov’s Question

Anyone with even a passing acquaintance with modern physics knows (or at least ought to know) that, by the Law of Conservation, no particle, however tiny, comes into being without its antipode, the corresponding antiparticle.

There is an antiquark for the quark, an antielectron for the electron, and antihydrogen for the hydrogen atom.

Now for the question: how far does this principle extend? How far along the elements in Mendeleev's periodic table can we assume that corresponding antistructures exist? And how about a hydrogen molecule, for example, instead of an atom? What about still more complex conglomerates of them?

In short, as Kozma Prutkov once asked: "Where is the beginning of the end with which the beginning ends?" More to the point, why should there be an end at all? For what conceivable reason?

"Long Live Antiworlds, Oases amid the Nonsense!"

The Pope praised the popular cosmological theory of the Big Bang for agreeing perfectly with Church dogma: there was nothing, and then the good Lord set off that very Big Bang.

But to a real scientist, God is *persona non grata*. Even if we grant the existence of a Creator, science will immediately ask what this God is, exactly, and where he himself came from.

In other words, the concept of God as something unknowable is a dead end for science.

Unfortunately, the Big Bang theory cannot tell us where all that colossal energy came from in the first place.

After the Bang, though, from the very first milliseconds, it is more or less clear how the Universe acquired its structure. Nothing supernatural about it. Entropy rises overall, but thanks to the store of energy supplied at the moment of the Bang, the system develops nonlinearly and gives rise to self-organising "subsystems", from atoms and galaxies to living organisms, thinking ones included.

The very beginning can only be explained as follows: the Universe and the Antiuniverse exist eternally. This is a dynamic state. They periodically appear, develop and disappear, at the moments corresponding to the Big Bang and the Big Crunch.

Quantum mechanics describes a similar process in the vacuum, the continual appearance and disappearance (annihilation) of virtual particle–antiparticle pairs. Since energy cannot be created out of nothing, one partner has positive energy and the other negative energy, as Hawking points out.

The same thing presumably happens in the Universe–Antiuniverse system. Gravity is due to the force of attraction between these two worlds, or antipodes. It “pulls together” all structures, curving space and time into a loop and ultimately bringing about the inevitable Big Crunch at the end of each cycle, in accordance with Friedmann’s first model.

But that is only a new beginning. The energy of annihilation makes it possible for the Universe (and the Antiuniverse) to develop again in a new cycle, and so it goes, *ad infinito*.

This idea, “staring us in the face”, was evidently dismissed for the simple reason that astronomers have never confirmed the presence of objects made of antimatter in the cosmos.

Why should that be?

Let me remind you that we can observe only celestial bodies that either emit light themselves or reflect light from other sources.

Suppose, however, that some object in space travels faster than light. Then a light wave cannot catch up with it and consequently cannot reflect off it. So in principle the object cannot be detected by astronomers. It will always be in what is known as the “reverse light cone” (see above).

It is well known that a material body cannot attain the speed of light, much less exceed it. As for antimatter, though… why not? If mass is a measure of inertia, might “antimass” have precisely the opposite properties and serve as a kind of “measure of mobility”?

Quantum mechanics tells us that matter and antimatter, spatial symmetry and the two directions of time are related in a particular way. If we replace particles with antiparticles and spatial coordinates with their mirror reflections, and reverse the direction of time (from the future into the past), the physical laws governing the resulting object remain unchanged.

That is presumably the very relationship within the Universe–Antiuniverse pair. The Antiuniverse consists of antimatter, has negative energy, is a “mirror image” of our Universe and exists in reverse time. This preserves all the physical laws within the system. The total amount of matter and energy stays constant (equal to zero), in accordance with the first law of thermodynamics.

The second law of thermodynamics, which concerns the increase in entropy, also holds, of course, for the material Universe, where irreversible processes run in the positive direction of time, from the past into the future.

In the Antiuniverse, where time runs backwards, entropy, as a function of time, gradually decreases. All the parameters of the dual system thereby remain perfectly symmetrical at any moment.

So time, which seems to stand still at the moment of the Big Bang (so to speak, “ending and beginning” at one and the same instant), takes on two directions as it moves away from that moment, one on either side of zero, just like the spatial coordinates. Time is positive in the Universe and negative in the Antiuniverse (for faster-than-light objects). Speed is the only barrier between these “parallel worlds”.

For bodies in the Antiuniverse, the curvature of space takes on a “negative” value. Space seems to turn “inside out”, curving away from the shape of a sphere and taking on the shape of a saddle, or, if you prefer, the inner surface of a sphere, the mirror image of the Universe’s shape.

Interestingly, this is exactly the kind of “saddle-shaped” curvature of space that some authors say is needed for travelling backwards in time.

In the past proper (in negative time), the only objects that exist are those that exceed the speed of light, that is, those that can “outrun themselves”.

Objects that can reach the speed of light, or a speed close to it, can “catch up with themselves”. For them time seems to stop. They “double” and can be in two (several, many) places at once.

That is precisely what happens to particles in the double-slit experiment when they begin to exhibit wave properties.

Acknowledging the existence of the Antiuniverse makes it possible to resolve practically all the outstanding questions of modern cosmology. In particular, it lets us find the supposedly “missing” antiparticles, which, as it turns out, never went missing at all (of course they didn’t: that would run counter to the first law of thermodynamics).

They simply joined with others of their kind to form more and more complex structures, just as their material counterparts did. As they gained the necessary speed, they vanished… from our field of view.

Finally, I would like to add that the entropy of the Antiuniverse, like energy there in general, evidently has a negative sign. That is precisely what is meant by “negentropy”, or information in Shannon’s sense.

So what does that leave us with? Cosmic fields of information? A higher intelligence, perhaps? The indestructibility of our consciousness, which is the sum of what we know about ourselves and the world around us, or in other words, information? Perhaps the immortality of our souls? I do not know. I would only echo Spinoza: “God is Nature.” There is no point looking for anything outside it.

A Picture of the World with a Formula for Everything

1. There are equal numbers of particles and antiparticles in the world.
2. The total sum of matter and energy is zero.
3. We cannot see cosmic objects made of antimatter. They move faster than light, so light does not reflect off them.
4. In microparticles, faster-than-light speeds affect the characteristics of their spin.
5. The system of “twins” exists eternally, periodically appearing and disappearing through annihilation. For the antipodal worlds, these are the moments of the Big Bang (the Big Crunch).
6. Counterbalancing entropy is **negentropy**, equal in absolute value but opposite in sign: cosmic fields of information.
7. The existence of a virtual, “imaginary” world justifies the use of imaginary numbers. “Zero” then becomes a perfectly definite quantity expressed as a complex number.

The formula for everything:

$$\mathbf{E - E = E + i^2E}$$

Or, since E equals m c squared, we can also use an analogous formula with the corresponding equivalent in place of E:

$$\mathbf{mc^2 + (-mc^2) = mc^2 + i^2mc^2}$$

God and Systems Theory

The Creator of **everything that exists** cannot be outside the system, because then he … **does not exist** (whatever is **outside what exists** does not exist). Nor can he belong to everything that exists **alongside** its other constituents, because then he is **part of the system**, and a part is always **simpler** than the whole. At the same time, **systems theory** tells us that the **creator of a system** cannot be simpler than the system he has created. The conclusion (the only possible one) is that the **Creator** must be **identified** with **Creation**: **God** is the **world**. A note: our knowledge of the world is far from perfect. Don’t get stuck on that.

The Intimate Connection Between Space and Time

Time is the distance a photon travels in that time.

Distance is the time it takes a photon to travel that distance.

Old News

To anyone who understands physics, I am saying nothing new when I say that every journey we make into space goes **backwards** in time. Backwards in time, I stress, in the sense that its “increment” is negative, as distinct from being in reversed, negative time. In other words, we always fly into the past, and only into the past. A flash is registered only after a certain time has elapsed from the moment it occurs. That interval corresponds to the distance between the source of the flash and Earth, and is the time it takes the photons to cover it at the speed of light. Remember how “the light of dead stars reaches us”? And if we set off to meet it, we’re flying into the past, towards light that is coming to us from the past into the present. We meet it sooner. I never thought today’s “educated folk” would have trouble with this. Oh, well.

The Tale of the White Bull

A light-year is the distance light travels in a year (ten thousand billion kilometres). A year, accordingly, is the time it takes light to cover a distance of ten thousand billion kilometres.

Time is measured by distance. But distance is a vector. If distance and time are so closely related, time, too, must have the properties of a vector and take on negative values.

Scientists today put the limits of the visible universe at about 14 billion light-years. This means that the light from the farthest points we can observe, the light we register now, began its journey to Earth 14 billion years ago.

In other words, photons travelling from the depths of space to us (or, more generally, from any event to any subluminal observer) travel from the past into the future, in the normal, positive flow of time.

Photons travelling in the opposite direction (out into space towards the limits of the universe, from observer to event) travel from the future into the past, in reversed, negative time. We cannot see anything at all except photons. And we see only photons in positive time that have reached the retina. We do not see photons moving away from us (that is how the optics of the eye and other detecting instruments work).

There are important subtleties here, however.

Material bodies travelling below the speed of light can get in the path of receding photons (in the reverse light cone), reflect them off their surfaces and send them back to the observer. We will see those photons (and thus the bodies that reflected them), albeit after a delay. In other words, despite the "negative increment", the resulting time remains positive.

If the object that same flash is overtaking travels at the speed of light, the negative increment to ordinary Earth time will be so great that the resulting value will read "time zero". Time will stop and the object will be in several places at the same instant (it will exhibit wave properties). It may "catch up with itself" and become "its own antiparticle", become radiation. To paraphrase a familiar saying, "If an object looks like a wave and acts like a wave, it is a wave."

Finally, in the third case, when an object moves faster than the photons, it stays ahead of them and cannot reflect them. It remains invisible, in time whose sign has changed. A subluminal observer does not register negative time or the objects in it. According to quantum mechanics, it is precisely in negative time that objects made of antimatter are permanently located.

In the microworld, where the observer is outside the system, the spin helps tell an antiparticle from a particle. On a cosmic scale, this is "another dimension", a world invisible to us. And every bit as real as ours. Processes in that world are reversible,

as is characteristic of time running backwards and of faster-than-light speeds. Boltzmann already warned of the existence of such regions.

So there is nothing “seditious” about it, yet physicists are still arguing.

Hawking thinks reversed time is even more real than ours. Penrose thinks it does not exist at all… “Think for yourselves, decide for yourselves, to have or not to have.”

On the Ether, with a Smile

Water waves are in the water.

Particle waves are in the particles.

A particle counts as a wave when it is in many places at once at a given instant. Is a special “carrier” really needed for something that can of itself double, multiply and occupy different places in space at the same time? So, a field. An entire “global ocean” of massless (energy-carrying) particles, plus fluctuations, with virtual particle–antiparticle pairs forever appearing and disappearing in the vacuum. There’s your medium of propagation.

What ether are you talking about, eggheads?

Still, call this ocean ether if you like. It’s a lovely word. “I’ll call myself Gantenbein.”

That Strange Duality

In some experiments electrons behave like waves. They exhibit reflection, diffraction and interference. They show a speed-dependent wavelength equal to Planck’s constant divided by the product of the electron’s mass and speed.

They acquire the ability to be in many places at once.

In other experiments electrons behave exactly like particles.

But never like both waves and particles at the same time.

The same happens with light.

In some experiments photons behave as waves, and in others exclusively as particles.

The Principle of the Cosmos: The Law of Antipodes

In mathematical logic there is Kurt Gödel's incompleteness theorem, which says that any complex formal system must necessarily be inconsistent to be complete.

Inconsistency is naturally a feature of both the systems that do the describing (formal systems) and those being described (real systems) if they lay claim to completeness.

Thus in the cosmos, alongside the material Universe that everyone knows, there must be its antipode, an invisible, nonmaterial world made of antimatter and an exact mirror image of our Universe.

Time in this Antiuniverse runs backwards, its objects consist of antiparticles, and its energy is negative. The part of that energy equal in magnitude to the Universe's entropy but opposite in sign, namely negentropy, takes the form of cosmic fields of information, a kind of "World Mind".

An Antiuniverse with these characteristics is subject to the known laws of physics, in particular the law of conservation. This guarantees the perpetual existence of information as part of the system's negative energy, allowing us to suppose … that our consciousness is eternal. Consciousness is simply the information individuals have accumulated about the world around them and about themselves.

Since decreasing entropy corresponds to the reversed flow of time, it is logical to conclude that all processes in the Antiuniverse go from chaos to order, “from worse to better”. Highly organised systems therefore have no need to struggle with their surroundings to survive, and natural selection has no role in their evolution.

An individual must experience the way such a world is organised as a kind of perfect “paradise” in which the very conditions for evil are absent.

Recognising that the Universe has a nonmaterial antipode makes a supernatural Creator superfluous. This dual world is self-sufficient, with its own “built-in” mechanisms for endless cyclic emergence and development, from one annihilation to the next (from the Big Bang to the Big Crunch), in a perpetually repeating cycle.

One might say that this dual world can itself be regarded as God, an eternal, indestructible entity that includes fields of information, the “World Mind”, as part of a single, all-embracing organism.

All existing kinds of field, all material structures and all nonmaterial structures made of antimatter, every object and phenomenon without exception, are simply components and life processes of this contradictory, continually changing and transforming, dynamic and eternal organism.

Objects in the Antiuniverse cannot be registered by an observer because of the limitations of our senses. Evolution has adapted them to perceive only the material world that produced us as a biological species.

In particular, we can see celestial bodies only if they emit or reflect light. When an object exceeds the speed of light, as is natural for motion in negative time running backwards from the future into the past, a light wave cannot reflect off it because it

cannot catch up with it. So structures in the Antiuniverse remain invisible to us.

Still, knowing that even the tiniest material particle cannot come into being without its corresponding antiparticle and that they too can disappear only in pairs, through mutual annihilation, we must conclude with confidence that antiparticles did not disappear. Like their material counterparts, they must inevitably have formed an antimatter world of their own, the exact antipodal counterpart of the material Universe.

Recognising the dual nature of the existing world makes it possible to resolve many of the contradictions that have accumulated in modern physics and cosmology.

Recognising the dual nature of the existing world allows us at last to dispense with the primitive, anthropomorphic “God” proposed by ancient religious scriptures and give the only possible definition of the Divine that agrees with scientific data and the logic of things:

God is everything that exists.

We need only bear in mind that the World, Nature, is exactly twice as complex as we used to think, the difference being the Antiuniverse. It is possible, however, that negative mass takes the form of the square of an imaginary mass, supplying the necessary “minus one” at the moment of annihilation. (A stray thought about quark triads belongs here.)

There is nothing supernatural about this system. God’s laws are the very laws of Nature, which human beings, as biological creatures, simply cannot break. A person is only the tiniest cell or particle of an all-embracing organism, living first in its material part and then in its antimaterial part, subject to the physical conditions that prevail in each.

The Multiplicity of Worlds and the Divine Hierarchy

Gödel's theorem also states that for any system of sets there is always some superset.

Yet even in this sense my proposed term God = world, nature = everything that exists, takes in everything.

It even allows for a possible "hierarchy of Gods", that is, for a multiplicity of worlds on many levels.

As we know, virtual particle–antiparticle pairs continually appear in the vacuum. Such a pair is, in fact, the only eternally existing system known with certainty to modern science.

Imagine for a moment that thinking beings live inside one such pair.

Would it matter to such "intellectuals", living, say, inside a structure belonging to a particle, that there are vast numbers of such pairs in our Universe? I assure you, it wouldn't matter in the slightest!

This is where the principle known as "Occam's razor" should be applied: "Do not create needless complications without dire necessity."

Our hypothetical creatures could not care less how many such pairs there are, or whether all of them belong to a superset, our Universe. For them, their world, their God, is the only one.

What matters here is simply the principle, regardless of the number of sets: the law of antipodes. Without it, a system cannot be complete or self-sufficient and cannot claim to be an eternal and infinite, divine entity.

Likewise, when we speak of ourselves, the inhabitants of the material universe, we should understand that for us, everything that exists is the eternal pair, Universe plus Antiuniverse,

which cyclically appears and disappears at the moment of annihilation (the Big Bang / Big Crunch). This is our dual world, our one and only God.

There might, of course, be many pairs of worlds analogous to our dual world of antipodes. They might in turn be part of some “Hyperuniverse”. There is no need to think about that, by the same principle of Occam’s razor.

A Plain Explanation for the Quick-Witted

If an object moves faster than light, a light wave will never catch up with it and thus will not reflect off it. We see celestial bodies only when they either emit light themselves or reflect it.

That is, if there are objects in the cosmos capable of moving faster than light, in principle they cannot be detected by astronomers.

Could such objects consist of the “missing” antiparticles? If mass is a measure of inertia, might “antimass” be a measure of mobility? Might the cosmos contain, alongside the Universe, an Antiuniverse that we cannot see?

Could this be the “crazy idea” that helps physicists out of the dead end they are in?

I should note here that, as faster-than-light objects cannot become “secondary” light sources (by reflecting light), they evidently do not generate photons themselves either. That seems to be the prerogative of stars, thoroughly material bodies in a thoroughly material world, where a thoroughly material combustion reaction rages. Otherwise, faster-than-light speeds would long since have been discovered in the cosmos. As it is, “dark matter”, gentlemen.

Second Mass

The Creator of everything that exists cannot fail to be included in the system, since he himself exists. Yet he cannot be included as one of its components, because a part is always simpler than the whole, and a system cannot be more complex than its creator. The only way out of the logical contradiction is to identify the system with the Creator.

God is the world, nature.

But nature is more complex than modern science's conception of it. Besides the visible world, the Universe, there is an invisible antipodal world made of antimatter and living in reverse time. The coordinates of its objects are mirror images of the coordinates of their material counterparts. Because time in the antiuniverse runs backwards, its processes are reversible: entropy decreases and negentropy (information) increases. This world is invisible because its objects move faster than light, a necessary condition for existing in negative time. We see an object only when the light reflected from it reaches the observer's eye. Anything moving faster than light outruns the rays and ends up in negative spacetime, in "another dimension", a parallel (or, more precisely, symmetrical) world, if you like, beyond the reach of our observations.

Since cosmic time is determined by the speed of a photon and nothing else, time is the distance a photon travels in that time.

And there is nothing supernatural about it. The Law of Conservation alone is enough to rule out the idea of an "external", "supernatural" Creator.

Eternity cannot be "at one end only". An eternal world has neither end nor beginning.

The World is the All-Embracing and Only perpetual-motion machine, the absolute.

Its existence is a cyclic process analogous to the only eternal system known to science: the particle–antiparticle pair, which constantly appears and disappears in the vacuum in an endless succession of annihilations.

The analogy is complete, except that in the case of antipodal worlds the interval between annihilations is measured in billions of years, rather than in microscopic intervals of time.

Counterpart Antipodes Are Reciprocal

As we know, a particle and an antiparticle have a reciprocal relationship. Each is an “antiparticle” to its partner. In other words, we can call the electron, for example, the particle, in which case the positron is its antiparticle. But we can just as well regard the electron as the positron’s antiparticle. I should point out that in a sense the latter option is preferable. Of the two partners, it is the electron that has the negative charge (negative energy?). The conclusion may be wrong, however. We should take into account the particle’s spin, energy level, whether it has rest mass, and other parameters. Its speed, in particular, may determine whether it behaves like a particle proper or like a wave. In any case, we should be exceedingly cautious about classifying “elementary” particles. Consider the photon, if nothing else: it can be its own antiparticle. That is, a particle’s parameters at a given moment may matter more than its “name”.

Stephen Hawking’s definition: “An antiparticle is a particle that lives in reverse time.”

I will add one more: a wave is a particle whose time has stopped.

Nothing but Annihilation

The only way for matter to be converted fully and equivalently into radiation, or, conversely, for matter to arise from radiation, is the process of annihilation.

Direct or reverse, respectively.

The antipodes involved in these processes may be as simple or as complex as you like, from quarks to universes.

The more complex the structures taking part, the greater the interval between any two annihilations in their endless chain.

Laughing for No Reason Is a Sign of a Cheerful Disposition

When a quantity is measured by distance (and that is how we measure cosmic time), there are no grounds for thinking it cannot take negative values.

The direction of motion of photons (and other objects) from “event” to observer may be taken as positive, and the opposite direction as negative motion through spacetime. If the “event” we are considering takes place at the observer’s location, the motion of photons (and everything else) away from the observer is always motion in the negative direction along the spacetime axis.

That is precisely why time for a subluminal cosmonaut slows compared with the time measured by the observer. If a moving object reaches the speed of light, its time stops. Exceeding the speed of light reverses time: it runs backwards, changing from positive to negative. The faster-than-light “cosmonaut” will end up in the past and “reach his destination at seven in the morning — yesterday”.

It is regrettable that even physicists who count themselves adherents of the Theory of Relativity apparently do not understand these regularities and say, for example, that physicists can supposedly talk about negative time “only in jest”.

What can one say to that… Physicists certainly joke often, but only the most easily amused can laugh for no reason.

Enough Spaghetti for Your Ears! Here’s Food for Thought

Cooked Up by Dirac, Einstein, Steve Hawking and Yours Truly

1. Cosmic time is measured by distance. These two quantities are inseparably bound by the speed of light: “Time is the distance a photon travels in that time.” (Or vice versa: distance is the time it takes a photon to cover that distance.)
2. Points in spacetime are “events”. For the most part these are simply flashes of light.
3. The direction of motion through spacetime from event to observer is positive.
4. The direction of motion from observer to event is negative.
5. That is why, on a flight into space, time for the cosmonaut slows compared with time for the observer who stayed at home (the result of a “negative increment” in time).
6. If (!) the cosmonaut were to reach the speed of light, his time would stop (!).
7. If (!) the cosmonaut were to exceed the speed of light, his time would be reversed (!). It would become negative and run from the future into the past.

8. This is impossible for material objects (objects with mass). We are material inhabitants of a material world, of positive spacetime.

   But!

9. There are objects in nature that do live in reverse time.

   These are antiparticles (and structures made of them). Such objects obey the rule of “CPT symmetry”: they are made of antimatter (antimaterial substance), have coordinates that are mirror images of the coordinates of their material counterparts, and exist in negative time.

10. In the world of antimatter, in negative time running backwards (at speeds above that of light), the principle of causality is violated.

11. Whatever exists in negative spacetime lies beyond the reach of our observation. Photons cannot catch up with objects that travel faster than light. Consequently, those objects do not reflect light. And we see only sources of light, emitted or reflected.

12. The only way to “look” at an antiparticle is to pull it out of negative time and into “our dimension”, into positive spacetime.

13. But if the rule of CPT symmetry is violated, the laws of physics are violated. So in positive time the antiparticle will exhibit positive mass, in violation of the law of conservation (!).

Ideas About the Ideal

A particle is something that is in a particular place at a particular time. For material particles the speed of light is an unattainable upper limit. These particles exist only in positive time.

A wave is something that is in many places at once at a particular time. Energy-carrying (massless) particles travel at (or near) the speed of light.

As a particle acquires wave properties, it may also enter the negative spacetime continuum and become its own antiparticle.

When an antiparticle is in a quantum state rather than a wave state, its coordinates at a particular instant correspond exactly to those of its material antipode. They are its exact “mirror image”: likewise “in a particular place at a particular time”. Antiparticles travel faster than light. Objects that travel faster than light live in negative time.

For antiparticles, the speed of light is a lower limit.

And when an antiparticle attains the speed of light, it naturally becomes a wave too, capable of being “everywhere at once” (in both positive and negative time) and of serving as its own antipode.

That is, there appear to be three fundamental categories in nature: radiation (energy), particle and antiparticle. (Plus their conglomerates: objects and “clumps”.)

All these categories change into one another (matter into energy, for instance), either partially or completely, at moments of direct or reverse annihilation.

Entropy, a measure of chaos, is part of positive energy. Its antipode is negentropy, negative entropy with the opposite sign, a measure of order, or information.

Human consciousness, too, is nothing other than information about oneself and the surrounding world, accumulated by the individual over a lifetime.

As a physical substrate, information is always equal to entropy in absolute value. The measure of chaos is the same as the measure of order, with the sign reversed.

That is, at a particular point in the cycle, when all positive energy turns into entropy through dissipation, everything that exists is a kind of clump of information, the seed of the dual world to come.

“The ideal” is information.

Its possible “forms of embodiment” are matter, antimatter and energy (positive and negative).

All these categories change form and exist cyclically for ever, in accordance with the Law of Conservation, the fundamental law of nature.

Further Considerations

A particle travels below the speed of light (an antiparticle travels above it). A wave, or radiation, travels at the speed of light. A particle and a wave are one and the same thing: it shows now the properties of a wave, now those of a particle. Which means that a particle (or its antipode) becomes a wave when it attains the speed of light. Or: a wave is a particle travelling at the speed of light.

As for the mathematics… There is only one thing I can say with confidence. A mathematical description of the absolute (the system of “everything that exists”) must include all numbers known to mathematics. It must certainly include imaginary and complex numbers as well as the positive scale of natural numbers.

It is clear that from a purely mathematical point of view, quantities that cancel one another must have opposite signs, at least at the very moment of annihilation (for universes, the moment of the Big Bang).

All physical categories go through the whole spectrum of states, taking positive, imaginary and negative values, or zero.

(I hope it comes as no secret to anyone that a particle's mass depends on its speed.)

Errors in Textbooks

In many textbooks we find the following formulation of the rule of CPT symmetry:

> If the particles of an object are replaced by antiparticles, its coordinates by their mirror image, and time is reversed to run from the future into the past, the physical laws governing the object will remain unchanged.

That is wrong. Or rather, it is not quite right.

When time changes direction, so does the direction of entropic processes. Entropy, as a function of time, will change its sign too, and the processes will exhibit reversibility.

Thus the Second Law of Thermodynamics will hold for such an object "exactly the other way round": entropy will decrease as time goes on instead of increasing. This is precisely what is required if the First Law of Thermodynamics, the fundamental law of nature or Law of Conservation, is to remain unshaken.

> Not everything in textbooks, all the time,
> is the plain truth. Got it? Fine.

Excuse Me

Pardon me, my good man, but frankly, you would have to be a complete idiot not to see that the law of conservation does not allow a single particle, not one, however wretched, to come into being without its antipode, an antiparticle.

And it is intolerable that, in physics, scientists still do not know where those antiparticles that inevitably formed in the early universe went or why they "disappeared", since what we see today is exclusively a material world.

And my idea, the very one you dislike so much, answers precisely that question.

Namely: the antiparticles did not disappear anywhere at all. It is simply that structures made of antimatter, unlike their material counterparts, travel faster than light. In accordance with the rule of CPT symmetry they are always in negative spacetime. Anyone with even a passing acquaintance with the subject ought to see that time reverses precisely and exclusively for objects capable of outrunning light.

The region of negative time (or, more precisely, spacetime) arises in the part of each and every flash of emitted or reflected light where the photons move away from the observer.

But we, products and inhabitants of our own positive dimension, can see photons only at the moment they draw near enough to reach the retina. Certainly not the ones rushing “out of sight”.

That is how we are built. So everything belonging to the negative spacetime continuum lies beyond our observation. No optics can help with that.

I do hope some people have the brains to grasp these features of their own make-up and of the way the universe is built, and to understand their consequences.

Urfin Jus, the Ex-Wizard

Dear Wizard Urfin, do stop being sly! I've known your tricks for ages, till they set my teeth on edge. Here we meet and once again you have your questions. Yet any explanations go for nothing just the same. You come to pay a visit and to flatter your own vanity, to spout solemn inanities without listening to replies. You come in different guises: now a physicist of many years' standing, now an esteemed mathematician, now a witless, spiteful thief who always dresses up as a pious parishioner and charges me with every mortal sin. He's sure that he can slyly win God's favour by reproaching and reviling those who haven't been trained out of thinking.

Once, speaking as a physicist, you declared that science would have absolutely nothing to do with speeds beyond the limit. Objects in the Universe don't travel faster than light, you said, so most of my ideas must be wrong. My answer is this: I have no idea where you found the claim of mine you've just refuted. I never spoke of material bodies. They behave themselves, meek and mild, "without any funny business". Only in the objects of the virtual Antiuniverse, hidden from view, does antimass cast off every prohibition in flight. In its headlong rush it always outruns light, preserving the balance, the beauty of the Cosmos.

You insist your colleagues have lately found all masses, even an antiparticle's, to be unconditionally positive, or so the story goes… Well, what am I to do? How shall I put it gently so as not to hurt your feelings? In your pride in the experiment, you seem to have forgotten to interpret the new facts with care. Pardon me, but what would make a positive mass suddenly annihilate with a particle's mass? The trouble, dear friend, is that virtual worlds are reluctant to reveal their nature to us. Looking positive is an old "trick", and not just for little particles. The rules of the game are clear: positive time, positive mass, positive force, or energy, that is. These are all we can see and

all we can measure… Yet an immutable law cannot simply be ignored. You might care to recall the Law of Conservation: no structure suddenly pops up out of nowhere. Only as part of a pair with its antipode, whose every parameter has the opposite sign. *C'est la vie*, that's life!

Yes, that is nature's law… If a tiny particle suddenly wants to "dance" with a mass like its own, their antipodes can only join together and wait out the "dance". For any large structure the "dance" goes on longer, the more masses it draws in. For particles, milliseconds, for universes, billions of years. Yet the end of the dance is settled in advance. A flash brought them into being, and a flash will see them out…

Ah, now you're a mathematician, and poor you, you can't find the equation that would let a pair of antipodes conserve momentum, though you've tried this way and that! Why, I once offered you a simple, perfect formula. But for some reason, my dear Urfin Jus, you are so determined to get by without imaginary numbers… And complex ones, incidentally, aren't to your taste either. You scoffed that you couldn't see the point of them. Square "i", you said, and it is absolutely, simply minus one. So the equation I offered you is nothing but a tautology.

I reply patiently: there is a place in the world for numbers that aren't real. Their dance steps don't please blinkered pedants whose training is sound but whose logic… is dull. In the imaginary world the numbers are imaginary. "i" is there, squared, and one day it will wipe out "one" in the flash of an explosion. From free energy, eternally, in each new cycle, the World's complex structure creates itself.

If these numbers are used, then they are needed somewhere. There must be something in the world that can't be described without them. Perhaps one day I'll meet a mathematician with some sense who will put hypercomplex numbers to work… Is anybody there? Let me know! As for you, Jus, don't come

calling, don’t put yourself out. How long can this go on? There’s no use arguing with you. I’m tired of it. I give up. Better count your profits, or cram the Bible, or go and meet some ladies, Urfin Jus, you ex-wizard!

A Reply to Mr Z

“The positron is an antiparticle, but this particle, like the electron, exists while time flows positively,” you write.

Not at all. Finding antiparticles in the material universe does not automatically mean they are in positive time. They are constituents of a small system in its own right, the atom, where time of its own flows. Being in the negative part of that time may be expressed … simply by the particle’s spin and nothing else.

As Hawking noted, a particle’s spin can reflect which side we see it from (which light cone it is in).

The same applies to this microscopic object’s faster-than-light speed. To find itself “face to face” with its material counterpart, perhaps the antipode “slipped round the loop twice” instead of simply “moving away” to the corresponding distance from the axis.

Now for the mass of antiparticles… Or rather, mass in general.

It looks very much as though there is some mutable entity in nature which takes on different signs according to its speed, from positive through zero to imaginary and negative values. With only a positive “yardstick” at our disposal, we are apt to pronounce it “ordinary positive mass” if and when we manage to catch and measure it. That is all.

Besides, the relationship between particle and antiparticle is reciprocal, so we can also consider the electron the positron’s

antiparticle. We don’t know these processes quite as thoroughly as we think. But all that is by the way.

You see, I have been reproached for citing the opinions of famous people which they expressed in private conversations or popular publications, outside their scientific papers.

As for me, all I can do is put my “unkempt thoughts” in parentheses. Sorry.

Next. Let us turn to an important point which, alas, showed that you had completely misunderstood what you read in my work.

You say:

> Increments of time can in fact be positive or negative, but it does not follow that the integral characteristic of time can be negative. A change in the direction of an object’s motion is in no way evidence of a change in the direction in which time moves.

An entirely fair statement, except that it leaves one thing out: it would never have occurred to me to claim the contrary!

Moving away from the observer means only that the object is moving in a “negative increment” of time.

For a “cosmonaut” travelling below the speed of light, time will stay positive but will slow down (a well-known phenomenon). The cosmonaut will “grow younger”.

If the cosmonaut could attain the speed of light while moving away (which he cannot, being material: mass is a measure of inertia), his time would stop. But for small particles this is possible. At the speed of light they turn into a wave and “double” (the double-slit experiment). Time at a standstill means that an object is in two places at once. That distinguishes a wave from a particle, which at a particular

moment is in a particular place. At the speed of light the object does a kind of “wheelspin” in time and space.

Only when its speed exceeds that of light does time reverse and begin to run from the future into the past, becoming negative.

And it is precisely in negative time that antimatter lives (in accordance with CPT symmetry, as is well known, in particular, to all the physicists in Geneva who have studied antihydrogen).

The passage you wrote reminded me of an objection raised earlier by another esteemed Samizdat author. He said:

> By your account, all it takes is to step behind a candle to find yourself in negative time.

Well, here’s my answer to him and to you:

Manage to do it faster than light and that’s exactly where you’ll be.

The Principle of Causality, or What Makes Us So Principled

Surely to acknowledge that alongside the material universe there is a world made of antimatter, with time running backwards, is itself equivalent to acknowledging that a signal can travel faster than light.

Many people mistakenly think the Theory of Relativity imposes a ban on faster-than-light speeds. That is quite wrong.

The ban on motion faster than light comes neither from special relativity nor from Maxwell’s or Lorentz’s mathematical derivations. It comes solely from … the principle of causality. Einstein himself, considering faster-than-light signalling, wrote:

> We are compelled to regard as possible a mechanism for transmitting a signal whereby the resulting effect precedes its cause.

And then: “Although this result contains no logical contradiction, it is nevertheless so contrary to the character of all our experience that the impossibility of the assumption V > C seems sufficiently well founded.”

Notice that after the word “although”, the subject is simply ordinary everyday experience. That once served as the basis for the idea that Earth was flat. It has no direct bearing on science in general or on the Theory of Relativity in particular.

That is, if anything, it is this statement by Einstein that conflicts with special relativity. The theory itself, on the contrary, “compels” us to recognise the possibility of faster-than-light speeds.

The belief that a faster-than-light signal supposedly contradicts the Theory of Relativity is so deeply rooted, however, that even L. Wang, who carried out the impressive experiment in which the speed of light was exceeded by a factor of three hundred (!) in a laboratory, believes the experiment … is at odds with relativity!

The scientist reasonably points out, at the same time, that the claim that speeds above that of light are unattainable apparently applies only to objects whose rest mass is greater than zero.

“Light, however, can be represented either as waves, to which the concept of mass does not apply at all, or as photons, whose rest mass, as is well known, is zero,” writes Wang. “So the speed of light in a vacuum is no limit at all.”

Incidentally, a photon, strictly speaking, is never “at rest” in the first place.

So let’s sort out what’s what.

First of all, we must understand that the principle of causality is violated at and above the speed of light, and its scope is therefore confined to the world of matter.

It does not apply to the field, that is, to energy-carrying particles. Upon reaching the speed of light, massless particles can occupy many places at once, in keeping with wave properties, and can serve as their own antiparticles.

Still less can it apply to objects made of antimatter, which reside permanently in reversed time. There entropy, a function of time, decreases instead of increasing as in the world of matter. Consequently, processes are reversible, and it is precisely the cause that gets ahead of the effect!

So I repeat: the principle of causality is inviolable in the material (visible) universe. But the world of antimatter is just the medium that allows faster-than-light speeds to become a physical reality. In the words of researchers studying antihydrogen, it is “a key to another universe, where time changes direction and runs the opposite way, from the future into the past, everything left becomes right, and everything positive becomes negative”.

Indeed, we need only know that the masses of a particle and an antiparticle can annihilate with each other to understand that there is something radically different about those masses. At least at the moment of annihilation, it is unlikely that they have the same sign. Otherwise they would add up rather than cancel each other.

There seems to be a single fundamental entity in nature which changes so much according to its speed that it takes on different signs, from positive through zero to imaginary and negative values (the square of an imaginary value is one possibility), passing into another dimension of spacetime.

As inhabitants of a strictly material world, with only a positive "yardstick for taking measurements", we are apt to pronounce it "ordinary positive mass" if and when we manage to catch and measure it, violating CPT symmetry in the process. As a result, what is found to have been violated is … the first law of thermodynamics (!). And we like to repeat that an antiparticle's "ordinary positive mass" is an "established fact". Yet even the mass of a particle at rest is an elastic concept, since strictly speaking a particle does not exist at rest either.

As Einstein wittily remarked, if the fact contradicts the theory, so much the worse for the fact.

Then again, anyone can get the nuances of interpretation wrong by yielding to the weight of received opinion.

What matters is that the scientific method enables us to perceive the world around us as accurately as possible and to correct our mistakes as new knowledge accumulates.

By now even the latest radio observations by astrophysicists give grounds to suspect faster-than-light motion in some cosmic objects, and an impressive number of experiments have been carried out in laboratories around the world to slow light down and speed it up. We can be sure that Einstein, with his remarkable ability to think unconventionally, would never now make such a sweeping remark about the inadmissibility of the faster-than-light speeds his own theory "compels us to talk about".

As for those who are so excessively principled in their concern to keep outdated postulates inviolate, who miss the point of the Theory of Relativity and want to be more Catholic than the Pope… we must admit that they're a long way from being Einsteins. Unfortunately.

“No Trails Have Been Blazed to the Antiworld Yet”

It so happened that I discovered another physical reality, one that calls for new mathematical tools. I do, however, propose the final formula myself. It is simple, as a “Formula for Everything” should be.

It is simple because it is made up of the mathematical symbols for two perfect counterpart antipodes capable of annihilating:

Universe = $E = mc^2$
Antiuniverse = $-E = -mc^2 = (-1)E = (-1)mc^2$

Given that **$-1 = i^2$** (the imaginary unit squared),

Antiuniverse = $(i^2)E = (i^2)mc^2$

Then:

Universe + Antiuniverse = $E + (-E) = mc^2 + (-mc^2) = 0$ = Nothing

or:

$E + (i^2)E = mc^2 + (i^2)mc^2$ = Everything (a definite quantity expressed as a complex number).

In fact, we could put “Particle” and “Antiparticle” in place of “Universe” and “Antiuniverse” (or indeed any pair of antipodes with any values of E and m).

I claim a **complete analogy** between the only **eternal** system known to modern science, a pair of antipodal particles that periodically appear and disappear in the vacuum, and … antipodal universes, which, like particles, come into being and disappear in the cosmos in an **annihilating** Big Bang. Like particles, they arise from the energy of annihilation. Like particles, they do so cyclically and eternally.

Frankly, it is hard for me to understand why almost every educated person, with rare exceptions, bristles at the mere mention of negative energy. Even gravity is a perfectly good example of negative energy, though it is measured by the positive energy expended in overcoming it.

Hawking, in particular, has no doubt that when a particle–antiparticle pair appears in the vacuum, the energy of the latter is negative, simply by virtue of the law of conservation, which does not allow energy to come from nothing. Only antipodes whose combined “increment is zero” may come into being.

But mass and energy are equivalent, so an antiparticle’s mass must also be negative (remember Dirac). I am prepared to allow that it might be negative only at the moment of annihilation, when it and the particle cancel each other out. From a purely mathematical point of view, quantities that cancel each other must have different signs. But since the instant of annihilation is, properly speaking, “time zero”, we may suppose that at that instant of stopped time the particles are no longer particles at all, but waves that inhabit two spacetime dimensions simultaneously, that is, radiation.

In negative time, however, mass is more likely imaginary:

The starting formula:

$$\mathbf{E = mc^2}$$

If the particle is moving,

$$\mathbf{E = \gamma m_0 c^2}$$

where m_0 is the rest mass.

$$\mathbf{\gamma = 1 / \sqrt{(1 - v^2/c^2)}}$$

In words, gamma equals one divided by the square root. The expression under the root is one minus v squared divided by c squared.

When the speed is greater than the speed of light, gamma is imaginary. With real (albeit negative) energy, the rest mass will be imaginary. The same is true when time is reversed, since speed appears squared. Such hypothetical particles were called tachyons. I would regard them simply as a state of the antiparticle. And if they didn't exist, they would be worth inventing. So the mass is imaginary. The momentum of such a particle is usually equal to

$P = \gamma m_0 v$

which is real, and since it is a vector, its sign depends on the frame of reference.

Nothing seditious, as you can see.

Now I'll try to outline what I think about entropy.

Entropy is **part** of a system's energy, the part that cannot be converted into work. The dissipation of energy increases the system's entropy. Entropy, as part of the universe's positive energy, is naturally positive.

As for information or negentropy… The relationship between entropy and probability was established by Boltzmann. In 1948 Wiener observed that the amount of information, being the **negative logarithm** of a quantity that can be regarded as a probability, is in essence a kind of **negative entropy**.

Which is just what I affirm: yes, negentropy, **negative** entropy, information, is part of the negative energy found in the antiworld, where all physical categories change sign.

From the science news: "Geneva physicists consider antihydrogen atoms a 'ticket' … to **another mathematical**

Universe, where ‘positive becomes negative and left becomes right’.”

More precisely, a ticket to a different physical reality described by different mathematics, where on crossing “through the looking-glass” (CPT symmetry), everything **positive** becomes negative: space, its curvature, time (inseparable from space, since cosmic time is measured by distance), mass (possibly the square of an imaginary mass, with one imaginary mass on either side of the observer along the same line of sight, which will supply minus one in real terms), and energy in all its constituent parts, including entropy (which changes sign because time running backwards makes entropic processes reversible).

And again, nothing seditious:

Perpetual-motion machines do not and cannot exist … in half the system, in the material world. But the combination of antipodes, the principle on which a complete (contradictory) system is built, is precisely the only possible Perpetual-Motion Machine: the Absolute.

That is, the world owes its indestructibility and eternity to its duality and nothing else, to the Law of Antipodes. Like particles, worlds arise only in pairs and disappear in pairs: direct and reverse annihilation of two exact counterpart antipodes. Annihilation alone is the means of converting matter entirely into energy, into radiation, and vice versa.

I want to stress that I regard information as a full-fledged physical category, the antipode of the universe’s entropy. Its existence does not require an “observer”. Cosmic fields of information are an objective reality. Moreover, because waves (massless particles, quanta) can serve as “their own” antipodes, we can call entropy itself information: the measure of chaos is the same as the measure of order, with the opposite sign.

So a negative logarithm, used in place of the positive one used in entropy calculations, is the mathematical principle for calculating negentropy.

In general, the principle of antipodes is “easy to use”: do not ignore negative and imaginary numbers, which, after all, keep turning up in cosmological calculations. “Algebra is wiser than the natural scientist,” as one clever mathematician said.

And when dealing with cosmological problems, we should put logic first, ahead of experiment.

As I have already said, there are no genuine antiparticles in our world. Ours is a world of positive time. When we obtain antiparticles to examine them, we … extract them from negative time. And the experiment affects its own result: CPT symmetry is violated, and the antiparticle changes its properties, notably by exhibiting positive mass. Strictly speaking, this violates the law of conservation, which is no surprise but must be understood correctly.

It is the “border crossing” itself that brings about the change in parameters, which matter more than the particle’s “name”.

Once more: objects made of antimatter obey the rule of CPT symmetry. One of its requirements is that the time in which such an object exists must run backwards, must be negative. Antimatter lives in reversed time, and only objects travelling faster than light can exist in that kind of negative time.

Think of S. Hawking’s joking limerick “on the subject”:

> A lady they nicknamed Miss Bright
> Once wanted to ride down the street.
> But she raced out of sight,
> Past the speed of light,
> And here was the price of that feat:
> Back at her own front door she arrived,
> Long before she had left for the ride!

Ordinary positive mass, as we know, cannot attain even the speed of light, let alone exceed it. So the mass of antimatter is no “ordinary positive mass”. In the antiworld physical quantities change sign. It is quite likely, though, that mass becomes imaginary (tachyonic?), which is characteristic of virtual particles.

A Devil’s Dozen of Damnably Interesting Thoughts

1. The antipode of a particle is an antiparticle.
2. Particle and wave are two names for the same thing.
3. The antipode of a wave is an antiwave.
4. Energy-carrying (massless) particles (waves) are their own antipodes.
5. A quantum of energy is its own antipode.
6. Entropy is a form of energy.
7. The antipode of entropy is called negentropy.
8. Negentropy is synonymous with information.
9. Entropy can also be considered information, since it is “its own” antipode.
10. Entropy is a measure of the chaos, the uncertainty, of a system.
12. Information is a measure of order.
13. In measuring chaos we measure order, but with a minus sign.

If it still isn’t clear what that means, here’s a hint:

Both categories, entropy and negentropy, are information, its dual code (plus one / minus one). Note: minus one, as mentioned earlier, can be the square of an imaginary quantity.

And at this point “Scheherazade ceased her permitted discourse”.

To tell the truth, I’ve given up hope of persuading my opponents. The history of science tells us it’s a “lost cause”, but those of us on friendly terms with paradoxes have a habit of stepping on the same rake again.

For the Three Hundredth Time, and a Little Edgily

One of the greats put it this way: “If an idea cannot be expressed in simple words, it is wrong and should be discarded.”

My idea is stated plainly:

The antiparticles that formed in the early universe by the law of conservation, as the indispensable partners of material particles, did not, by that very law, disappear anywhere except from our field of view. Unlike material objects, the objects made up of those antiparticles, the counterparts and antipodes of the objects in the material universe, move faster than light and therefore cannot be detected by optical instruments.

Forever outrunning light, they cannot reflect it, and all we see are sources of emitted or reflected light.

That is the simple idea you have to refute, “if you’re really itching to”.

I don’t think anyone will manage it, though.

My Greatest Mistake, Which Is Also My Chief Achievement

“God is everywhere, present in all his creations, so he is everything. Some call that nature, others God. Aren’t we talking about the same thing?” the theists ask.

Well, then… I’m prepared to agree.

All the more so since it makes a good deal of sense in terms of systems theory. The creator of everything that exists belongs to the system because he himself exists. Yet he cannot be a part of it, because as a part he would necessarily be simpler than the system, and that will not do for its creator.

So you’ve persuaded me: “creator” and “creation” are synonyms. Let’s call nature, the world, God.

In principle, it makes no difference what word we give this entity.

But let us stress that agreeing to call something by a word used in religious scriptures hardly means it behaves as those texts say it does.

A thinking person should stop and sort out what’s what. Discard the unsubstantiated, naïve inventions of our ancestors about unnatural miracles supposedly performed on Earth by biblical characters. Discard the absurd regulations governing people’s everyday conduct, down to supposedly God-given diets, rules for sexual intercourse, styles of dress and even the way to defecate, to say nothing of certain “extremes” in dealing with other people (calls to mercilessly “wipe out” unbelievers and apostates by the will of Allah, for instance).

In other words, we must condemn clerics who foist their own beliefs and rules upon God.

We should radically rethink prevailing ideas about the divine so that the alternative we propose satisfies both uneducated fanatics and people of a critical turn of mind.

At the same time we must try to understand that nature, the world, is more complex than present-day orthodox physics makes it out to be.

Science has, of course, yet to say its last word. It would be absurd to suppose that it ever could, however distant the future.

There are, however, a few clarifications that are overdue, and that your humble servant invites you to consider.

We need, for instance, to understand that our consciousness (the “spirit” or “soul”, in theists’ terms) is nothing other than the organism’s information about the world around it and about itself in that world.

As a physical entity, information is a nonmaterial, massless substrate of the physical world, the antipode of entropy, that part of a system’s energy which is a measure of chaos.

I should mention that I use “matter” and “material” here in a narrower sense than is often allowed. That is, I take the world to contain three distinctly separate physical categories: matter, antimatter and energy (massless, energy-carrying particles). Those accustomed, for convenience, to lump all these categories together as matter should bear this point in mind.

So, as the antipode of entropy, information or “negentropy” is “the other side of the coin”. It measures order and harmony rather than chaos. Its absolute value is equal to that of entropy, but with the opposite sign, of course. You may call it the consciousness of the world or the cosmic mind, if you like.

Then the sum of the information an individual has acquired, the mind of a single person, is a small part of the cosmic mind. As

part of the system's total energy it obeys the law of conservation and so exists eternally, changing form. It follows that "existence" may be a broader concept than "life" (which does not altogether take the meaning out of talk about … the immortality of the soul: "cogito, ergo sum", so to speak).

But God encompasses more than the cosmic mind. God is all that exists, all of nature, which includes information as a physical substrate alongside the other categories.

For instance, in carrying out the informational programme of their own DNA during ontogeny, human beings encompass their own mind and consciousness. They are more than a mind alone: they are a whole thinking organism.

Similarly, God (the world, nature), depending on the phase of the cycle, takes form in different structures as the balance of information and energy changes, "embodying" itself in the universe. This includes the visible, material world, which we perceive with our senses, plus an invisible world, which we do not sense directly but which makes itself known by a number of indirect signs: the virtual world of antimatter (which appears to be what "dark matter" actually is).

Objects made of antimatter, quantum mechanics tells us, obey the rule of "CPT symmetry".

This means that in the macroworld the coordinates of the antipodes are "mirror images" of the coordinates of the corresponding ordinary celestial bodies, material planets and galaxies. More precisely, virtual structures are always in negative spacetime running backwards relative to any "event-point" in the material world and, accordingly, relative to any "subluminal" observer.

This effect is due to the faster-than-light motion of bodies made of antimatter. Because of this property a light wave cannot catch up with them, and they do not reflect light, unlike all

material objects (for the properties of negative time, see above).

And in the reversed time “through the looking-glass”, processes are reversible. Entropy declines as the proportion of information, order and mind increases. A kind of invisible, nonmaterial “paradise”, an “abode of souls”, a repository of the world’s collective consciousness.

But the world is a changing, dynamic system. At the moment of the Big Bang (“time zero”), the entire mass of the universe annihilates with all the antimatter in the world. From the energy of annihilation (from radiation), the antipodal worlds begin to take shape in a new cycle. This is how the world exists eternally in accordance with the Law of Conservation, in an endless succession of cycles in which matter and energy change into one another.

The system is self-sufficient and, as is characteristic of every complete system, contradictory. It is a unity of counterpart antipodes, like the eternal “particle–antiparticle” pair that appears and disappears in the vacuum.

A world of just this kind can be called a divine entity without contradicting any of the facts known to physics and, paradoxical as it may seem, while agreeing with certain religious claims.

And finally, a few words about that unfortunate slip, my “greatest mistake” mentioned in the title, which is presumably what I should have been talking about all along.

The “mistake” is precisely this. Scientists are wary of encroaching on the clerics’ particular terminology and so leave the most important questions about the cosmos to a fraternity for the most part ignorant of science. I, on the other hand, took the theists up on their proposal and called nature God, setting

out beforehand the points that might stand in the way of identifying the two.

And the result of my quite commendable resolve was this: people, who for the most part are notoriously “lazy and incurious”, simply turn away from the idea. They don’t want to bother with the carefully explained particulars that justify what I did. Theists and scientists alike would sooner try to prise open a door that is already open, each wielding their own trusty crowbar.

But is that my mistake? And is it a mistake at all?

Let me just put it on record: I’ve done my job. The door is open. The only sparks are from your crowbars scraping on the threshold, giving the illusion that there is something in the way.

Five Sensations

Sensation One

The law of conservation, which affirms the eternity and indestructibility of matter and energy and their conversion into one another, embraces every physical category without exception. It encompasses matter, antimatter and energy of every kind and form, including entropy as part of the system’s positive energy and negentropy, that is, information: in Shannon’s terms, entropy with the opposite sign.

Sensation Two

We cannot in principle see cosmic objects made of antimatter. Because they move faster than light, they are always in the reverse light cone relative to the observer. Their presence is confirmed by indirect evidence, however, by the unmistakable effect they have on bodies in the material universe. This is what makes astrophysicists look for the notorious “dark matter”.

Sensation Three

The Big Bang is nothing other than the annihilation of all the matter in the world with all the antimatter it contains. The energy of the Bang sets the antipodal worlds developing again, until their next annihilation. The process is cyclic and eternal, like the process for a “particle–antiparticle” pair.

Sensation Four

Whether a quantum-mechanical object is a particle or a wave depends entirely on its speed. Below the speed of light it is a particle. At or close to the speed of light it is a wave. On reaching speeds above that of light the object presents itself as an antiparticle, with corresponding changes in all its other parameters.

Sensation Five

Time, like the spatial dimensions, has two directions along the coordinate axis, one on either side of zero, which is the flash, the “event”. There is positive time in the forward light cone and negative time in the reverse cone.

The motion of any object towards an observer is always motion in the positive direction. Away from the observer it is motion in the negative direction, both in space and in time.

A Quintessence as Easy as Steaming Turnips

We know from quantum mechanics that the coordinates of objects made of antimatter are mirror images of those of their material counterparts and that these objects exist in negative time, running from the future into the past (the rule of CPT symmetry). It is logical to suppose that antimatter obeys the same rules on the cosmic scale and so is also in negative time. A reminder: we can see only the light that falls on the retina, which is to say we register positive spacetime exclusively. Regions of negative time and space arise on the cosmic scale wherever photons from a flash-event are heading away from the observer, rather than towards them. Material objects that temporarily find themselves in a region of negative time will inevitably be overtaken by the receding photons. By reflecting the “fugitives” off their surfaces and sending them back to the observer, the objects reveal their presence as secondary sources of (reflected) light. Bodies made of antimatter, however, move faster than light and remain in negative spacetime for ever. Photons do not catch up with them and do not turn them into sources of light. Reverse (negative) spacetime is a different dimension, a world invisible to us. Yet it is every bit as real. Processes in this world are reversible, as is characteristic of time running backwards and of faster-than-light speeds. The reversibility of processes in a system of exact counterpart antipodes is the only way the law of conservation, which guarantees the eternity of the world, can be fully realised.

The Theory of Relativity: Dotting the i’s

Photons take time to reach an observer, so particles are always emitted before the emission is registered. It follows that photons move from the “event” to the observer in ordinary positive time, from the past into the future. Motion away from the observer, by contrast, takes place in “reverse” time, from the future into the past. Backwards motion in time can decrease the absolute value of positive numbers or increase the absolute value of negative ones, depending on the coordinates of the starting point. Conversely, motion from the event to the observer increases the magnitude of positive values or decreases that of negative ones. Speed plays only a supporting role in the sense that the greater the speed, the more quickly these absolute values of time increase or decrease. If the object under observation is moving below the speed of light and the increment in magnitude is negative, we can register only the compression, the slowing down, of time. At the speed of light (characteristic of energy-carrying particles), the negative increment “cancels out” the entire advance of positive time. Faster-than-light objects are always in negative time.

The ABCs

Light moving away from the recipient always travels in “reverse” time, from the future into the past. The farther from the observer the receding photons encounter an object that can reflect them and send them back to the visual receptors in the eye that perceives the light, the later they are registered.

Backwards motion in time can decrease the absolute value of positive numbers or increase the absolute value of negative ones, depending on the coordinates of the starting point.

If the object under observation is moving below the speed of light and the increment in magnitude is negative, we can register only the compression, the slowing down, of time. But at the speed of light (characteristic of energy-carrying particles), the negative increment “cancels out” the entire advance of positive time. And faster-than-light objects are in negative time altogether (they are not registered visually).

To reject negative time is tantamount to rejecting the theory of relativity (which even Einstein himself is not allowed to do).

All that “Cyrillic” set out above allows us to identify and/or explain some of the most important physical phenomena:

- the nature of the Big Bang,
- quantum–wave duality,
- the cause of the curvature of space,
- the operating principle of the Universe, the unique Perpetual-Motion Machine,
- the meaning of time as a category,
- the Twin Paradox,
- the reasons to prefer Friedmann’s First Model,
- the mathematical Formula for Everything,
- the inevitability of faster-than-light speeds,
- the presence of a CPT-symmetric world,
- the tachyonic mass of antiparticles,
- the breakdown of cause and effect “through the looking-glass”,
- the complementarity of entropy and information and the eternity of these physical concepts,
- the existence of a single fundamental entity that undergoes a “phase transition” depending on speed,
- the nature of gravity (and electricity),

- the composition of dark matter and energy,
- and all that sort of thing, including … “the rather difficult character of the Lord God Almighty”.

Schrödinger’s Cat’s Memorandum

An observer sees only light (emitted or reflected), and only at the moment the photons reach the sensitive cells of the organ of sight.

Since every flash occurs before it is seen, photons travelling towards the observer always move from the past into the future. That is the normal, positive flow of time.

Photons and other objects moving away from the observer move backwards in time, from the future into the past, giving time a negative increment.

Time for objects travelling below the speed of light simply slows down and stays positive.

At the speed of light, the negative increment is so great that time stops. The object acquires wave properties, the ability to be in many places at the same moment.

Objects travelling faster than light enter truly negative time, with entropic processes and cause-and-effect relationships reversed.

Cosmic objects in negative time are beyond the visual reach of a subluminal observer, but they are every bit as real for all that.

Antimatter “hides” in negative time, annihilating with all the matter in the universe at every successive “Big Bang”.

To sum up: any physicist who refuses to recognise that time runs backwards is a sorry physicist indeed.

Nota Bene

For objects travelling at the speed of light, cause and effect occur at the same moment. An object is in two (several, many) places at once and thereby acquires wave properties (it sets out and reaches its destination simultaneously).

For faster-than-light objects the effect precedes the cause. Antiparticles, for instance, live in reverse time (an object arrives before it sets off).

And only for bodies with positive mass does the law of causality work as we observe it in our everyday world of matter, where time goes from past to future.

A Quiver for the Arrows of Time

Whenever I come across solemn discussions of the “mysterious and unknowable” nature of time, I want to object: there’s … no mystery to it, ladies and gentlemen! The universal measure of space and time in the Universe is the speed of light. In a vacuum it is constant, at 300,000 kilometres per second. A second, then, is quite simply the time light takes to cover 300,000 kilometres. The photons have gone 300,000 km? A second has passed.

A minute, accordingly, is the interval in which photons cover sixty times as much ground, or 18,000,000 km. To express an hour in kilometres, multiply by sixty again. Want to know what a year is? By all means! An astronomical year is the time light takes to travel roughly 9.5 trillion kilometres. Half a year, consequently, is 9.5 trillion km divided by two. A month is one twelfth of the same figure, and so on. Fourteen billion years, the postulated age of the universe, is roughly ten to the twenty-third power in kilometres. It is also the size (radius) of the visible Universe.

To generalise: time is the distance light covers in that time. In other words, any time can be expressed in kilometres. An important consequence follows. If a quantity is expressed in units of length, it is a vector and can have positive as well as negative values. Inexplicably, the existence of negative time is still disputed. How did negative time “slip past” everyone? Wherever an observer may be, all they see is light that is caught by and makes contact with the receptors in their organ of sight. The observer does not register receding photons except if and when they catch up with an object on their way, reflect off it and return to the observer, albeit after a delay. Optical aids do not change the principle. We register an emitting source or a reflecting (secondary) source, but always by means of light that has reached the observer. Since every flash occurs before it is seen, photons travelling towards the observer move from the past into the future. And movement from past to future is precisely the normal, positive flow of time. Conversely, photons travelling in the opposite direction, away from the observer, go backwards in time, from the future into the past. This is the negative flow of time, in which each successive point is deeper in the past than the one before it. Photons (and other objects) moving in any direction from the observer’s own location are always moving away from the observer. From any other point in the universe (the so-called “events”), movement can be either towards the observer, into the future (the forward light cone), or away from the observer, into the past (the reverse light cone). Photons moving backwards, away from the observer, wherever they start, necessarily give time a negative increment. The size of that increment depends on the speed of objects moving in the photon streams that might reflect the photons and “redirect” them towards the observer. An object moving below the speed of light will sooner or later be overtaken by the light wave and become a source of reflected light. In other words, it will show up in positive time. Ordinary celestial bodies that start in the

reverse light cone of a flash will thus inevitably be registered by the observer later, if after a certain delay. Whether a cosmonaut notices time slowing or the space traveller in the “twin paradox” grows younger as he moves away from Earth, both remain in our familiar positive time despite a certain negative increment. Positive time is our world, our reality, our dimension, since every object in the material world (the world of matter with positive mass), including the observer, moves below the speed of light. Their unattainable upper limit is the speed of light. Photons, the visible part of electromagnetic radiation, and all “massless”, energy-carrying particles move at that speed. When a particle attains the speed of light, the magnitude of the negative increment of its time exactly matches the positive time elapsed since it began moving. If a microscopic cosmonaut were inside a photon, he would have to wind his clock back as he went, making sure its hands never moved from their starting position. The result for such an object is “time zero”. The particle starts out and reaches its destination at the same instant of “stopped” time. It seems to “catch up with itself” and “double” (remember the “double slit” experiment). At the speed of light an object can be in several places at once and thus acquires the properties of a wave. A wave differs from a particle precisely in being observed at a given moment in many places at once, rather than in one particular place. Incidentally, it is the ability of photons to be in both positive and negative time simultaneously that allows them to be “their own antiparticles”. Finally, if an object exceeds the speed of light, the negative increment in its time is so large that its time reverses completely. It becomes genuinely negative, takes a minus sign, and cause and effect are inverted in turn: it will “reach its destination at seven in the morning — yesterday”. Because bodies travelling faster than light cover any stretch of space sooner than photons do, they always outstrip the corresponding light wave even when heading towards the observer. The extra distance they inevitably cover

beyond the light rays will then lead away from the observer, which means negative time again. So objects that exceed the speed of light are always in negative time, whatever direction they are travelling in. Light never catches up with them. They cannot reflect it and cannot become “secondary” sources of light. Negative time is therefore a world invisible to a subluminal observer, another dimension, a different reality. What are the entities in this reality? We know from quantum mechanics that antiparticles live in negative time. Their characteristic spin indicates that they are in the reverse light cone, rather than the observer’s own cone. It is logical to suppose that there are structures made of antimatter on the cosmic scale too, moving faster than light and hence remaining in negative time, invisible to us. Indirect signs of their existence are apparently what make scientists talk about the notorious “black holes” that absorb even light (or, more accurately, fail to reflect it!) or some mysterious “dark matter”. Such phenomena are only hypothetical, like the “tachyon”, a hypothetical microscopic object in negative time. But if invisible faster-than-light cosmic objects are possible, why assign the role to something conjectural? It is perfectly real antimatter that has “disappeared”, that has been “lost”. In reality there ought to be just as much of it in the universe as there is ordinary mass, because, by the law of conservation, what appeared were pairs of antipodal particles. They could vanish altogether only in pairs, too. If ordinary particles are still with us today in material objects, we should expect the same of their counterpart antipodes. Structures made of antiparticles are “nonmaterial” structures: antimatter, invisible to us. Incidentally, there may well be more of this virtual, imaginary substance in the cosmos than there are material objects. For it to cancel a positive unit, it takes the square of the imaginary unit itself. That can supply the “minus one” of mass needed at the moment of annihilation. It seems that “tachyons”, “black holes” and the other “mysterious invisibles” are simply states of

objects made of antimatter, the states they take in their natural habitat, negative time. As we know, transferring antimatter into our time so that we can “observe and measure” it is done by … cooling faster-than-light structures down almost to absolute zero (one way of slowing particles down). Is it any wonder that the object’s parameters turn out to differ from what we expected? It is perfectly natural that we cannot directly observe imaginary, “tachyonic” mass and can only grasp it conceptually, since the mass of “fast” particles depends on their speed. As one would expect, the experiment affects its own result. Perhaps quantitative changes bring about a qualitative change: depending on its speed, a particle can become a wave or even an antiparticle, and vice versa. A single fundamental entity in different “incarnations”. This suggests that all discrepancies between the interpretations of classical theories and quantum mechanics arise simply because it is difficult for the researcher to interpret processes on the cosmic scale in strict accordance with those of quantum objects. On the cosmic scale the researcher is inside the system and cannot act upon the objects in that macroworld. Taking these considerations into account could help resolve many of the contradictions and problems that have come to a head in physics, particularly in understanding the nature of the Big Bang. It seems to be nothing other than another annihilation of all the matter in the universe with all its antimatter. This annihilating explosion is quite capable of providing the colossal energy from which the antipodal worlds begin to form again, in an endless cycle of matter changing into radiation and radiation into matter. This is exactly how the physical world, governed by the law of conservation, logically works. It is the unique and all-embracing Cosmic Perpetual-Motion Machine, analogous to the only eternal system known with certainty to present-day orthodox science, the “particle–antiparticle” pair in a vacuum.

A Paradigm for a Pair of “Diggers”

Physicists have long wondered what could have become of the antiparticles that formed in the early universe in accordance with the law of conservation. Annihilation did not take place, as the very existence of the material world shows. So there can be only one satisfactory answer. If they “disappeared” at all, they vanished only from our field of view. In reality, the invisible cosmic bodies made of the supposedly “lost” antimatter (antipodal copies of familiar celestial bodies) are alive and well to this day in reverse, inaccessible, negative time. What is negative time? A human being sees only light that has reached the retina. A flash of light always happens before it is seen, because photons take time to reach the observer. Thus photons, and indeed all objects approaching the observer, move “from earlier to later”, from the past into the future, in ordinary, forward-flowing positive time. It is logical to conclude that objects (photons and others) moving away from the observer, conversely, travel in reverse, “backwards” in time. For example, at any given moment, the farther a celestial body is from Earth, the deeper in the past it is relative to the observer.

Our eyes cannot register receding photons (those that do not strike the retina). But an object that happens to find itself in a photon stream may be able to reflect the receding photons and send them back to the observer, although later than the original flash. Then we will see the photons, now reflected and with a certain delay, indicating a negative increment in their motion through time. If the object travels below the speed of light, the photons will catch up with it fairly soon. Time will slow a little but stay positive. The object will become a source of reflected light, and the observer will register it as such. At the speed of light the negative increment cancels out the entire advance of positive time. Time “goes to zero”, stops, and the object displays wave properties, the ability to be in many places simultaneously (all “massless”, energy-carrying particles travel

at exactly that speed). Finally, for an object travelling faster than light, time reverses altogether. It begins to run from the future into the past and takes negative values. Objects in genuinely negative time are completely beyond our visual reach. Travelling faster than light, they cannot in principle be overtaken by photons, whatever direction they are moving in. They give off no reflected light and remain invisible to us for ever. It is no use trying to “lay eyes on” the universe’s dark matter. Only indirect signs will point to its undoubted presence. Evidently this is the “missing” antimatter. The conclusion is borne out by quantum mechanics: antimatter lives in negative time (the rule of CPT symmetry). Reversed time goes hand in hand with completely reversible entropic processes. That is precisely what ensures the eternal existence of the physical world and makes it a Cosmic Perpetual-Motion Machine, a cyclic process of formation, development and involution of the antipodal worlds from one annihilating Big Bang to the next, “ad infinito”.

“If all you’re armed with in a scientific dispute are facts and logic, your concept will be rejected.” Unless two or three “diggers” who chance to read it dig down to the heart of the matter. You can dig, or you can leave it. Up to you.

Once Again on the Evolution of the Universe

The “Big Bang”, currently the most popular cosmological theory, does not quite work. The Pope praised it for agreeing perfectly with Church dogma: there was nothing, and then God set off that very Big Bang. But to a real scientist, God is *persona non grata*. Even if we grant the existence of a Creator, science will immediately ask what this God is, exactly, and where he came from. In other words, the concept of God as something unknowable is a dead end for science. Yet the Big Bang theory cannot tell us where all that colossal energy came from. After the Bang, though, from the very first milliseconds, it is more or less clear how the Universe acquired its structure. Nothing supernatural about it. Entropy increases overall, but thanks to the store of energy supplied at the moment of the explosion, the system is driven away from equilibrium and, developing nonlinearly, gives rise to self-organising “subsystems”, from atoms and galaxies to living organisms, thinking ones included. The very beginning, however, can be explained only as follows. The Universe and the Antiuniverse exist eternally. This is a dynamic state: they periodically appear, develop and disappear at the moments corresponding to the Big Bang and the Big Crunch. Quantum mechanics describes a similar process, the continual appearance and disappearance (annihilation) of virtual “particle–antiparticle” pairs. Since energy cannot come from nothing, one partner has positive energy and the other negative energy, Hawking notes. The same holds for the “Universe–Antiuniverse” system. Gravity comes from the force of attraction between these two worlds. It “pulls” all objects together and curves space and time, ultimately bringing about an inevitable Big Crunch at the end of each cycle, in accordance with Friedmann’s first model. But that is only a new beginning. The energy of annihilation makes it possible for the Universe (and the Antiuniverse) to

develop again in a new cycle, and so it goes, “ad infinito”. Perhaps this idea, which “stares us in the face”, has occurred to scientists more than once and been dismissed for the simple reason that astronomers have never confirmed the presence of antimatter objects in the cosmos. Antiparticles are a subject exclusively of quantum mechanics. Why should that be? Let me remind you that the only celestial bodies we can observe either shine themselves or reflect light. Suppose, however, that some object in space travels faster than light. A light wave cannot catch it and therefore cannot reflect off it. In that case, astronomers cannot in principle detect the object. It will always be in the “reverse cone” of light. It is well known that a material body cannot attain the speed of light, much less exceed it. As for antimatter, though… why not? If mass is a measure of inertia, might “antimass” have precisely the opposite properties, serving as a kind of “measure of mobility”? When studying the objects of quantum mechanics, a researcher is, as it were, “outside the system” and can sometimes “take a peek” into the reverse light cone as well. Might this explain a property such as a particle’s spin, whereby the particle looks different in different directions (like … the phases of the Moon)? Quantum mechanics tells us there is a particular relationship between matter and antimatter, spatial symmetry and the two directions of time. If we replace particles with antiparticles and spatial coordinates with their mirror reflections, and reverse the direction of time (from the future into the past), the physical laws governing the resulting object remain unchanged. Similar relationships apparently hold in the “Universe–Antiuniverse” pair. The Antiuniverse consists of antimatter, has negative energy, is a “mirror image” of our Universe and exists backwards in time. This preserves all the physical laws in the system. The total sum of matter and energy remains unchanged, equal to zero, in accordance with the First Law of Thermodynamics. The Second Law of Thermodynamics, concerning the increase of entropy, also holds, of course, for

the material Universe. Its processes are irreversible and run in the positive direction of time, from the past into the future. In the Antiuniverse, where time runs backwards, entropy as a function of time gradually decreases. In this way all the system's parameters remain perfectly symmetrical (and stable) at any instant. Thus time, which stops at the moment of the Big Bang, runs in two directions as it moves away from that moment, on either side of zero, just like the spatial coordinates. In the Universe time is positive, and in the Antiuniverse it is negative. In the Antiuniverse the curvature of space takes on a "negative" value. Space seems to turn "inside out", curving into the shape of a saddle instead of the surface of a sphere. Or, if you like, it resembles the inner surface of a sphere, a mirror image of the Universe's shape. Interestingly, some authors say that it is precisely this "saddle-shaped" curvature of space that is needed to travel backwards in time. Only objects travelling faster than light travel into the past, which is to say they can "outrun themselves". But objects that can travel at or near the speed of light can "catch up with themselves". For them time seems to stop. They "double" and can be in two places at once. This is exactly what happens to particles in the double-slit experiment when they begin to display wave properties. Recognising the existence of the Antiuniverse resolves practically all the outstanding questions of modern cosmology. In particular, we can find the "missing" antiparticles, which evidently never went missing at all (of course they didn't: that would contradict the First Law of Thermodynamics). They merely combined with their own kind into increasingly complex structures, following the example of their material partners. As they acquired the requisite speed, they disappeared … from our field of view. In conclusion, I would add that the entropy of the Antiuniverse, like its energy in general, evidently has a negative sign. This is precisely "negentropy", or, what amounts to the same thing, information in Shannon's sense. So what does that leave us with? Cosmic fields of information? A higher

intelligence, perhaps? The indestructibility of our consciousness, which is nothing other than all we know about ourselves and the world around us, in other words, information? Perhaps the immortality of our souls? I don't know. I would only echo Spinoza: "God is Nature." There is no point looking for anything outside it.

The Creator's Lot (Systems Theory Leaves Him No Choice)

1. The creator of the system "All That Is" cannot be outside the system. If he is not included in a system defined as "Everything That Exists", he simply does not exist.
2. Nor can he belong to the system "All That Is" as one of its parts, alongside his creation. A part is always simpler than the whole, and a creator cannot be simpler than the system he created.
3. There is only one way out: identify the Creator with Creation. The Creator is All That Is.

If you find the reasoning above doubtful, we are unlikely to find a common language.

The Mistake

She swept into the lecture hall, stopped at the blackboard and, oblivious to our astonished looks, launched straight in:

Quantum mechanics has established that the coordinates of objects made of antimatter are mirror images of their material counterparts' coordinates, and that these objects exist in negative time running from the future into the past (the rule of CPT symmetry). It is logical to suppose that antimatter on the cosmic scale obeys the same rules and therefore exists in negative time too. What is negative time in the cosmos? As we know, scientists today put the limits of the visible universe at

about 14 billion light-years. That means the light we register now from the farthest observable points began its journey to Earth 14 billion years ago. Photons from that distant past made it all the way to us, in the present, at the speed of light. That is, their motion through spacetime has a distinctly positive direction: from past to future. Motion in the opposite direction, accordingly (whether of photons or indeed any objects at all), goes from future to past. A reminder: we can see only the light that falls on the retina, which is to say we register only positive spacetime. Regions of negative time and space arise on the cosmic scale wherever photons move away from the observer instead of towards them. Material objects that find themselves in a region of negative time will inevitably be overtaken by the receding photons. They will reflect the “fugitives” off their surfaces and send them back to the observer, thus revealing their own presence as secondary sources of (reflected) light. Bodies made of antimatter, however, move faster than light and will remain in negative spacetime for ever. The photons will not catch them, so they will not turn them into sources of light. Reversed (negative) spacetime is another dimension, a world invisible to us. Yet it is every bit as real for that. The “dark matter” astrophysicists have been looking for in vain is evidently precisely this antimatter inaccessible to observation. Processes in the “mirror” world are reversible, as is characteristic of time running backwards and of faster-than-light speeds. It makes perfect sense, really. When a quantity is measured by distance (and that is how we measure cosmic time), there is no reason to suppose it cannot take negative values. The speed of the moving object is what decides everything. Sometimes time stays positive and shows only a negative increment. That is precisely why time for a subluminal cosmonaut slows compared with the time measured by an observer on Earth. At the speed of light, time for the moving object would stop. It would then be in two places at the same instant. It would display the properties of a wave. Exceeding

the speed of light would reverse time: it would run backwards and change from positive to negative. The “cosmonaut” would end up in the past and “reach his destination at seven in the morning — yesterday”. It is regrettable that even physicists who count themselves adherents of the Theory of Relativity sometimes turn out not to understand these regularities and say, for instance, that negative time can supposedly be talked about “only in jest”. Physicists do often joke, of course, but the only ones who can laugh for no reason are … the most easily amused.

She fell silent and looked round the hall. Seeing the philology students, their mouths open in astonishment as they waited for a lecture on “The Linguopoetic and Lexicographical Aspects of Palindromes”, she frowned.

“Excuse me… which building is this? Oh, number twelve? My mistake, I’m sorry… Don’t go away. Your lecturer will be along in a moment, I’m sure.”

Just then Ella Sergeyevna’s tousled blond curls appeared in the half-open door. The stranger apologised once more and slipped past the philologist on her way in. We started chattering about what had happened, but Ellochka rapped her knuckles impatiently on the blackboard and everyone fell quiet. Only Kirill, making his way hurriedly between the desks towards the exit, was muttering under his breath: “Let me through, let me through… I’m off to the physics department!”

Negative Time

> It is indeed a dreadful crime
> Not to admit the backward time.

A human being sees only light that has reached the retina. A flash of light always happens before it is seen, because photons take a certain amount of time to reach the observer. It follows

that photons, and indeed all objects approaching the observer, move “from earlier to later”, from the past into the future, in ordinary forward-flowing positive time. Conversely, photons and other objects that move away from the observer go backwards in time. For instance, at any given moment, the farther a celestial body is from Earth, the deeper in the past it is relative to the observer. Our eyes cannot register receding photons (those that do not strike the retina). But an object that finds itself in a photon stream may be able to reflect the receding photons and send them back to the observer, albeit later than the original flash. Then we will see those photons in reflection, with a certain delay indicating a negative increment in their motion through time. If the object is travelling below the speed of light, the photons will catch up with it fairly soon. The object will become a source of reflected light and sooner or later the observer will register it as such. Its time will slow down a little but stay positive. If the speed is equal to that of light, the negative increment cancels out the entire advance of positive time. Time “goes to zero”, stops, and the object displays wave properties: the ability to be in several (many) places at the same instant. All “massless”, energy-carrying particles travel at exactly that speed. Some are visible and some are “invisibles”. Finally, for an object moving faster than light, time reverses altogether. As it runs from the future into the past it takes negative values. Objects in genuinely negative time are completely beyond our visual reach.

The Law of Antipodes (Nobel Lecture)

Esteemed members of the Nobel Committee! Esteemed ladies and gentlemen!

In mathematical logic there is Kurt Gödel’s incompleteness theorem, which means that any complex formal system must necessarily be inconsistent to be complete. If our perception of the world is accurate (otherwise we simply would not have

survived in it), we must conclude that inconsistency is a feature of both the systems that do the describing (formal systems) and those being described (real systems) if they lay claim to completeness. Thus, alongside the material Universe that everyone knows, the cosmos undoubtedly contains its antipode, an invisible, nonmaterial world made of antimatter and an exact mirror image of our Universe. Time in this Antiuniverse runs backwards, its objects consist of antiparticles and its energy is negative. The part of that energy equal in magnitude to the Universe's entropy but opposite in sign, namely negentropy, takes the form of cosmic fields of information, a kind of "World Mind". An Antiuniverse with these characteristics is subject to the known laws of physics, in particular the law of conservation. This guarantees the perpetual existence of information as part of the system's negative energy, allowing us to suppose … that our consciousness is eternal. Consciousness is simply the information individuals have accumulated about the world around them and about themselves. Since decreasing entropy corresponds to the reversed flow of time, it is logical to conclude that all processes in the Antiuniverse go from chaos to order, "from worse to better". Highly organised systems therefore have no need to struggle with their surroundings to survive, and natural selection has no role in their evolution. An individual must experience the way such a world is organised as a kind of perfect "paradise" in which the very conditions for evil are absent. Recognising that the Universe has a nonmaterial antipode makes a supernatural Creator superfluous, since this dual world is self-sufficient. It has its own "built-in" mechanisms for endless cyclic emergence and development, from one annihilation to the next (from the Big Bang to the Big Crunch), in a perpetually repeating cycle. One might say that this dual world can itself be regarded as God, an eternal, indestructible entity that includes fields of information, the "World Mind", as part of a single, all-embracing organism.

All existing kinds of field, all material structures and all nonmaterial structures made of antimatter, every object and phenomenon without exception, are simply components and life processes of this contradictory, continually changing and transforming, dynamic and eternal organism. Objects in the Antiuniverse cannot be registered by an observer because of the limitations of our senses. Evolution has adapted them to perceive only the material world that produced us as a biological species. In particular, we can see celestial bodies only if they emit or reflect light. When an object exceeds the speed of light, as is natural for motion in negative time running backwards from the future into the past, a light wave cannot reflect off it because it cannot catch up with it. So structures in the Antiuniverse remain invisible to us. Still, knowing that even the tiniest material particle cannot come into being without its corresponding antiparticle and that they too can disappear only in pairs, through mutual annihilation, we must conclude with confidence that antiparticles did not disappear. Like their material counterparts, they must inevitably have formed an antimatter world of their own, the exact antipodal counterpart of the material Universe. Recognising the dual nature of the existing world makes it possible to resolve many of the contradictions that have accumulated in modern physics and cosmology. Recognising the dual nature of the existing world allows us at last to dispense with the primitive, anthropomorphic “god” proposed by ancient religious scriptures and give the only possible definition of the Divine that agrees with scientific data and the logic of things: God is everything that exists. Any attempt to place God, the creator of All That Is, outside His Creation is entirely senseless. Whatever is outside everything that exists simply … does not exist. On the other hand, God cannot be part of “Everything That Exists”, since the creator of a system cannot be simpler than the system he created and a part of a system is by definition always simpler than the whole. In other words, God is Nature itself

rather than its supernatural creator. God and Nature are synonyms. We need only bear in mind that Nature is exactly twice as complex as we used to think, the difference being the Antiuniverse. It is possible, however, that negative mass takes the form of the square of an imaginary mass, supplying the necessary “minus one” at the moment of annihilation. There is nothing supernatural in this system. God’s laws are the very laws of Nature, which human beings, as biological creatures, simply cannot break. A person is only the tiniest cell or particle of an all-embracing organism, living first in its material part and then in its antimaterial part, subject to the physical conditions that prevail in each. Those, in brief, are the consequences of the Law of Antipodes I have postulated, the First Law of the Cosmos.

Thank you for your attention!

Five Principles of the Cosmos

First Principle

The law of conservation, which affirms the eternity and indestructibility of matter and energy and their conversion into one another, embraces every physical category without exception: matter, antimatter and energy of every kind and form, including entropy as part of the system’s positive energy, and negentropy, that is, information, or entropy with the opposite sign in Shannon’s sense.

Second Principle

In principle we cannot see cosmic objects made of antimatter. Because they move faster than light, they are always in the reverse light cone relative to the observer (in negative time). Their presence is confirmed by indirect evidence, however, by their unmistakable effect on bodies in the material universe. This is what makes astrophysicists look for the notorious “dark matter”.

Third Principle

The Big Bang is nothing other than the annihilation of all the matter (mass) in the world with all the antimatter it contains. The energy of the Bang sets the antipodal worlds developing again, until their next annihilation. The process is cyclic and eternal, like the process for a “particle–antiparticle” pair.

Fourth Principle

Whether a quantum-mechanical object is a particle or a wave depends entirely on its speed. Below the speed of light it is a particle. At or near the speed of light it is a wave. On reaching speeds above that of light the object presents itself as an antiparticle, with corresponding changes in all its other parameters.

Fifth Principle

Time, like the spatial dimensions, has two directions along the coordinate axis, one on either side of zero, which is the flash, the “event”. There is positive time in the forward light cone and negative time in the reverse cone. The motion of any object towards an observer is always motion in the positive direction. Away from the observer it is motion in the negative direction, both in space and in time.

Twins Without a Paradox

> I have to say, no physicist is worth a dime
> Who won’t admit the backwards flow of time.

Einstein once declared: “We are compelled to regard as possible a mechanism for transmitting a signal whereby the resulting action (the effect) precedes its cause. This result contains no logical contradiction whatsoever!” He caught his colleagues’ murmurs of disapproval and hastened to add: “However, since all our experience offers no evidence of a violation of cause and effect, we can consider the impossibility

of faster-than-light speeds proved.” Everyone sighed with relief. Only the unruly M. Gell-Mann stubbornly grumbled: “Everything that is not forbidden is compulsory!” Einstein nodded understandingly and spread his hands in an apologetic gesture. At that unfortunate moment, unnoticed by anyone, the “Twin Paradox” loomed into view.

“There are no real paradoxes in physics,” R. Feynman insists. “All paradoxes are just confusion in our own understanding.” Rejecting faster-than-light speeds (and with them negative time) runs counter to the logic of the theory of relativity. That rejection led to the idea that, after making a loop through outer space and returning to Earth, the travelling twin would be younger than his stay-at-home brother. A paradox still regarded as unresolved to this day. Yet the answer is simple. The human eye registers photons only when they strike the retina. A flash of light always happens before it is seen, because photons take a certain amount of time to reach the observer. Thus photons and indeed all cosmic objects approaching the observer move “from earlier to later”, from the past into the future, in ordinary positive time, with its absolute values increasing. Conversely, as they move away from the observer, they move backwards, from the future into the past, going deeper into it the farther away they get. In other words, when the traveller changes direction in space, he also changes direction in time. On a round trip, then, the difference between the traveller’s clock and his stay-at-home twin’s, which first appears and grows on the outward leg, diminishes on the way back and disappears altogether when the brothers meet. When they embrace, they will be the same age again, as twins should be. The logical error that led to the paradox, as we can see, was to ignore the vector of motion and take only the magnitude of the velocity into account. Unfortunately, negative time running backwards (from the future into the past) is still a long way from being accepted by all researchers. But once you allow for it, “there was a paradox, and now there’s none”.

It should be stressed that for speeds beyond the ordinary limits (relativistic speeds), the direction of motion is in fact irrelevant. For objects travelling at the speed of light, cause and effect occur at the same instant. As these objects acquire wave properties, they set out and reach their destinations simultaneously, as if they “multiplied”, being in several places at once. Faster-than-light objects live permanently in negative time, since they are always ahead of the light wave they were “on the same starting line” with at the outset. In principle they cannot be observed visually and make themselves known only through indirect signs. In negative time entropic processes are reversible, which is why effect precedes cause. Just don’t say that this refutes the theory of relativity. More precisely, it takes the theory further.

The Music of Time: Crescendo and Diminuendo

The theory of relativity, properly speaking, means that time should be different for objects at different points in the spacetime continuum and the same for objects at one and the same point. So the meeting of a traveller who has grown “disproportionately young” with a brother who has grown “prematurely decrepit” in their shared kitchen is a most vexing paradox.

After all, as Feynman rightly pointed out, the existence of paradoxes merely means that there is something we do not quite understand. Why tolerate such an inexcusable inconsistency and try at all costs to justify the paradoxical situation with calculations of “acceleration and deceleration”? As Einstein ought to have said, “If the calculations contradict the theory, so much the worse for the calculations!” Quite right: if the calculations help explain why paradoxes … occur in reality (!), so much the worse for them. And it is fairly simple to clear up the confusion about the twins: acknowledge that there

is time that runs backwards, from the future into the past. In fact, the slowing down, the compression, of time is precisely a negative increment in time. It arises because the object is moving away from the observer rather than as a consequence of the speed of its motion. Any emission of photons happens before the observer can register it, since the photons take time to get from the flash (the “event”) to the observer. In other words, in this case the photons move from the past into the present. But motion in the opposite direction, away from the observer, is motion backwards in time, from the present into the past. So on a round trip (a loop), the difference between the two brothers’ clocks, which first appears and grows on the traveller’s outward leg, diminishes on the way back and disappears altogether when the twins meet. When they meet at a single point, they will be the same age again, and there is no paradox! Naturally, a higher speed allows the twins to move away from each other more quickly, or to draw closer more quickly. At a higher speed the gap in the brothers’ time therefore grows or shrinks more quickly. That is, the logical error that led to the paradox lies in taking the magnitude of the velocity into account while ignoring the vector of motion, the cosmic “time zones”.

A Freakish Fixation on the Multiverse

> “Real science is done by freaks.” (A. Nevzorov)

God was invented to explain how the world came into being. Saying that God made the world raises the question: who made God? Creationists answer that God is eternal. Scientists wonder whether, in that case, this mysterious God is an unnecessary entity. For according to the Law of Conservation, the physical world itself is eternal! Not everyone can follow the scientific reasoning. People don’t like delving into things that are complicated and hard to understand. The clerics had a ready answer: faith needs no proof at all. You must believe any

nonsense unconditionally. The result: believers on Earth still far outnumber atheists and agnostics. A stupid idea catches on more readily with the masses. But devotees of science trivia won't give up. What if we stopped taking things on faith and looked for scientific evidence that God exists? What would cybernetics, that paid whore of imperialism, have to say, for example? The physical world unquestionably exists. Suppose God, the creator of that world, also exists. The Creator and Creation together would then make up a system that we could call "everything that exists". It takes in everything. Whatever is outside this system does not exist, by definition. So the world and God are two components of a single cybernetic system. But Systems Theory tells us that the Creator of a system cannot be simpler than the system he created. A part is always simpler than the whole. Even if the world later grows more complex by improving itself, the creator remains part of "all that is". He doesn't stick his head outside, lest he "fall out of existence". Here is the paradox: he must belong to the system, yet cannot be one of its parts. The solution is obvious. We must identify the Creator with Creation: God is everything that exists. That is, God is the world. In this sense alone can we still speak of a divine entity. Why? Good question! But, as the song goes, "is it worth talking about?" Where were we? God is the world, and it is eternal. How, then, can this Perpetual-Motion Machine exist, given that it is perhaps the only possible one of all "perpetual-motion machines" as such? So we move smoothly on to the conditions for eternity.

The question arises: is there any known eternal system in the world that has been studied sufficiently well? Imagine that, there is! It is the virtual "particle/antiparticle" pair, which appears and disappears in the vacuum through annihilation, owing to fluctuations in the equally eternal physical field. Everyone knows that matter and energy can turn into each other. No formula in the world is more famous than Einstein's equation describing these transformations. It is hard to find

anyone who has never heard that “E equals m c squared”. The equation describes the reciprocal transformations of mass and energy and states that energy is equivalent to mass multiplied by the square of the speed of light. And if, God forbid, energy is negative, the mass will be negative too, since the speed of light is squared. Annihilation is the only way matter can be converted entirely into radiation. More precisely, it converts matter and antimatter into radiation and back again. Annihilation requires antipodes, quantities of the same kind but of opposite sign. The term “annihilation” (*nihil*, nothing) is used for the phenomenon in which a particle and its corresponding antiparticle “disappear” and turn into photons or still lighter quanta of the electromagnetic or another field. Pair production is the reverse process, in which a particle and an antiparticle arise at the same time from the interaction of physical fields. It is worth remembering here that field quanta (energy-carrying particles) can be their own antipodes and are therefore capable of annihilation. Because “the large repeats itself in the small”, antipodes that cancel each other can be quite “elementary” or infinitely complex, from quarks to universes. The Big Bang (which is also the Big Crunch) is an ordinary annihilation of matter with antimatter too. Only the scale differs. Naturally, the more complex the structures taking part in this endless cyclic process, the longer the interval between annihilations. For entire worlds (universe and antiuniverse), it is billions of years.

The sum of all parameters in the antipodes is always zero in pairs, because the physical quantities of one mirror copy have signs opposite to those of the other. Physicists in Geneva regard the study of antimatter as a way into another mathematical universe, where everything positive becomes negative and left becomes right. They should say **another physical reality**. It is a world where, in accordance with CPT symmetry, space, its curvature, time and the other parameters acquire the minus sign characteristic of the “world through the

looking-glass". Mass in the symmetrical reality is represented by the square of an imaginary, tachyonic mass. (Allow for the familiar baryonic mass of the same magnitude, and "three quarks for Muster Mark" begin to come into view.) But squaring an imaginary number gives a negative number. Thus at the moment of the "Big Bang" the quantities that annihilate readily acquire precisely opposite signs. Otherwise they would add up instead of cancelling out. Energy in the antiworld is negative accordingly, including entropy, because time running backwards necessarily means that entropic processes and causal relations are reversible. Negative entropy is information, in the definition given by Brillouin, whose name is seldom heard in "broad circles of narrow specialists". How can we represent such a dual system mathematically? It is simple enough, since for exact antipodal copies the sum of their masses, in particular, is zero.

It has long been known (algebra is wiser than the natural scientist) that zero is a complex number with real and imaginary parts. The imaginary unit is written "i". To the first power it is the square root of minus one, and raised to the second power, "i^2", it equals minus one. Hence the **formula for everything**:

$0 = \text{Nothing} = x - x = x + (-x) = x + (-1)(x) = x + (i^2)(x) = \text{Something}$

For the annihilating interaction of worlds, we can put the total baryonic mass of the universe in place of "x", for instance. We can use the left- or right-hand side of Einstein's equation… though in fact the magnitude of the quantity itself is immaterial, so long as we remember its exact antipode. And on the scale of the universe, "something" is synonymous with "everything". Even if there are many such pairs of worlds (we can disregard them out of respect for old Occam), the principle is the same: a dual, dialectically contradictory entity is self-sufficient and eternal. The most important prerequisite, the

indispensable condition for eternity, is the existence of antipodes. According to the Law of Conservation, the physical world is eternal precisely because it includes an imaginary ("virtual") component, the Antiuniverse made of antimatter, alongside the Universe made of matter.

Why has the idea of an antiuniverse, which has occurred to scientists more than once, fallen out of favour? It was rejected because even seemingly perfect optical instruments failed for so long to register any of its objects. The learned gentlemen ran out of patience. But how did they fail to return to the antiuniverse hypothesis even after finding dark matter? Quantum physicists speak of negative mass, faster-than-light speeds and such things in passing, furtively and with a guilty look. Discredited and laughed at, the idea of a CPT-symmetric antiworld "hangs in the air" and … waits for its hour.

"Dark" matter, which cannot be registered visually, is precisely the accumulation of supposedly "missing" antimatter. Only the speed barrier separates the antipodes. Objects in negative time travel faster than light. "When you travel far faster than light, you'll reach your destination at seven in the morning — yesterday." In the broadest sense, everything that exists is information, with its own binary code. Yes, one and zero. But we must allow for the complex nature of zero, which lets us use the imaginary and negative numbers we badly need in order to describe the metamorphoses of the single fundamental entity whose existence Lee Smolin suspects. A system that lays claim to completeness must be contradictory. It is naïve to suppose that it can be described within, say, the positive natural numbers alone. So when you use renormalisation, think of a yellow monkey!

The cybernetic system of "all that is" can take the form, first of all, of a field of information, a dual one, naturally (the measure of chaos is simply the measure of order with the opposite sign). Second, it can take the form of any other energy field whose

quanta act as their own antipodes. And finally, the energy of annihilation can take form in the structural elements of the antipodal worlds, which gradually develop and then slowly undergo involution. This goes on until the next annihilation and the whole cycle repeats “ad infinito”. Information, then, is a kind of “genetic matrix of the world”. How can we help recalling the disconsolate Avery’s “DNA double helix”?

The most painstaking thinkers have long puzzled over what happened to the antiparticles. The early universe was teeming with antipodal pairs. Annihilation plainly did not take place: look, here’s the material world! What’s more, if we make the effort, we can even extract, say, antihydrogen from nonbeing. “Know-how”! So the right answer is: we can’t see the gopher, but it’s there! Antimatter is hiding from us somewhere. Let’s get our brains in gear. The only light we see is light that has reached our rods and cones. Photons take time to get from the light source to the receptors of the retina. Any flash (an emission of photons) is registered after it occurs. Thus photons, and everything else, moving from the “event” towards the “observer” go from the past into the future (the normal flow of time). But motion away from the observer goes backwards, into the past. Motion from future to past can either reduce the absolute value of positive numbers or increase the absolute value of negative ones, depending on the value of the initial instant. With a negative increment in a positive magnitude, speeds below that of light allow us to register only the compression, the slowing down, of positive time. At the speed of light, the negative increment “cancels out” the entire advance of positive time. Time goes to zero and stops. Energy-carrying particles “spin their wheels” in time and multiply as they display wave properties. Can you guess what happens to faster-than-light objects? Exactly! For those objects time always has a minus sign. That is why they are invisible bodies in the cosmos.

To reject negative time and its speeds beyond the limit is tantamount to rejecting the theory of relativity. Einstein himself bears some blame for this state of affairs. He once declared: “Since all our experience offers no evidence of a violation of cause and effect, we can consider the impossibility of faster-than-light speeds proved, although the existence of a mechanism for transmitting a signal whereby the effect precedes the cause contains no logical contradiction.” Only the unruly Gell-Mann, author of the “law of conservation of strangeness”, grumbled in reply: “Everything that is not forbidden is compulsory!” All credit to the stubborn fellow!

The Sigma Paradigm

A reply sent to the journal Nature by prior arrangement with Sir Richard Whitemere, former chairman of the Club of Undaunted Geniuses (translated from Esperanto).

Physicists have long wondered what could have become of the antiparticles that arose in the early universe in accordance with the law of conservation. Annihilation did not take place, as the very existence of this world shows. There can therefore be only one satisfactory answer: if they “disappeared”, it was only from our field of view. In reality, the invisible cosmic bodies made of the supposedly “lost” antimatter are alive and well to this day in reverse, negative time that we cannot access. What is negative time, then? An observer sees light only at the moment the photons reach the sensitive cells of the eye. Since every flash occurs before it is seen, photons travelling towards the observer always go from past to future, in the normal, positive flow of time. Photons and other objects moving away from the observer, on the other hand, go backwards in time, from future to past, and give time a negative increment. Time for objects travelling below the speed of light simply slows down and stays positive. At the speed of light the negative increment is so great that time reaches zero and stands still. The object therefore

doubles, multiplies, acquires wave properties, which is to say the ability to be in different places at the same moment. Objects travelling faster than light enter truly negative time, with entropic processes and cause-and-effect relationships reversed. Cosmic objects in negative time are entirely beyond the visual reach of a subluminal observer. Those of us who are subluminal need not even try to “lay eyes on” the universe’s dark matter, but there is no doubt that it is none other than the supposedly “missing” antimatter. The conclusion is borne out by quantum mechanics: antimatter lives in negative time. And the complete reversibility of entropic processes that goes with reversed time ensures the eternal existence of the physical world, turning it into a Cosmic Perpetual-Motion Machine. The cyclic process of formation, development and involution of the antipodal worlds continues from one annihilating Big Bang to the next, “ad infinito”.

Schrödinger’s Cat,
Kh. B. B.-L.
37 Mayvember 2050

Memo

The rule of CPT symmetry helps us see that the symmetrical world is the world of antimatter. Alongside the real world there is unquestionably a symmetrical world (also called the mirror, virtual, imaginary, tachyonic or “holographic” world), on both sides of the baryonic counterpart along the line of sight. If space curves on relativistic scales, the arrangement will resemble “globules” of three units treated as identical, rather like quark triads. Imaginary particles create an imaginary, virtual, invisible world in the reverse cones of every event in the spacetime continuum, the counterpart and antipode of the material universe (the universe made of matter). A world of faster-than-light speeds and reversed causal relations. The Antiuniverse.

The Path of Progress

> “The only path of progress is to refute the existing theory and create an alternative.”
> R. Feynman

The bulk (the inert mass) of today’s physicists prefers to consider the twin paradox resolved. But as the saying goes, “In reality, everything is different from how it really is.” The twin paradox remains unresolved and cannot in principle be resolved within the framework imposed by conservatives, who are bent on making it an insoluble problem for the theory of relativity as a whole. So what is the essence of the paradox? The twin paradox is a thought experiment involving two brothers. One twin sets off on a journey through space at close to the speed of light, after which he returns to Earth. By the relativistic effect, the traveller can just as well suppose that it is his stay-at-home brother who is moving, and each twin will observe the other’s clock running slowly compared with his own. Nevertheless, according to special relativity, the brother who left and returned will be younger than his twin. Here is the paradox: why do both observe time slowing down, yet only the traveller ends up younger than the one who stayed home?

The accepted “solution” boils down to saying that we cannot, supposedly, neglect the traveller’s accelerations. His frame of reference is not inertial and consequently there is no symmetry in their movements. But is that really so? To begin with, there is no need to take forces of inertia or gravitational fields into account, if only because the “ideal”, near-light-speed brothers in the thought experiment have no mass. But that is of little importance. What matters more is that simply projecting one’s own motion onto one’s brother’s movements cannot in principle selectively change particular details of that motion. When the cosmonaut accelerates relative to the other twin on parts of the journey, his acceleration must be just as relative as his velocity. In other words, if the travelling twin recedes, approaches, stays

at rest, slows down or speeds up relative to his brother, the stay-at-home brother likewise recedes, approaches, stays at rest, slows down and speeds up relative to the traveller. Every feature of the twins' trajectories in space and time is perfectly symmetrical from each other's point of view, as the principle of relativity requires. In other words, symmetry is preserved and there should be no paradox. The "calculations of acceleration and deceleration" offered in reply merely confirm the paradox by looking at the situation from some third, irrelevant viewpoint, in which Earth (and its earthling) stands still and the traveller moves, thereby creating the asymmetry. No wonder Einstein lamented, half in jest: "Once the mathematicians pounced on my theory, I stopped understanding it myself!" Yes, the twins are not on equal terms in the "outside observer's" frame, but who gave us permission to jump from one frame of reference to another at will? The very term "relativity" implies that what is observed depends on which frame we choose. And the conservative "solvers" are plainly looking for the viewpoint that shows them what they want to see: "the carriage will move and the platform will stay put". But there are only two twins in the thought experiment, and all we should consider is how each perceives their relative motion. Apart from the two brothers, there is simply nothing! In essence there is one motion, "attributed" to two different objects as relativity allows. Where, pray, does the asymmetry come from? And even if… One painstaking philosopher made an interesting point: "It's far from obvious," he said, "that periods of acceleration so short in duration can produce such tectonic differences in time." Yet none of this is the main point. What matters, when we judge by the strictest standard, is that it makes no difference how these hypothetical brothers move. It makes no difference even whether the stay-at-home twin moves at all or whether we can treat him as stationary. Indeed, we can disregard their trajectories altogether. What matters is that the brothers eventually meet at one point in space. And one of the

conclusions, or postulates, of the theory of relativity states that time should be different for objects at different points and the same for objects at one and the same point. Now there's a paradox if ever there was one! How are we to understand the meeting of a traveller who has grown "disproportionately young" with a brother who has grown "prematurely decrepit" in their shared kitchen? Why tolerate such an inexcusable inconsistency with relativity and try at all costs to justify the paradoxical situation with calculations of "acceleration and deceleration"? As Einstein ought to have said, "If the calculations contradict the theory, so much the worse for the calculations!" Quite right: if the calculations help explain why paradoxes … occur in reality (!), so much the worse for them. In Feynman's entirely sound view, there are no real paradoxes in physics. All paradoxes are just confusion in our own understanding.

It is fairly simple to clear up the confusion about the twins: acknowledge that there is time that runs backwards, from the future into the past. In fact, the slowing down, the compression, of time is precisely a negative increment in time. It arises because the object is moving away from the observer, rather than as a consequence of the speed of its motion. As we know, an observer registers photons only when they strike the retina. Any emission of photons occurs before it is seen, because photons take a certain amount of time to reach the observer (the light of dead stars "reaches us"). Thus photons, and indeed all objects, moving from any flash (an "event") towards the observer (wherever the observer may be, on or off Earth) move from past to future in ordinary positive time, with its absolute values increasing. Conversely, movement away from the observer goes backwards, from the future into the past, going deeper into it the farther it gets from the recipient. In other words, when the traveller changes direction in space, he also changes direction in time. On a round trip (a loop), the difference between the readings of the two brothers' clocks,

which first appears and grows on the traveller's outward leg, diminishes on the way back and disappears completely when the twins meet. When they meet at one point, they will be the same age again, and there is no paradox! Naturally, a higher speed allows the twins to move away from each other more quickly, or to draw closer more quickly. At a higher speed the gap in the brothers' time therefore grows more quickly or shrinks more quickly, depending on which way the traveller is going. That is, the logical error that led to the paradox lies in taking the magnitude of the velocity into account while ignoring the vector of motion. Correct the error and the paradox vanishes! What's more, this simple and elegant solution actually confirms, clarifies and improves the theory of relativity, rather than refuting it, in keeping with Bohr's correspondence principle. So read on and understanding will come later. For Einstein's main idea was: "The platform will move and the carriage will stay put!"

Monitoring Time in Space

> "The use of the term 'spacetime' is due to a poverty of terminology. We might just as well call water 'hydrogen–oxygen'."
> M. Kamenarovich

> "Where nothing happens, there is no time. For time is not a quantity introduced into a system from outside, but an immanent property of the world, bound up with the nature of the changes that take place in it. Several times may be possible, moving in different directions. Some of them are reversible, others are not. From the observer's point of view … only one time flows, of course. This is because the observer measures time by their own clock, because they have immersed all the various streams of time in the single time given to them by Nature." (Stanisław Lem)

All classical scientific theories laid claim to describing the world objectively, but did not take into account the influence of the observer on the object of inquiry. The new physics is meant to break with the tendency to underestimate the role of the observer, while still allowing for the possibility of objective knowledge. Both relativity and quantum mechanics have shown how necessary this approach is. To remain objective, we must clearly establish the circumstances in which observers assess the information they receive. We need to understand why our assessments are involuntarily limited and anticipate the consequences. For instance, an observer can see only emitted or reflected light, at the moment the photons strike the receptors in their eyes (optical aids do not change the principle). Photons take time to reach the observer, so particles are always emitted before the emission is registered. That is, photons move from the "event" to the observer in ordinary positive time, from the past into the future. Present-day orthodox science, with the exception of a few outstanding scientists, considers positive time the only real time. Nevertheless, let us try to look more deeply into the situation. Obviously, if the emission of the photons is taken as the initial moment, time zero, then as the particles move towards the observer their time coordinates will take positive values of increasing magnitude. If instead we set the origin ("time zero") at the observer's location, it makes sense to conclude that the farther the moving point is from the observer in space, the greater the magnitude of its negative time value. We can, of course, count from any "nonzero" point, bearing in mind that motion away from the observer always takes place in "reverse" time, from the future into the past. Backwards motion in time can decrease the absolute value of positive numbers or increase the absolute value of negative ones, depending on the coordinates of the starting point. By contrast, in motion from the event to the observer, as already noted, the changes correspond to an increase in the magnitude of positive values

or a decrease in that of negative ones. This holds for any subluminal observer, wherever they may be. It is evidence both for the isotropy of the universe and for the objective character of the knowledge obtained. Speed plays only a supporting role in the sense that the greater the speed, the more quickly the absolute values of the time readings increase or decrease. If the object being observed is travelling below the speed of light and the increment in magnitude is negative, we can register only the compression, the slowing down, of time, whose values nevertheless remain positive.

Because the backward motion of photons (away from the recipient) cannot be observed, genuinely negative time remains “elusive”. This has given people grounds to speak of a supposedly one-way “arrow of time” in the Universe, since our world of positive mass is a world of speeds below that of light. At the speed of light (characteristic of energy-carrying particles), however, the negative increment “cancels out” the entire advance of positive time. Time for photons and other particles travelling at the speed of light goes to zero. The particle acquires the properties of a wave, as if it “multiplied” in several places at once. This offers a logical explanation of the quantum–wave duality of particles. Faster-than-light objects, which are beyond our visual reach, get a kind of “overshoot” of reverse time because of their speed beyond the limit. That keeps them permanently in genuinely negative (“minus”) time, whatever the direction of their motion. Figuratively speaking, a faster-than-light “traveller” grows younger than his brother and even younger than himself, entering his own past. The world of speeds above that of light is a world of “cosmic invisibles”, a world of reversed causal relations. In quantum mechanics, antiparticles live in reverse (negative) time (the rule of CPT symmetry). The “dark matter” of the Universe is apparently made of antimatter as well. That is, if antimatter has “disappeared”, it has disappeared only from our field of view, which makes any talk of a violation of the law of conservation

quite senseless. Human beings cannot perceive the “symmetrical” world through their senses. That hardly means the world is unreal: indirect signs and the logic of things point unequivocally to its existence. We should also understand that the idea of negative time does not refute the theory of relativity. It develops and improves it, making relativity a special case of a new, more general theory that can resolve many outstanding problems of modern physics. The very problems whose solutions have long been “in the air”.

The Antiuniverse

An article sent to the journal Nature by prior arrangement with Sir Richard Whitemere, former chairman of the Club of Undaunted Geniuses (translated from Esperanto).

The “Big Bang”, currently the most popular cosmological theory, is not much of a success. No wonder the Pope himself praised it: apparently it fits Church dogma perfectly. There was nothing, and then God set off that very “Bang!” Easy for the Pope. But not all scientists are happy with it. Even if we grant the existence of a Creator, the question immediately arises: what is God himself, then, and where did he come from? So for science, God is *persona non grata*. After the explosion, though, from the very first microseconds, we have a fair idea of how the Universe acquired its structure, apart from one small “but” which I will come back to (S. Weinberg, *The First Three Minutes: A Modern View of the Origin of the Universe*, New York: Basic Books, 1993). Nothing supernatural about it, on the whole. Entropy increases overall, but the store of energy supplied at the moment of the “explosion” drives the system away from equilibrium. It develops nonlinearly (P. Coveney and R. Highfield, *The Arrow of Time*, 1990) and gives rise to self-organising “subsystems”, from atoms and galaxies to living organisms, thinking ones included. Yet where the colossal energy of the Big Bang came from remains unclear. It looks as

though things happen like this: the Universe and the Antiuniverse exist eternally. This is a dynamic state, in which they periodically appear and disappear (annihilate, “cancel each other out”) at the moments corresponding to the Big Bang and/or the Big Crunch. A similar process is well known in quantum mechanics. S. Hawking writes, for example, that in so-called “empty space” pairs of virtual particles and antiparticles continually come into being and disappear. They appear together, move away from each other and come together again to annihilate, releasing a certain amount of energy (S. Hawking, *The Illustrated A Brief History of Time*, Bantam Books, 1997). “Since energy cannot be created out of nothing,” Hawking goes on, “one of the partners in a particle–antiparticle pair has positive energy and the other negative energy.” Apparently this is also the case for the “Universe–Antiuniverse” system. Gravity comes from the force of attraction between these two worlds. It opposes the expansion of the Universe that began with the Big Bang, “pulls together” all its objects and curves space and time, ultimately bringing about an inevitable Big Crunch at the end of each cycle. Annihilation again supplies the colossal energy from which the Universe begins to develop in a new cycle, and so it goes, “ad infinito”. So of Friedmann’s three models, hitherto considered equally probable, Model I, the closed Universe, is the right one. But the Big Crunch it foretells is only a new beginning.

Let us return to quantum mechanics. There is a complex relationship between matter and antimatter, spatial symmetry and the two directions of time. If we replace particles with antiparticles and spatial coordinates with their mirror images, and reverse the direction of time (from future to past), the physical laws governing the resulting object remain unchanged (C, P, T invariance) (P. Coveney and R. Highfield, *The Arrow of Time*, 1990). Similar relationships apparently hold in the “Universe–Antiuniverse” pair. The Antiuniverse consists of

antimatter (for the definition, see S. Weinberg, *The First Three Minutes*), has negative energy, is a “mirror image” of our Universe and moves (develops, exists) backwards in time. This preserves all the physical laws in the system (whereas all laws break down at the “singularity” prescribed by the Big Bang theory). Indeed, the “total sum” of matter and energy in the “Universe–Antiuniverse” pair, as in a “particle–antiparticle” pair, remains unchanged, equal to zero. That is, in accordance with the First Law of Thermodynamics, the fundamental law of nature, energy (and matter) cannot come from nothing or disappear. The system exists eternally and needs no Creator. As for the Second Law of Thermodynamics, concerning the increase of entropy in the Universe, it too holds, of course, for the Universe, where irreversible processes run in the positive direction of time, from past to future. “Irreversibility is the development of a system in one direction, with its own inherent direction of time” (P. Coveney and R. Highfield, *The Arrow of Time*, 1990). As we know, classical and modern physical theories alike allow both directions of time, the “law of time symmetry” (P. Coveney and R. Highfield, *The Arrow of Time*, 1990). Entropy is a function of time. When time goes from past to future, entropy rises as time goes on. When time “changes sign” and goes from future to past, entropy falls accordingly. That is, in the Antiuniverse, which consists of antimatter and exists backwards in time, entropy gradually decreases. This maintains perfect symmetry across all the parameters of the “Universe–Antiuniverse” pair at any instant. So time, which according to Einstein can “slow down”, evidently “stops” at the Big Bang / Big Crunch and goes in the opposite direction in the Antiuniverse. In other words, time, like the other coordinates, has two directions, one on either side of zero. The curvature of space, too, may (and logically could), on reaching its maximum at the “singularity”, take on a “negative value” in the Antiuniverse. Space then seems to turn “inside out”, curving like a “saddle surface” rather than the surface of a sphere, or

more precisely like the inside surface of a sphere, a “mirror image” of the configuration of the Universe. Interestingly, some authors (S. Hawking, for example) say that it is precisely this “saddle-shaped” curvature that is required for travelling “backwards in time”.

Here is my translation of Hawking’s limerick:

> “A lady they nicknamed Miss Bright
> Once wanted to ride down the street.
> But she raced out of sight,
> Past the speed of light,
> And here was the price of that feat:
> Back at her own front door she arrived,
> Long before she had left for the ride!”

Since material objects cannot exceed the speed of light, such “effects” are presumably the prerogative of the Antiuniverse, its virtual particles and the structures made of them. However, some quite interesting things happen in the Universe too at speeds close to the speed of light. As we know from quantum mechanics, wave and particle are two different definitions of the same entity. The dual nature of particles is demonstrated by the “double-slit experiment”. If light (electromagnetic waves) passes through two narrow slits in a barrier to reach a recording screen, interference fringes appear on the screen, as is natural for wave processes. It was also found that if the source of light is replaced by a source of particles, say electrons sent one at a time, interference fringes appear on the screen again, although we might reasonably expect each electron to go through one slit or the other. Yet the interference pattern shows that the electron goes through both slits at once! This led to the conclusion that the electron is both a particle and a wave. What might explain this phenomenon? As we know from “thought experiments”, at speeds above that of light, an object can “outrun itself” (“A lady they nicknamed…”). Apparently at the speed of light it is possible to “catch up with

oneself". "Any particle whose rest mass is zero," writes S. Weinberg, "such as a photon, a neutrino or a graviton, moves at the speed of light. Material particles approach the speed of light if their energy is comparable in magnitude to their rest energy mc^2, where m is the mass of the particle" (S. Weinberg, *The First Three Minutes*). Perhaps such a particle, travelling close to the speed of light and "catching up with itself", seems to "double", being in "two places at once", right next to itself. It can therefore pass through both slits simultaneously and create an interference pattern, earning the "status" of a wave. As we know, wavelength is inversely proportional to the mass of the particle, which makes perfect sense since it is harder to "speed up" a heavier particle to the speed of light. As for the quantisation of energy, the fact that energy (electromagnetic waves, light for instance) is emitted in distinct portions, or "quanta", rather than in a continuous stream, it follows naturally from the fact that the photon is first and foremost a particle, even though with its zero rest mass it travels at the speed of light, at which wave properties are most pronounced.

Back to cosmology. The existence of an Antiuniverse does not seem quite so unlikely if we recall the description of conditions in the early Universe, in the first moments after the Big Bang. "At first," write I. Nicolson and P. Moore (*The Universe*, Equinox, Oxford, 1985), "particles and antiparticles were continually created and destroyed by one another. Then … a tiny excess of particles over antiparticles arose, for reasons that are not yet entirely clear." This is the small "but" mentioned on the first page. To understand the reasons for that excess, let us turn again to quantum mechanics. Quantum mechanics says that each particle has a partner, an antiparticle. Obviously, what is meant here is each "class of particles". But if we take the statement more literally, to mean that each individual particle has a "partner", an antiparticle, and this is exactly what was happening initially in the "homogeneous soup" of the early Universe (see, for example, S.

Weinberg or I. Nicolson and P. Moore), we can suppose that if an excess of particles arises in the Universe, an equal excess of antiparticles must accumulate somewhere. That is, when the Universe began to acquire structure as the temperature fell (after the Big Bang), and material particles joined into increasingly complex structures, antiparticles likewise congregated into antimatter, into the Antiuniverse. Such structuring (joining with “their own kind”) prevents particles and antiparticles from annihilating immediately and delays the process for billions of years, from the Big Bang to the Big Crunch!

Given the above, it seems logical to conclude that the Universe, the sum of all particles, must inevitably have a partner, the sum of all antiparticles: the Antiuniverse. There is nothing “supernatural” about such an object. Virtual particles, of which it is made, are not picked up by detectors, as we know. Yet the existence of antiparticles is confirmed by their effect on material particles, and quite a conspicuous effect it is, right up to the annihilation of the latter. In the same way, the Antiuniverse, the cause of the coming Big Crunch, is real because it exerts a decisive influence on our Universe: on its origin, on its evolution down to the smallest detail, on the motion of all its objects (along geodesic baskets in the curved, four-dimensional spacetime continuum), on its end and on the subsequent repetition of the cycle. The Universe and the Antiuniverse are tied by strict reciprocal relations. Because time runs in opposite directions in the two worlds, an event that is due to happen in the Universe “tomorrow” has already happened in the Antiuniverse “yesterday”, and vice versa. That is, our world determines the past of the virtual world just as inevitably as the virtual world determines our future.

In this virtual, “imaginary” world, which exists backwards in time, the use of imaginary numbers in calculations is justified. “Imaginary time is time measured using imaginary numbers”

(S. Hawking, *The Illustrated A Brief History of Time*). Mathematicians have long used imaginary numbers to solve cosmological problems (S. Hawking, R. Penrose and others), but until now this was regarded as an artificial device, although Hawking does say, “Perhaps what we call imaginary time is actually more fundamental than real time” (S. Hawking, *The Illustrated A Brief History of Time*). Given the above, it would be more accurate to conclude that imaginary time is at least as fundamental as real time. It is simply that the world turns out to be exactly “twice” as complex as we have imagined it until now, the difference being the Antiuniverse. It is then eternal, indestructible and self-contained, needing no Creator and simply repeating an endless cycle of states. Perhaps we may regard the Antiuniverse, where entropy falls and negentropy, in other words information, consequently increases (C. Shannon, I. Prigogine, L. Brillouin), as a world of information. Some may find it tempting to call this virtual world of information a “divine” substance, a Higher Intelligence which supposedly has our future fixed in it. Well, that is a matter of taste. I would only point out that there is no use praying to such a God. Reciprocal ties like these between the “world” and the “antiworld” are simply a consequence of the system’s way of existing. They are determined by clear, immutable physical laws and admit of “no appeal”. We may, of course, regret that our “free will” is strictly speaking only an illusion in this world so rigidly determined by cause and effect. But because we cannot possibly predict coming events, we must take the “illusion” quite seriously. I would only echo Spinoza: “God is Nature.” There is no point looking for anything outside it.

Why, then, have scientists not yet registered the bodies of the Antiuniverse? Let us try to make sense of it. As we know, on cosmic scales the concepts of space and time are inseparably linked. We measure the distance between celestial bodies by the time it takes light to cover it. We also know that a light wave spreading from a source travels both towards the

observer (the forward light cone) and away from the observer (the reverse light cone). If we take the moment of a particular flash as the origin, it is natural to count the direction of the photons' motion as positive in the forward cone and negative in the reverse cone. Time undergoes corresponding changes. In the reverse cone, unlike the forward cone, it takes negative values, because each successive point lies further in the past than the preceding one, at a greater number of light-years from the observer. That is, in the reverse cone photons move backwards in time, from the future into the past, on a negative time scale. In fact, we can regard any object moving away from the observer as moving backwards in time. That is precisely why time slows down for the cosmonaut. But does the cosmonaut enter negative time?

Let us conduct a thought experiment. Suppose the cosmonaut travels at half the speed of light (anything is possible in a thought experiment). Then, travelling backwards in time for, say, ten years, he covers a distance of five light-years. That is, he finds himself five years in the past compared with Earth time and must therefore set his clock back five years:

10 − 5 = 5

This means that the cosmonaut's time has remained positive, although it has slowed by half. If instead the cosmonaut travels at the speed of light, he must set his clock back, while travelling backwards in time, by exactly as many years as he spent on the journey:

10 − 10 = 0

In other words, time has as good as stopped for him. At the end of his flight the cosmonaut will be at the point where the tips of the forward and reverse cones meet, at the epicentre of the wave. Finally, if the cosmonaut exceeds the speed of light, say by a factor of two, he enters the reverse light cone, into negative time! The cosmonaut's time is reversed and runs

backwards, since in ten years of travel he sets his clock back twenty years:

10 − 20 = −10

Apparently the notion that time has only one direction persists simply because material objects cannot attain the speed of light, much less exceed it. Any object moving at less than the speed of light will sooner or later be overtaken by a light wave, which will reflect off it and return to the observer. The only light we can perceive is light coming towards us and striking the retina. Light from the reverse cone can be registered by the observer only if it reflects off an object it meets along the way. Then it will once again be in the forward light cone, but coming from a different source. As we know, all celestial bodies shine by their own light or by reflected light, and in that sense they are distinct sources of light for the observer. Is it any wonder that when we consider only light travelling in one direction, we also take time to be a quantity with only one direction?

Does that mean, however, that there are no objects in the cosmos capable of moving in negative time? Not at all. It merely means that astronomers cannot detect them. Since their speed exceeds that of light, a light wave will never catch them and reflect off them, however long ago they set off and however long they keep going. The existence of such objects is confirmed only by indirect evidence (see above). There is no doubt, however, that the antiparticles that formed in the early Universe cannot disappear. They evidently could not have annihilated with their partners, material particles, since the latter still exist in objects in the material Universe. The only thing left for the antiparticles was to follow their example and group together into bodies of the Antiuniverse. But whereas mass is a measure of inertia, antimatter objects have apparently acquired the mobility they need to evade the keen attention of astronomers tirelessly searching the Universe for dark matter. Objects that move faster than light always remain

in the reverse light cone, so in principle they cannot be registered by an observer on Earth. Yet it is the sum of those very objects that constitutes the Antiuniverse.

It should be stressed that when we deal with objects of quantum mechanics, where the observer is in a sense outside the system and, under certain conditions, can register what happens in both the forward and reverse cones, we should be particularly cautious in classifying particles. Clues here may include whether or not the object has rest mass, and properties such as its charge and spin. Finally, could it be that photons, this substance on the boundary, are their own antiparticles precisely because they can visit both worlds?

Kh. B. B.-L.
15 December 2000 – 12 Marchvember 2050

A Notch in the Bark and Timber of the Brain

1. An observer sees only light that has struck the receptors of the eye. Since every flash happens before it is seen (photons take time to reach the observer), it is clear that photons travelling towards the observer go from “earlier to later” (from the past into the future).
2. When light from a source heads away from the observer, however, the receding photons are not registered visually at all, except if they overtake an object along the way and return to the observer after reflecting off it.
3. The farther away the photons travel before reaching the point (or moment) of reflection, the farther they must travel to get back to the observer. In other words, the return takes longer.

4. So when photons move away from the observer they move backwards in time (from the future into the past), each successive point lying deeper in the past than the last.

Einstein, a Violin and a Touch of Nerves

Dear ladies and gentlemen, hear me out once more! I stand by my words now as I did before. We are compelled to regard as possible a mechanism for transmitting a signal whereby the resulting action precedes its cause. This result contains no logical contradictions! Your humble servant's theory proclaims the relative nature of time, whose dimension is inseparably linked to the speeds at which objects move through space. Echoes of my ideas have entered common parlance in reflections and jokes about what we might expect if the speed of light were exceeded. Take this charming limerick from my colleague Hawking:

> "A lady they nicknamed Miss Bright
> once wanted to ride down the street.
> But she raced out of sight,
> past the speed of light,
> and here was the price of that feat:
> back at her own front door she arrived,
> long before she had left for the ride!"

I've never written verse myself, but I took great pleasure in the wit of those lines. And today, watching you argue without end, I think it my duty to stress that an outcome preceding its cause occurs precisely at faster-than-light speeds: the object reaches its destination before it sets out. It lands in the past, moving in reversed, negative time. To say that Einstein's theory supposedly forbids faster-than-light speeds is both inaccurate and thoroughly absurd. As with the speed of light itself, they are impossible for objects with positive rest mass, but I assure

you that this is only part of a much more complicated picture. For instance, all massless, energy-carrying particles move at the speed of light. And they have wave properties, that is, the ability to be in several places at once. Cause and effect are simultaneous, synchronous, at the speed of light. Antimatter, by contrast, lives in reversed time in accordance with the rule of "CPT symmetry". It most probably has the imaginary, tachyonic mass Dirac predicted. The tremendous speed of an object travelling faster than light makes cause and effect trade places. I am very sorry that you modern scientists are so timid that, on encountering unmistakable though indirect signs in your work that the speed of light has been exceeded, you lapse into a complete stupor and disown the results of your own experiments. Or you dream up explanations so preposterous they're laughable, imagining that otherwise you might inadvertently refute old Albert. I am deeply touched by such tender regard for me, but the truth matters more. Remember: finding faster-than-light objects does not refute the Theory of Relativity in any way! On the contrary, I'll be delighted to see my cherished brainchild, the idea of the relativity of time, finally develop as expected and reach a worthy logical conclusion. And now, my young friends, your humble servant would like to play you a little sonata on the violin. Listening to this charming piece is a good deal more pleasant than the long, tangled disquisitions of an old physicist. What matters most in the world, after all, is simplicity and elegance!

My Picture of the World

1. There are equal numbers of particles and antiparticles in the world.
2. The total sum of matter and energy is zero.
3. We cannot see cosmic objects made of antimatter. They move faster than light, so light does not reflect off them.

4. In the case of microparticles, which we can observe in both the forward and the reverse light cone, their faster-than-light speeds affect the characteristics of their spin.
5. The system exists eternally, periodically appearing and disappearing through the annihilation of the antipodal worlds at the Big Bang (the Big Crunch).
6. Counterbalancing entropy is negentropy, equal in absolute value but opposite in sign: cosmic fields of information.
7. The existence of a virtual, “imaginary” world justifies the use of imaginary numbers. “Zero” then becomes a perfectly definite quantity expressed as a complex number, that is:

The formula for everything:

$$E - E = E + i^2E$$

God from the Standpoint of Systems Theory

If we speak of All That Is, everything that exists, we should not place God outside Nature, since God would then fall under the concept of All That Is as part of the system:

EVERYTHING = GOD + NATURE (THE WORLD)

A part of a system is by definition always simpler than the whole system. Yet the Creator of a system cannot be simpler than the system he created. Thus we must either exclude God from the concept of “Everything That Exists”, in which case He … does not exist, or, as seems the only correct option, identify the Creator with His Creation:

EVERYTHING = GOD = NATURE (THE WORLD)

God and Nature are synonyms, and nothing lies outside this concept. In other words, nothing is supernatural. In studying

Nature, we come to know God. On this view, we should seek the mechanism by which the Universe came into being in Nature itself, and it must obey the known laws of physics. Amen.

Cosmo- and Theo-logies, or a Test for Divinity

> “For the believer, only what is sacred is true. For the scientist, only what is true is sacred.”
> L. Feuerbach

An American priest sadly admitted: “We make our parishioners park their brains at the church door.” At the risk of drawing the wrath of many believers, I must say that I agree with him entirely. All existing religions present a false, distorted picture of God, far too primitive for someone of a sober, critical turn of mind to believe in such a God. When a genuine scientist says they believe in God, we must understand that their image of God differs greatly from the one an illiterate peasant woman, for instance, might imagine. As the person is, so is their God. In that sense, we make God “in our own image and likeness”. I need hardly say that the savage medieval customs, cruelties and prejudices that abound in fundamentalist readings of religious texts are unacceptable to a civilised modern person. People with sufficient intelligence and education will also shake their heads doubtfully at the fairy tales about miracles, the legends and fantasies that fill scriptures written centuries and even millennia (!) ago. Naturally, the God of the past corresponded to the knowledge, ideas, traditions and circumstances of the time. To survive in an enlightened modern society, religion has to modernise. It is no secret, though, that any modernisation of faith is very painful. Remember the scholars of the Middle Ages who paid with their lives for their scientific convictions, because their picture of the world contradicted the Church dogma of the time. Yet as science

opens up new horizons, one after another, the Church, forced to reckon with undeniable facts, announces … a new boundary to knowledge. For instance, discussing the “Big Bang”, currently the most popular cosmological theory, at the Vatican, the Pope told the scientists present: “It is permissible to study the events that followed the Big Bang, but the Bang itself and what preceded it are God’s work. Human beings should therefore take no interest in those mysteries.” Naturally, only the most enlightened and educated religious figures, the elite, go so far in agreeing with science, all the way to the “Big Bang”. The middle ranks and, especially, the flock are much more conservative. In the religious press, for instance, the question of whether Darwin’s theory is harmful because it “contradicts the Bible” is still stirred up to this day. But in principle the Church, at least at the top, is prepared to agree with the natural course of the evolution of the Universe proposed by scientists and declare only the very beginning, the moment of the Bang itself, to be the creative act of the Creator. For the simple reason that science cannot (could not?) yet offer an even moderately acceptable alternative of its own as to the source of the energy of that very “Bang”. The religious “explanation” is ready at hand: God had a mind to set off a sort of “fireworks display”, and that was how everything began. Oddly enough, most people are satisfied with that explanation. But to an inquiring mind it borders on the absurd. Suppose God created the Universe. What, then, is this God, and how did he himself arise? In other words, to science God is only a new riddle, rather than an explanation of all that exists. One can, of course, declare God incomprehensible to the human mind, as the churchmen have done with admirable cunning. But in that case why not simply be satisfied, with as much success, with the claim that the Universe itself is incomprehensible? Fortunately, the human mind refuses to recognise artificial boundaries to knowledge and pays no heed to prohibitions.

There are people to whom only the truth is sacred. What follows is addressed to them.

Let us put the question before us this way: could the Universe have arisen spontaneously, or did its origin require the intervention of a supernatural force from outside, a Maker, a Creator? Unfortunately, it is doubtful that the material Universe we know could have arisen spontaneously. That would contradict the First Law of Thermodynamics, the Fundamental Law of Nature, which says that matter (or energy, which amounts to the same thing, since the two quantities are equivalent) cannot arise from nothing. Does that leave us with an unknowable Creator? Let's not jump to conclusions. Before accepting the supernatural, let us try to remember whether we know of any natural system capable of arising spontaneously. Yes, such a system exists. From quantum mechanics we are familiar with the continual, spontaneous formation of virtual "particle–antiparticle" pairs in the vacuum. This does not contradict the fundamental law of nature, because the particle comes into being … paired with its antipode, the antiparticle, so that the total amount of matter (energy) in the Universe does not change. According to modern cosmology, the same process occurred in the early Universe, where during the first milliseconds after the Big Bang "particle–antiparticle" pairs constantly formed and disappeared through annihilation. Only later, as the temperature fell, did material particles begin to form increasingly complex structures, from atoms and molecules to planets and galaxies, while the antiparticles … "disappeared somewhere". But they could not have disappeared! That, too, would contradict the First Law of Thermodynamics, the "law of conservation". Antiparticles can "disappear" only together with their antipodes, material particles, by annihilation (mutual destruction), when both are converted into an amount of energy equivalent to the mass that disappeared. And yet material particles … continue to exist to this day in the objects of the material world.

Does it not make sense to suppose that the antiparticles did not vanish either? Like their material partners, they would have formed comparable structures with their own kind, from antimatter atoms and molecules to antimatter planets and galaxies made up of them, objects of the antimatter world, the Antiuniverse. Then we must rephrase our question. Instead of "Where did the antiparticles disappear to?" we must ask "Why can't we see the objects made of them, the antimatter planets and galaxies of the Antiuniverse?" Here we ought to ask ourselves why we see celestial bodies in the first place. The answer is simple. We see them if, and only if, they emit light themselves or reflect it. Now imagine that some object in the cosmos can move … faster than light. A flash (a wave) of light would be unable to catch up with it and would therefore never reflect off it. A person could not in principle see such an object. Material bodies cannot exceed the speed of light, but the suggestion that objects consisting of antiparticles, of antimatter, can do so hardly seems so absurd. Matter is a measure of inertia, and its antipode, antimatter, could logically serve … as a measure of mobility. The conclusion is that besides the known material world we observe, the Universe, there may well be an invisible, nonmaterial world in the cosmos whose objects move faster than light and so lie beyond our powers of observation.

Isn't it the presence of an invisible, nonmaterial world that theologians talk about? Apparently there really is such a world, and there is nothing "supernatural" about it. We can speak with a fair degree of confidence about the existence of the Antiuniverse, because acknowledging it makes it possible to resolve practically all the outstanding questions of modern cosmology and lead physics out of the dead end in which it currently finds itself. But all these aspects are the subject of other works (see, for example, "The Antiuniverse: An Equal Partner of the Universe in the Cosmos", "The Tamers of Time", "Once Again on the Origin of the Universe", "Catching Hold of

the Good Lord's Tail", and others). Here we will confine ourselves to the central question for the problem under discussion, the first cause of the Big Bang. In the view set out above, it is nothing other than … the result of the annihilation of two worlds, the Universe and the Antiuniverse. This annihilation supplies the colossal energy from which the Universe (and, naturally, its antipode) begins developing in every new cycle. Everything happens just as it does in a "particle–antiparticle" pair, except that the time from one annihilation to the next is measured in milliseconds for particles and in billions of years for universes. So what the Universe needs in order to arise is an equal partner in the cosmos, the Antiuniverse, rather than an unknowable supernatural Creator. Its existence follows logically from the requirements of the First Law of Thermodynamics, the "Law of Conservation", or Fundamental Law of Nature. Does this mean we should dispense with the idea of "God" altogether? Or should we instead regard this invisible, virtual, nonmaterial world itself as the Divine, as God, since it is the sufficient condition for the origin and eternal, cyclic existence of our Universe? Again, let's not jump to conclusions. Let's try to get to the bottom of the question more thoroughly. What properties, then, might an object such as the Antiuniverse have?

Specialists know that if particles are replaced with antiparticles and spatial coordinates with their mirror images, and the direction of time is reversed, from future to past, the physical laws governing the resulting object remain unchanged. The same relations should hold in the Antiuniverse. The Antiuniverse consists of antimatter, has negative energy, is a "mirror image" of our Universe and exists backwards in time, as does any object travelling faster than light that, figuratively speaking, "outruns itself" and travels into its own past (a term familiar to specialists). Accordingly, the entropy of the Antiuniverse, as a function of time, also changes to the opposite

sign. There is nothing strange about this. The Universe's entropy is part of its positive energy. Entropy in the Antiuniverse is part of that antimatter structure's negative energy and therefore takes a negative sign: negative entropy, negentropy, or … information in Shannon's sense. Rather than a "measure of chaos", as the Universe's entropy is, it is a measure of order. Its absolute value is naturally equal to the sum of all the entropy in the Universe. Objectively existing … fields of information, then? A "Higher Intelligence", perhaps? Well, why not? There is nothing "criminal" about the term. Moreover, as we can see, there is nothing "supernatural" about it either, since everything in the system obeys the laws of physics we know. Now, if fields of information exist eternally in the cosmos, does that not open up the possibility of immortality for our "soul", our consciousness? Higher organisms acquired the capacity to absorb information as a cosmic substrate in the course of evolution, when signalling, the higher nervous activity of living systems, emerged. Our consciousness, our memory, is simply information: the sum of our knowledge and ideas about ourselves and the world around us. And perhaps when a person dies, the physical body remains in the material world, while consciousness, our informational essence, like radio waves when a receiver breaks, does not disappear without trace. That would violate the First Law of Thermodynamics. Instead, it joins the fields of information in the cosmos and continues to exist in one form or another. Here we should note that to a consciousness born under the conditions of the material, earthly world, an antiworld where entropy decreases and everything therefore moves from chaos to order, from worse to better, must seem a place of perfect goodness, a kind of "paradise after death". In the Antiuniverse the harsh law of natural selection ceases to apply. In a world of decreasing entropy, there is no need to "swim against the current" by struggling against the environment, which for an individual is everything outside that individual. So it seems the

theologians are right about the immortality of the soul too. The only thing missing is … punishment for sinners in the antiworld, where conditions are apparently favourable to all its inhabitants. And again, nothing “supernatural”, nothing that violates known physical laws. Those laws are only more fully realised if we accept the idea of the Antiuniverse.

We must note, however, that the fields of information in the cosmos, this “Higher Intelligence”, cannot be identified with “God”. Here is why. Let me recall two simple rules from cybernetics which, in this connection, seem to me worth our attention and logically impeccable:

First: a part of a system cannot be more complex than the system as a whole.
Second: the creator of a system cannot be simpler than the system they created.

Both rules will help us conduct the “test for divinity”. Let us reason it through. Since negentropy, the fields of information, is only a component of the Antiuniverse, these fields cannot count as “God” under the rules of systems theory set out above. That is, the “Higher Intelligence” is not God, but perhaps only something inherent in him, just as our human intelligence is only one attribute of a person as a thinking being. Would the Antiuniverse as a whole qualify as God, then, since it includes the cosmic fields of information and is a sufficient condition for the origin of the Universe? As usual, let us avoid hasty conclusions. Before answering that question, let us try to define the Divine correctly, in a way that does not contradict logic and that might satisfy someone to whom only the truth is sacred. I hope no one will object if I define God as Something eternal and self-sufficient, requiring no outside force to come into being or to exist. Does the Antiuniverse fit this definition? The links between the Universe and the Antiuniverse are reciprocal, just like those between a particle and an antiparticle. Neither is self-sufficient. They can arise and disappear (annihilate) only

together, paired with one another. Only as a pair can they come into being and exist without violating the Fundamental Law of Nature or the other laws of physics. And only as a pair do they form a stable, viable dynamic system, since all the processes and phenomena in the two subsystems are mutually reversible. So, recalling the definition above, it seems that only both entities together, the Universe and the Antiuniverse as a single whole, can pass the “test for divinity”.

Let us look at this system more closely. The Universe and the Antiuniverse exist eternally, repeating an endless cycle of transformations from annihilation, through evolution, to the next mutual destruction. Although the system exists eternally, its existence is neither static nor continuous. It is dynamic, pulsating, alive. In accordance with the First Law of Thermodynamics, the total energy (matter) of the system does not change and is always zero:

Universe + Antiuniverse = E − E = $mc^2 - mc^2 = 0$

Yet we can also write this equation in another form if we note that −E is −1 multiplied by E, and that −1 in turn equals i^2:

Universe + Antiuniverse = E − E = E + i^2E

That is, a definite quantity expressed as a complex number, including “i”, the imaginary unit (its square). It should be noted that imaginary numbers are quite natural for describing the imaginary, virtual world in which time runs backwards, the so-called “imaginary time”.

Thus **0 = E + i^2E**, so that “Nothing” is at the same time “Something”, or “Everything”, if you like, since E equals all the energy (matter) in the Universe.

What have we got, then? “Everything” and “Nothing” at the same time, “Yes” and “No” at the same time. As if it were a clump of information, which is characterised by just such a binary code. And this “Something”, this “Everything”, exists

objectively and eternally, neither coming into being nor disappearing, only changing form, regardless of whether the information is “written” into material objects, entities of the virtual, invisible world, or both.

Does it not make sense to call this Something God, since it exists eternally, is self-sufficient and requires no outside force to come into being or exist? There is nothing “supernatural” about this concept, for God is Nature itself, rather than something outside it. Nature obeys immutable laws of physics and includes both the material Universe, its visible and fairly well studied part, and the Antiuniverse, a part inaccessible to our senses. The latter is no less real for that, since without it the material world simply could not have come into being or function normally in accordance with all the laws of physics. And everything in Nature, in accordance with the law of conservation, exists eternally: all material and virtual objects, all the fields and particles belonging to them, planets and galaxies, elephants and midges, and every last human being. And all of them are no more (and no less!) than cells, tissues and organs of this Dual World, this One God, who had no other material to make everything from but himself.

One interesting consequence follows. Human beings cannot break God’s laws, because breaking the laws of nature is simply beyond their powers as creatures of nature. People can break religious laws (which, incidentally, are by no means always humane), and they can break legal or moral laws, for which they should receive fair punishment from their fellow human beings. But they simply cannot break God’s laws, and God accordingly “punishes” no one, since we are all merely his own body.

God is All That Is, what we already know and what we have yet to learn. In studying Nature, we come to understand God, since God and the World, God and Nature, are simply synonyms. Thus when someone says “There is no God”, it means that

person simply has the wrong idea of God. On the other hand, when someone says “There is a God”, that does not mean they have the right idea of God. The only acceptable interpretation of the Divine is to recognise that the World itself is the way the Divine exists and manifests itself. Only if we grasp this idea, if we understand that God is everything, may the age-old dispute between science and religion finally come to an end, along with absurd talk of being “chosen” and of holding the “true faith”. At last the “witch hunts” and religious wars may disappear, those born of every religion’s and every denomination’s unproven certainty that it alone is right. Science will retain its inalienable right to think freely, and a renewed religion will keep its principal function: to comfort those who suffer, promise everyone a happy life “after life”, and preach love for one’s neighbour. For we are all cells of one organism, and the best way to prepare for eternal life is to harbour hatred for no creature on Earth.

I Demand an Exhumation!

Most honoured, dearest, most esteemed physicists! I beg you: do not bury the Fundamental Law of Nature, the Law of Conservation, alive! It is a dreadful, unforgivable mistake to claim that the Law of Conservation is sometimes violated in quantum mechanics because, supposedly, “out of nothing” there arise, for some microscopic intervals of time … two particles of the same mass! Understand this: not a single, even the tiniest, particle of the material world can arise “out of nothing”. Energy is merely converted into mass and vice versa. The Law of Conservation is eternal and immutable. Any structure can arise only in a pair with its own exact antipode, under conditions in which the total matter and energy of the pair remain unchanged, equal to zero. A particle arises in a pair with an antiparticle and disappears after a time, again together with it, through mutual destruction, or annihilation. Annihilation is possible precisely because the partners’ masses, like their other characteristics, are quantities with opposite signs. The particle’s and antiparticle’s masses do not add up: the result is zero, rather than two masses. In annihilation the particles are entirely converted back into the energy from which they originally arose. Naturally, both the direct and the reverse process are possible. The sum of all parameters for a pair of antipodes (and all structures in the world are paired) necessarily equals zero. Energy itself is dual too. Under certain conditions an energy-carrying particle (which is also a wave), a photon in particular, is its own antiparticle. A wave, by definition, is something that is in many places at once, and waves “in opposite phase” cancel each other out in interference. Do not let a misinterpreted experiment fool you. After all, to register an antiparticle and measure its mass, you have at the very least violated CPT symmetry by managing to “extract” it from the reverse light cone, in other words from negative time. That led to the apparent violation of the most

important law of physics, because the experiment affected its own result.

I repeat: if the Law of Conservation appears to have been violated, find your mistake! I demand an exhumation! The Law of Conservation is alive!

Nihil but An-nihil-ation!

The only way of converting matter into an equivalent amount of radiation, or, conversely, of making matter arise from radiation, is annihilation. Direct or reverse, respectively. The antipodes involved in these processes may be as simple or as complex as you like, from quarks to universes. The more complex the structures taking part, the greater the interval between two annihilations in their endless chain.

The Secret of Eternity

> only a cyclic process lasts for ever,
> one that can run the other way as well.
> two antipodes, the manifest and the imaginary,
> the world of the Absolute, a miracle without miracles,
> where relative time and space
> keep every category constant.

How can any system be eternal? The only eternal pair known with certainty to science, the “particle/antiparticle”, now “appears” or “manifests itself”, now “goes to zero” in the vacuum, existing alternately as particles and as a field. Apparently, the only thing that can in principle be eternal as a whole is something that, while existing, in a sense does not exist at all. Namely, a pair of counterpart antipodes whose characteristics, summed pair by pair, add up to zero. And zero, as we know, is a complex number. It contains a real part and an imaginary part.

Nothing = x − x = 0 = x + i^2(x) = Something

In other words, something can exist eternally only by being equal to “nothing”. And according to the Law of Conservation, the world too is eternal precisely because it includes an imaginary component, the Antiuniverse made of antimatter, alongside the material Universe. Only the barrier of speed divides them, and the antipodes located in negative time travel faster than light. All in all, everything that exists is information, with its own binary code. Sometimes it exists “in pure form” (entropy/negentropy), and sometimes it is embodied in the structures of the antipodal worlds, which develop and undergo involution in cycles. During “materialisation”, the code’s negative unit is the square of the imaginary unit. Attention: quark triads “are coming into view”.

Seven Circles of Paradise

Hello, Ludwig, how are you?
Surprised to see me? I was expecting you.
I knew for certain I would meet you here.
Here, let me give you my hand.
Put the kettle on, my friend.
I’ve dreamt of having tea with you, my dear.

There’s tea, a roll and a knife,
gooseberry jam, the good things in life…
Tell me, when will they issue me my wings?
You look an angel through and through,
though handsome may be too much for you.
At last our meeting is a real thing.

How lovely it is to be with you!
Ludwig, look round, do:
see the ignoramuses hiding in the bushes?
Do you mind if I drop the formalities?
Here, I’ve brought you flowers,
pure white flowers, and clothes as well.

You, my treasure,
show me somewhere I can live,
close by, so we can meet more often.
I've had quite enough of crooks!
And your company
is something I need, more than wings, perhaps.

How free and lovely it is here…
And Einstein, did he find you, dear?
You say he'll drop in towards evening too?
Ludwig, how glad I am
to welcome you all, my friends!
Did you save some ambrosia for the occasion?

Henrietta, where is she,
your dear wife?
Ask her to play us something on the harp!
Look, who's that coming there,
one step back and two steps ahead?
That tall gentleman with the bright scarf?

Oh, why, that's Dirac!
What's he doing, the odd fellow?
What does he need a walk like that for?
Then again, why not?
Give him my regards.
Tell him a beauty has arrived!

Towards evening the five of us
will sit round the table,
and by midnight Stephen will honour us with his presence.
And towards morning, across the threshold,
Comrade God Himself will step:
God can't bear a dinner party with an empty place.

God's Formula

God exists. And he doesn't.

Say whichever comes to you. You won’t be wrong.

For God (All That Is) is information, with a binary code: “yes” and “no” at the same time.

For eternity belongs only to counterpart antipodes whose parameters, taken together in pairs, can be described mathematically at every stage of formation and development, yielding a total dimension of zero. And zero is a complex number, whose expanded form implies the presence of an imaginary, “virtual” part. In that form it corresponds to both “everything” and “nothing” at the same time.

Here it is, God’s Formula:

$x + x(i^2) = \text{nothing} = \text{something}$

Difficult? Who said it had to be easy?

(“x” is a number expressing the total baryonic mass of the universe.)

A Tale of Lost Time

Photons take a certain amount of time to reach the observer, so particles are always emitted before the emission is registered. Thus photons (and anything else that moves) go from the "event" to the observer in ordinary positive time, from the past into the present (the future), with the magnitude of positive time values increasing or that of negative values decreasing. Motion away from the observer, by contrast, takes place in "reverse" time, from the future into the past. Backwards motion in time can decrease the absolute value of positive numbers or increase that of negative ones, depending on the chosen coordinates of the starting point. Speed plays only a supporting role in the sense that at greater speeds the absolute values of time decrease or increase more quickly. If the object being observed is travelling below the speed of light and the increment in magnitude is negative, we can register only the compression, the slowing down, of time. At the speed of light (characteristic of energy-carrying particles), the negative increment "cancels out" the entire advance of positive time. Faster-than-light objects are always in minus time, negative time proper. In negative time antiparticles live unseen. Researchers studying antihydrogen have reminded us of this once again. Objects consisting of large amounts of antimatter are so-called "dark matter". Negative time means faster-than-light speeds. Ordinary objects and those consisting of antimatter are separated solely by the speed barrier. This "antiworld" cannot be seen by us, but it affects objects in the visible world of "matter", right up to annihilating with it at the Big Bang, when a new cycle of formation of the antipodal worlds begins, *ad infinito*.

The Celestial Clock Face

Although the cosmos is an ocean of light from every sort of source, Earth at any given moment is washed by one single, perfectly definite wave: the light from a flash that occurred, say, 8 minutes ago at a distance of 8 light-minutes from Earth, or 5 hours ago at 5 light-hours, or a year ago at one light-year, or 40 years ago at 40 light-years, or a million (a billion) years ago at a million (a billion) light-years, and so on, all the way to the edge of the visible universe… The “crests” of this wave are intervals of space and time inseparably linked and “marked out” by the speed of light. Here the distance in kilometres corresponds exactly to the interval of time in which light covers it in a vacuum. This is the “celestial clock face”, the natural calibration of the cosmic scale that forms a kind of universal “time zone” system. All other “events” at any moment are entirely unobservable and therefore wholly irrelevant to a subluminal observer. They are in negative time, a different dimension of cosmic space, “not ours”. For rays in the forward light cone, the scale is calibrated with successive increases in the absolute values of the reading. In the reverse cone, they decrease. The point where the forward and reverse cones meet is zero on the scale. Naturally, when zero is the observer’s location, the magnitude decreases in every direction of motion away from it.

The Time Zones of the Universe

Cosmic time is determined by the distance light travels in that time. In other words, the farther the photons have travelled, the more time has elapsed. We see light after the photons were emitted, and the farther from us the emission occurred, the later we see it. The conclusion is that all “events” are in a different time, in the past relative to the observer’s own time. In other words, as “events” get farther from the observer, depending on the value we assign to the time coordinates of the observer’s location, either the magnitude of the positive time values for the “event” observed decreases, or that of its negative time values increases. The principle holds for any observer, wherever they may be. This is the basis, the source, of the theory of relativity. It is the physical meaning of the relativity of time. A law of nature governing the “time zones” of the universe.

Undoubtedly

Everyone knows that a particle’s mass depends on its speed. In answer to Lee Smolin’s question about whether there is a single fundamental entity, I say: undoubtedly. There is an entity that, depending on the circumstances, manifests itself in three possible modes or “incarnations”:

- Below the speed of light it is a material particle (structure) with ordinary positive mass.
- At the speed of light (or approaching it) it is a wave, a massless, energy-carrying particle, a “clump of energy”.
- At a speed above that of light it is an antiparticle, a structure that under certain conditions has an imaginary mass. Squaring that mass at a certain instant will yield the negative mass required for it to annihilate with its own material counterpart and antipode. It should be stressed

that antimatter lives in a “CPT-symmetric” world whose foremost characteristic is negative time.

Short and Clear

Rejecting faster-than-light speeds (and with them negative time) led to the idea that, after making a loop through outer space and returning to Earth, the travelling twin would be younger than his stay-at-home brother. A paradox still regarded as unresolved to this day. Recognising the reality of negative time resolves this paradox quite simply. But what exactly is time that runs backwards, from the future into the past? A person registers photons only when they strike the retina. Any emission of photons happens before it is seen, since photons need time to reach the observer. Thus photons, and indeed all objects approaching the observer, move “from earlier to later”, from the past into the future, in ordinary positive time, with its absolute values increasing. Moving away from the observer, accordingly, means going backwards, from the future into the past, and going deeper into it the farther one gets from the recipient. In other words, when the traveller changes direction in space, he also changes direction in time. On a round trip, then, the difference between the traveller’s clock and his stay-at-home twin’s, which first appears and grows on the outward leg, diminishes on the way back and disappears altogether when the twins meet.

Dissecting the Theory of Relativity

The possibility of a negative increment in time is what makes time relative. How far and how fast an object recedes from the observer affect the size of that increment. For objects above the speed of light it takes them into negative time proper. Below the speed of light it compresses time, and at the speed of light it makes time go to zero, with the object becoming a wave (multiplying itself).

Missionaries

“The creator of everything, if such a being exists, must be included in the system ‘Everything That Exists’. Yet he cannot be part of the system, because a part is always simpler than the whole, and a creator is not permitted to be simpler than what he himself created. There is only one way to reconcile this contradiction: identify the creator with creation. Everything that exists is God. Well, we are prepared to call the World, the Universe, God. But God is not as the scriptures portray him. We should discard the naïve inventions of our ancestors about unnatural miracles supposedly performed on Earth by biblical characters. Forget the absurd regulations for everyday conduct, down to supposedly God-given diets, rules for sexual intercourse, styles of dress and even the way to defecate. Condemn savage customs in dealing with other people, calls to mercilessly kill unbelievers and apostates, for example. In other words, we must stop clerics from foisting their own beliefs and rules upon God.” …

Indignantly flinging the pernicious magazine aside, the young emerald-green humanoids Gil and Liri sank into deep despondency. “Just imagine what they let themselves get away with! Such heresy! No, we’ll never manage to found the Church of the Almighty Little Green Man here!” Liri wailed agitatedly, wringing her chitinous hands and twitching her large, perfectly

shaped ears. “Calm down, darling,” Gil replied sensibly. “Apparently the specimen that wrote this can’t take in anything spontaneously without putting it through the cerebral cortex. But don’t be upset. We’ll find people who haven’t got that dreadful habit of thinking! Just don’t worry!” And Gil, trying to soothe his beloved, held his lips, stretched into a transparent tube, to Liri’s gelatinous lips in a long kiss.

Or Maybe…

it might do you some good
to go to some church or other,
or maybe not a church,
or maybe not to go.
without a church, you know,
we’d be plotting evil deeds,
stealing this and that,
or maybe killing someone.

without a church, as sure as two times two,
just about everyone would sin,
or maybe not everyone,
or maybe not at all.
he wouldn’t honour his father and mother,
he’d make himself some idols,
even if just the once,
but what a mess he’d make!

the cheeky devil would take
the Almighty’s name in vain,
and call him Vassya
when he’d had a drop too much.
or maybe he wouldn’t call him that,
and then there’d be another scandal:
the psalms and songs of praise
he’d forget to sing.

while making himself an idol
he wouldn’t drink his kefir,
he’d be sure to covet
somebody else’s ass…
he’d bear false witness,
or maybe not so false,
and he might go in for every
kind of a-dul-ter-y.

in short, to every soldier,

or maybe not to soldiers,
but to their commanding officers,
one might re-com-mend
some church or other,
or maybe not a church,
or maybe it’s not worth
going to any at all!

Catching Hold of the Good Lord’s Tail

> “There is a vast difference between a little post that isn’t there at all and a little post that can never be detected.”
> A. Shlensky

As a child I was a militant atheist. The younger I was, the more militant. Once, in the heat of a theological dispute with my grandmother (I hope I wasn’t even five at the time), I ran out of arguments and actually kicked her in the leg with my boot. She nearly cried, and I still can’t think of it without burning shame and remorse. But since God was plainly to blame for the incident, I didn’t apologise. I only nursed a greater dislike of God. It was bad enough that he didn’t exist, and now I had to fight over him too! God, however, behaved meekly and gave no sign of hostility. In time I stopped paying any attention to him. Well, if he wasn’t there, he wasn’t there! My education in the natural sciences naturally reinforced my contempt for this non-existent personage. I soaked up knowledge greedily, like a

sponge, but found no scientific evidence, not even a hint, of supernatural forces. I was surrounded by young sceptics and atheists, and in our merry company, with its debates and exams, hiking trips and parties, there was of course no room for God. From a beginning like this the reader is entitled to expect an account of my “conversion” next. Anticipating the reader’s disappointment, I hasten to say that there was no conversion. I still don’t believe in God. At any rate, none of the Gods on offer in the “divine marketplace” suits me at all. True, my atheism faded somewhat over the years. First it lost all its militancy. Then it gave way entirely before the thought that atheism is really another faith: faith that there is no God. And I wasn’t made to believe. I was made to know. Or so it seemed to me, at any rate. My self-assurance did not stretch so far as to let me know everything, though, so I chose a maxim: believe nothing, but allow for much. On those terms God and I made peace. For a time. But the day came when my withered atheism raised its head again. It even tried to become militant once more. As you will understand, this coincided with the religious revival in our dear homeland following “perestroika”. Former Party atheist functionaries suddenly began crossing themselves with sweeping gestures and fasting fervently (not too successfully, though, since their ample bodies did not lose any of their dimensions). And when I began to notice little gold crosses on the necks of the “leading theoretical physicists” of Muscovy, I really flew into a rage. At times I felt an irresistible urge to make the appearance of these “new converts” match the familiar line “a cross round the neck and blood on the feet”. If not on the feet, then at least on a face I’d scratched. I can’t stand hypocrites, and I couldn’t believe their conversions were sincere. Nor did I doubt that, watching their tears run over the fresh scratches, I’d feel no pangs of conscience, unlike in the case of my grandmother.

My resurrected atheism revelled in my new mood, but then came a snag. Among those at prayer I found some very

intelligent people whose sincerity, to my complete bewilderment, I could not doubt. I thought: what if they know something I don't? I lost my appetite for scratching and … surrounded myself with books on religion. Every people and every denomination. I won't say I read them all with equal care. Some I merely skimmed, reckoning that if there was anything for the eye to catch on, it would catch on it. But after spending a fair amount of time on these works, with a few works of philosophy along the way, I had to admit that "this place is empty", for me at any rate. There was undoubtedly beauty of language and there were pearls of human wisdom, but what I was looking for was absent: evidence that the Divine existed. There was only an urgent recommendation to believe. And as I said, I wasn't made to believe. So came the next stage. Having drawn a veil of "mourning taffeta" over the great works of the world's religions, I turned … to cosmology.

My reasoning went roughly like this. Religion has its origins in the distant past. Human thought has come a very long way since then, while religious dogmas have frozen, turned to stone, remained unchanged. As an American priest said, thereby earning a blow on the head with a "militant theist's" umbrella, "We make our parishioners park their brains at the church door." I didn't want to part with my brains. Not that I set such a high value on them, but you have to admit they're not entirely useless. And I'd grown accustomed to them. I hadn't been in the world only a year or two. Suspecting that the picture of the world was now being drawn by science rather than religion, I turned to the works of modern theoretical physicists. "Oh, what a little pug! She must be strong!" the kindly reader will think. The sceptical reader will merely snort: "Keep to your own sledge!" In my defence, let me say that I had long taken a keen interest in physics, particularly in those branches connected with cybernetics. Studying the physiology of higher nervous activity, the signalling activity of living systems, made it simply impossible to ignore information theory. Besides, my intentions

were as innocent as they could be. I merely wanted to understand for myself exactly what modern cosmologists thought. My son, a physicist by training and a programmer by trade, saw my new “toys”, books by various good physicists, and asked, intrigued, “What do you need those for?” Modestly but honestly I replied, “I want to catch hold of the Good Lord’s tail.” To which I received an approving “Well, well!” But even without encouragement my mouth was watering in anticipation of the pleasure awaiting anyone who contemplated the coherent, majestic picture of the world offered by modern science.

To my great bewilderment and dismay, however, it turned out that modern cosmologists had no coherent, majestic picture of the world. As for the picture they did have… The deeper I delved into all that book learning, the clearer it became how far it was from perfect. Here and there were classical patterns, elsewhere extremely abstract ones. There were dreadful “black holes” and a great many blank spots, interspersed with traces of crude retouching… It turned out that the picture did not entirely satisfy even its creators. Take the Big Bang, the most popular theory today. To begin with, it’s not clear what kind of explosion it was, why it happened all of a sudden, or where its energy came from. The physicists have nothing to offer on this score, and the Pope, overjoyed by that, declared: “Well, my lads, if you’re really itching to know, you can go into what happened after that Bang of yours, of course. But the Bang itself, and especially the events before it, are out of bounds! Keep your noses out: that’s God’s work!” Remembering the little Communist ditties of my childhood (“What did Mamma have to go and give birth to a Papa like that for?!”), I naturally ignored Papa’s advice. All the more so because the situation very much resembled a curious incident Hawking described. An old lady attended a lecture by a famous physicist and listened diligently to his explanation of the solar system. At the end she addressed the speaker: “That’s all extraordinarily interesting,

young man, but in actual fact the Earth rests on four elephants, which in turn stand on an enormous turtle!" The physicist, smiling ironically, asked, "Very well, but what's at the very bottom, then?" The answer came back confidently: "Why, the turtle's at the very bottom!"

So it is with the idea of God. If God is "at the very bottom", in my view that only substitutes one problem for another. A scientist must then ask what, from the standpoint of physics, this God is, and where he himself came from. So for a true scientist, God, like the turtle he angrily disfigured, is *persona non grata*. Having waved away this non-existent but exceedingly tiresome God, who, instead of meekly offering the tail I was after, kept trying to put a spoke in my wheel, I sadly went over in my mind the separate pieces into which the picture of the world I so desired had fallen on closer inspection. My son, visiting me, would ask cheerily, "Well, how's the research going?" and I'd mumble indistinctly, "Spinning my wheels." I felt cheated by God and by people alike. Many, apparently, were quite satisfied with this unfinished picture of the world! Perhaps if you don't get too close even a daub can look like a masterpiece. And if you're told that this picture has not two, not three, but at best four dimensions, perhaps many more, and it takes an enormous brain to imagine them… I'm afraid plenty of people will be eager to declare how deeply they appreciate and understand the masterpiece on offer! But this sounded suspiciously like the story of the emperor's new clothes my unforgettable grandmother read me as a child.

Still, even jumbled in a heap, the separate fragments of the picture were so delightful, so arresting! I wanted to rummage through them endlessly, turning them this way and that, fitting one to another, trying to make something whole and perfect out of the scattered pieces. I couldn't tear myself away from my favourite toy. Well, nobody could see me, could they? What harm was there in what I was secretly doing? And one day,

putting my hands into my treasures for the umpteenth time, I brought a charming piece to light. Not that I'd never come across it before, but suddenly I saw it in a new light. Here, look for yourself: "So-called empty space, the vacuum, is full of pairs of virtual particles and antiparticles. They arise together, move away from each other and, on coming together again, annihilate… Since energy does not arise from nothing, one partner in the pair has positive energy and the other negative energy, and this process goes on for ever…" For some reason my breath caught. I carefully put the book back in its place, rose from my chair and slowly walked away from my treasures, looking back. I didn't go near the desk for several days. In some corner of my mind, though, a thought glimmered: I'd found Ariadne's thread. Now if I went back to my favourite game… I was afraid to follow the thought through. And then… Then I was sitting at the desk, no longer reading but writing… The article "The Antiuniverse…" was written, as they say, in one sitting.

And the next time my son honoured us with a visit, inwardly jubilant, I asked, "Well, aren't you going to ask how my research is going?" Finishing his chop, he nodded obediently. "I'm asking." At which his "mumsy" sprang from her seat, dragged over pencil and paper and declared:

"Now listen. The Universe exists, right?"

"Uh-huh."

"And there's …" (a pause) "there's an Antiuniverse!!! Their annihilation is the Big Bang, or, what amounts to the same thing, the Big Fizzle! Do you understand?!"

"Uh-huh!"

"This system needs no Creator: the process repeats in cycles, for ever! The system is in equilibrium at any instant, because, owing to CPT symmetry, time in the Antiuniverse runs

backwards. That means there's no paradox of irreversibility! Macroscopic processes are reversible too, only we can't follow them because we ourselves are inside the system! The Antiuniverse is the mirror image of our Universe, its exact counterpart, but made of antimatter objects. Where else could all the antiparticles have gone, the ones that, as we know, formed in pairs with material particles in the early Universe?! Gravity is the pull of the Antiuniverse, whose energy is negative, which is why you have to spend energy to overcome gravity. And there's more. Entropy is a function of time, and since time in the Antiuniverse goes backwards, entropy there goes down and negentropy goes up. Information, in Shannon's sense! Fields of information, do you understand?! Information as a cosmic substrate!!!"

No "uh-huh" followed. What followed was a gloomy "Where?"

"What do you mean, where?" I faltered.

"Well, where is this Antiuniverse of yours? Why has nobody ever seen it?"

I took offence. "It isn't mine. It's ours. It belongs to everyone. It's just that the world is exactly twice as complex as we've thought until now. Do you understand?!"

But my son dug in and kept glumly silent.

"Come on, brain, do some thinking and I'll buy you a cap!" I promised.

"Where?" he stubbornly repeated.

"The cap, you mean?" I asked innocently.

"The Antiuniverse!" my offspring said, unresponsive to the joke.

We parted coolly that day. But I knew I'd struck gold. Now whichever piece of the scattered mosaic I picked up fitted instantly into its allotted place, and the picture of the world

came into view more and more clearly. “Ah, you want to know where?” I muttered vengefully. “Where the crayfish spend the winter! In the reverse light cone, of course. Where else?! Light from any cosmic body can be directed towards us observers (the forward light cone) or away from us, in the reverse light cone. In the latter case we don’t see it unless it reflects off an object it overtakes along the way. Then we see reflected light, which is in a sense the forward light cone again, only from a different, more distant source. But if an object travels faster than light, a light wave will never catch it or reflect off it, and so we cannot in principle see it! It is always in the reverse light cone. The bodies of the Antiuniverse, made of antimatter, have ‘negative inertia’, which is to say they are so mobile that they outrun light! A virtual world! That’s all. What’s so difficult to understand?”

And everything really did begin to make sense:

- the origin of the Universe “out of nothing” without violating physical laws;
- the uselessness of the concept of a “Creator” (that’s for the little turtle!);
- the source of the energy of the Big Bang;
- the whereabouts of the “missing” antiparticles;
- the nature of gravity;
- the “curvature” of spacetime postulated by Einstein (of course: the universe and antiuniverse, moving apart and then coming together to annihilate with one another, naturally move in closed orbits);
- the dual nature of particles: objects travelling at (or approaching) the speed of light catch up with themselves, that is, they “double”, and are in two places at once. Hence the interference in double-slit experiments.

In addition:

- the paradox of irreversibility fell away;
- time, like the spatial coordinates, acquired two directions from zero;
- the use of imaginary numbers in cosmological calculations was justified (imaginary spacetime is measured in imaginary quantities).

And finally, it became possible to predict the future of the Universe clearly: Friedmann's first model, of course. But instead of being the end of everything, it was only the beginning of a new cycle.

Yes, it seems I did catch the Good Lord by the tail: there is no God. Though we can equally well say that He exists. But not the absurd and pompous, fault-finding and malicious character people have invented for themselves. Knowing what I know now, I can agree to only one definition of God: God is Nature. Exactly. If you like, you can say that the World, Nature, is how the Divine exists. Nothing exists outside Nature, and Nature obeys strict physical laws. That includes the Antiuniverse: as long as CPT symmetry holds, all physical laws are preserved. And there is nothing "supernatural". It comes down to words, then. Some prefer to say "God" and others "Nature". There is really nothing to argue about.

Here, if you like, is the formula for everything:

Universe = $E = mc^2 = 1$
Antiuniverse = $-E = -mc^2 = -1$

Nature (God) = Universe + Antiuniverse
= $E - E = mc^2 - mc^2 = 1 - 1 = 0$ = Nothing

However, **$-1 = i^2$**, so:

Nature (God) = $1 + (-1) = 1 + i^2$ = Everything,

that is, a perfectly definite quantity expressed as a complex number.

That’s the picture of the world I came up with. Immodest of me, but I dare say it’s right. Some people, I suppose, will take a statement like that for a sign of megalomania, schizophrenia. Well, then. As in the joke: “Are you mad?!” “Yes, why?” At any rate, I can hardly apologise now for having understood this “machinery” before the people whose opinions are listened to. Anyway, “if you don’t want it, don’t take it!” I’ll enjoy the grand spectacle all by myself.

And another thing… Although I firmly prefer the word “Nature” to “God”, being absolutely convinced that there is nothing “supernatural” in the world and that strict physical laws govern it, I do want to point out that the Fundamental Law of Nature, the First Law of Thermodynamics, needs a little addition: “the law of conservation of matter, energy and … information”. This need not have been mentioned, since when we talk about the conservation of energy we should take it to include all its manifestations, entropy for example. It is clear from this that the part of the Antiuniverse’s negative energy known as negentropy or information must be covered by the law too. And if I mention it all the same, it is only because a thought arises, unbidden. What is our consciousness, our personality, our soul if you like, if not information, the sum of the knowledge and ideas about ourselves and the world around us that we accumulate over a lifetime? And if that is so, might our “informational essence” be immortal? And might it, once rid of the “ballast” of the material, physical body, merge with the eternal fields of information in the cosmos? The very fields whose absolute value must match the sum of the Universe’s entropy. Then, although I reject the “classical” God, do I not rule out … the immortality of the soul?

I’ll admit frankly that I don’t like the thought myself. Out of “concern for my own skin”. Who likes getting an umbrella over the head? And my wayward head had better get ready for blows from all directions. I need hardly say that this conception may

be decidedly unwelcome to traditional believers, for understandable reasons. The trouble is that many “militant atheists”, physicists included (and some of them, oddly enough, manage at times to “militate for atheism” with a cross round their necks), will call these reflections “esoteric”. Haughtily refusing me admission to their “narrow circle of narrow-minded people”, they will classify me as a kind of Blavatsky. To be honest, that’s the last thing I need. There is nothing “esoteric” in my conclusions. It’s physics pure and simple. The conception is no more “metaphysical” than the idea that antiparticles exist. I understand perfectly what a tasty morsel those words about the immortality of the soul are to the “militants”, but, as they say, you can’t cut the words out of a song. I don’t believe it, you see. I know it. By the logic of things. In accordance with eternal and immutable physical laws. Information, negentropy, is indestructible as part of the overall energy balance in nature. It would, on the contrary, be pure idealism to think that anything, no matter what (our consciousness, for example), could come into being and then disappear without a trace. Except from our field of view, like the antiparticles of which the faster-than-light bodies of the Antiuniverse consist. Nothing disappears. It only changes form, existing eternally. I might add that conditions in the Antiuniverse, where entropy decreases, should seem to us earthly creatures perfectly good. Everything goes from chaos to order, from worse to better. What is that if not paradise! So, friends, see you through the looking-glass, in the world “on the other side”! And nothing supernatural. That’s how it is.

P.S. People have also asked about the “practical application” of my idea. Well, what can I tell you? May I suggest making some dumplings out of antimatter? Bon appétit, everyone!

A Reply

Sent to the journal Science by prior arrangement with Sir Richard Whitemere, former chairman of the Club of Undaunted Geniuses (translated from Esperanto).

In mathematical logic there is Kurt Gödel's incompleteness theorem, which states that any complex formal system must necessarily be inconsistent to be complete. If our perception of the world is accurate, inconsistency is evidently a feature of both the systems that do the describing (formal systems) and those being described (real systems) if they lay claim to completeness. Thus, alongside the Universe, there must be its antipode in the cosmos, made of antimatter and constituting an exact mirror copy of our world. Time in such an Antiuniverse runs backwards, its objects consist of antiparticles and its energy is negative. The part corresponding in magnitude to the Universe's entropy but opposite in sign, that is, negentropy, takes the form of fields of information, a kind of "Cosmic Mind". An Antiuniverse with these characteristics obeys the known laws of physics, in particular the law of conservation. This guarantees the eternity of our consciousness, which is the information individuals have accumulated about the world around them and about themselves in it. Since decreasing entropy corresponds to time running backwards, it is logical to conclude that in the Antiuniverse all processes go from chaos to order. Highly organised systems therefore have no need to struggle with their surroundings to survive, so natural selection no longer has a role in their evolution. In other words, there are no conditions for evil at all. Recognising that the Universe has a nonmaterial antipode makes a supernatural Maker or Creator superfluous. The dual world is self-sufficient and includes its own "built-in" mechanisms for coming into being and developing, from one annihilation to the next at the moments of the "Big Bang" marking each successive cycle. One might say that just such a dual world can itself be regarded as God, an

eternal, indestructible entity that includes the “Cosmic Mind” as part of a single, all-embracing organism. Objects in the Antiuniverse cannot be registered by the observer because of the limitations of our senses. Evolution has adapted them to perceive only the material world, the world of matter that produced us as a biological species. In particular, human beings can see celestial bodies only if they emit or reflect light. When an object moves faster than light, light cannot reflect off it because it cannot catch up with it. So structures in the Antiuniverse remain invisible to us, existing “in another dimension”, which is what negative spacetime is for us. Still, knowing that not one material particle, even the tiniest, can come into being without its corresponding antiparticle and that they can also disappear only in pairs through mutual annihilation, we must conclude with confidence that antiparticles, like their counterparts, must inevitably have formed an antimatter world of their own, the exact antipodal copy of the material Universe. Recognising the dual nature of the existing world makes it possible to resolve many of the contradictions that have accumulated in modern physics and cosmology.

Schrödinger’s Cat, Kh. B. B.-L.
38 Mayvember 2050

No Problem!

> “As I have repeatedly stressed, an experiment means nothing at all until it is interpreted by a theory.”
> Max Born

The American scientist Lee Smolin, author of the theory of loop quantum gravity, singles out three of the problems in contemporary theoretical physics as particularly important:

1. Unify general relativity and quantum theory in a single theory that could claim to be a complete theory of physics.

2. Find out whether the different particles and forces can be explained as manifestations of a single fundamental entity.
3. Determine what “dark matter” (“dark energy”) is.

All these problems are easily resolved if we accept the exact analogy I propose between the pair of antipodal particles that exists eternally in the vacuum and the pair of antipodal worlds, universe and antiuniverse. This approach shows that the two systems need to be described, mathematically as well as otherwise, on a common principle, since they differ only in scale, not in how they function. Any unifying account must take into consideration the antimatter that we cannot observe on the cosmic scale, but can observe, with some errors of perception, when we work with the microscopic objects of quantum mechanics. It seems that antimatter is precisely what we call “dark” matter. Like virtual particles, it is not registered by optical instruments, yet it plainly affects objects in the material world. And there is undoubtedly a single fundamental entity that, depending on the circumstances (more precisely, on speed), manifests itself in three possible modes or “incarnations”. Below the speed of light it is a material particle (or structure) with ordinary positive mass. At the speed of light (or near it) it is a wave, a massless “energy-carrying” particle (a “clump”). At a speed above that of light it is an antiparticle (or structure) with an “imaginary” mass under certain conditions. The square of that mass will at a certain instant yield the negative mass needed for annihilation with its own material counterpart and antipode.

It should be stressed that antimatter lives in a “CPT-symmetric” world characterised above all by negative time. It is in “reversed” spacetime, in its reverse cone, that all faster-than-light cosmic objects are found. They remain invisible because light cannot catch up with them and they cannot reflect it. We, on the other hand, can see only sources of light, emitted or reflected, at precisely defined moments. The

distance between the flash and the observer must match the time it takes a photon to cover that distance (for more on negative time, see “the Almighty’s Clock”). The annihilation of all the universe’s matter with all its antimatter is the Big Bang (“Time Zero”), which provides the energy for the next cycle of formation of the antipodal worlds. The same clear analogy with particles again, except that the interval between annihilations is billions of years rather than microseconds.

And there are no “insoluble” problems: “Knock, and it shall be opened unto you!”

In the Briefest Possible Terms

A particle (of matter) is something that is in a particular place at a particular time (for material particles the speed of light is an unattainable upper limit). Particles exist in positive time. A wave (radiation, energy) is something that at a particular time is in many (“all”?) places at once. Energy particles travel at (or close to) the speed of light. An “elementary” particle, on attaining (approaching?) the speed of light, acquires wave properties, becomes a wave. An antiparticle’s coordinates at a particular instant correspond exactly to those of its material antipode. They are the “mirror image” of the coordinates of the material particle, likewise “in a particular place at a particular time”. Antiparticles travel faster than light (the speed of light is the lower limit). Faster-than-light objects live in negative time. At the speed of light an antiparticle becomes a wave, capable, as I have said, of being “everywhere at once” (in both positive and negative time simultaneously). That is, there are three fundamental categories: particle, antiparticle and radiation (energy). (The same applies to objects, or “clumps”, made up of particles.) All these categories turn into one another (matter into energy, for example), partially or, at moments of annihilation, entirely (direct and reverse processes).

Entropy, a measure of chaos, is part of energy. Its antipode, negentropy, negative entropy with the opposite sign, is a measure of order, or information. Human consciousness, too, is nothing other than information (about oneself and the world around one). Information is thus “the ideal”, “spirit”. In absolute value, information is always equal to entropy: the measure of chaos is the same as the measure of order, with the opposite sign. That is, at a certain point in the cycle, when all positive energy turns into entropy through dissipation, everything that exists is a clump of information, the seed of the dual world to come. “The ideal” is information. Its possible “incarnations” are matter, antimatter and energy. These categories change form and exist cyclically, for ever, in accordance with the Law of Conservation.

Some further explanation: a particle travels below the speed of light. A wave, a quantum, travels at the speed of light. Particle and wave are one and the same thing: it shows now the properties of a wave, now those of a particle. This means that a particle becomes a wave when it attains the speed of light. :))) Or: a wave is a particle travelling at the speed of light.

As for the mathematics… There is only one thing I can say with confidence. A mathematical description of the Absolute (the system of “everything that exists”) must include all numbers known to mathematics. It must certainly include imaginary and complex numbers as well as the positive scale of natural numbers. But I have yet to come across such mathematics. The mathematician N. Maksimov, who agrees with me that negative time exists, came up with a splendid line: “Algebra is wiser than the natural scientist.” It is entirely logical for antiparticles to have negative mass. From a purely mathematical standpoint, quantities that cancel one another must have opposite signs, at least at the moment of annihilation (for universes, the moment of the “Big Bang”). I firmly believe that all physical categories go through the entire range of states, taking on positive,

imaginary and negative values, or zero. I hope it is no secret to anyone that a particle's mass depends on its speed. However, it is quite possible that the negative mass required at the moment of annihilation to cancel positive mass is supplied by the square of an imaginary mass (I have written about this in other pieces and mention it here too for the sake of accuracy).

The Antillerymen (a Conversation with an Opponent)

> No trails have been blazed to the Antiworld yet,
> but brace yourself, it's like the front out here:
> we're bombarding nuclei with protons,
> so that makes us antillerymen!
> V. Vysotsky

I am certainly a friend to paradoxes. But I think a paradox arises wherever there is a misunderstanding. So I prefer to find paradoxes a solution, an explanation, a logical justification. And I too would very much like to see, as soon as possible, a mathematical model of processes taking place in negative time. But I am no mathematician. I've done my job: I discovered a new physical reality that calls for new mathematical tools. I did, however, propose the final formula myself. It is simple, as a "Formula for Everything" should be. It is simple because it is made up of the mathematical symbols for two perfect antipodes capable of annihilating:

Universe = $E = mc^2$
Antiuniverse = $-E = -mc^2 = (-1)E = (-1)mc^2$

Given that $\mathbf{-1 = i^2}$ (the imaginary unit squared),

Antiuniverse = $(i^2)E = (i^2)mc^2$

Then:

Universe + Antiuniverse = $E + (-E) = mc^2 + (-mc^2) = 0$ = nothing

or:

$E + (i^2)E = mc^2 + (i^2)mc^2$ = everything

(a definite quantity expressed as a complex number).

In fact, we could put “Particle” and “Antiparticle” in place of “Universe” and “Antiuniverse” (or indeed any pair of antipodes with any values of E and m). I claim a complete analogy between the only eternal system known to modern science, a pair of antipodal particles that periodically appear and disappear in the vacuum, and … antipodal universes, which, like particles, come into being and disappear in the cosmos in an annihilating Big Bang. Like particles, they arise from the energy of annihilation. Like particles, they do so cyclically and eternally. Frankly, it is hard for me to understand why almost every educated person, with rare exceptions, bristles at the mere mention of negative energy. Even gravity is a perfectly good example of negative energy, though it is measured by the positive energy expended in overcoming it. Hawking, in particular, has no doubt that when a particle–antiparticle pair appears in the vacuum, the energy of the latter is negative, simply by virtue of the law of conservation, which does not allow energy to come from nothing. Only antipodes whose combined “increment is zero” may arise. But mass and energy are equivalent, so an antiparticle’s mass must also be negative (remember Dirac). I am prepared to allow that it might be negative only at the moment of annihilation, when it and the particle cancel each other out (from a purely mathematical point of view, quantities that cancel each other must have different signs). In negative time, however, mass is more likely imaginary. If a particle is moving,

$E = \gamma m_0 c^2$,

where m_0 is the rest mass and

$$\gamma = 1 / \sqrt{(1 - v^2/c^2)}$$

For particles travelling below the speed of light, gamma is positive and the energy is positive. If the speed is greater than the speed of light, gamma is imaginary. Then, with real (albeit negative) energy, the rest mass will be imaginary. The same is true when time is “reversed”, since speed appears squared. Such hypothetical particles were called tachyons. I would regard them simply as a state of the antiparticle. “And if they didn’t exist, they would be worth inventing.” So the mass is imaginary. The momentum of such a particle is

$$P = \gamma m_0 v$$

which is real, and since it is a vector, its sign depends on the frame of reference. Nothing seditious, as you can see. One esteemed opponent agreed with this at the time.

Now I’ll try to outline what I think about entropy. Entropy is part of a system’s energy, the part that cannot be converted into work. (The dissipation of energy increases the system’s entropy.) Entropy, as part of the universe’s positive energy, is naturally positive. As for information or negentropy… The relationship between entropy and probability was established by Boltzmann. In 1948 Wiener observed that the amount of information, being the negative logarithm of a quantity that can be regarded as a probability, is in essence a kind of negative entropy. Which is just what I affirm: yes, negentropy, negative entropy, information, is part of the negative energy found in the antiworld, where all physical categories change sign. I have quoted this before, but let me repeat it. From the science news: “Geneva physicists consider antihydrogen atoms a ‘ticket’ … to another mathematical Universe, where ‘positive becomes negative and left becomes right’.” More precisely, a ticket to a different physical reality described by different mathematics, where on crossing “through the looking-glass” (CPT symmetry),

everything positive becomes negative: space, its curvature, time (inseparable from space, since cosmic time is measured by distance), mass (the square of an imaginary mass, with one on either side of the observer along the same line of sight, will supply minus one in real terms), and energy in all its constituent parts, including entropy (which changes sign because time running backwards makes entropic processes reversible).

And again, nothing seditious: perpetual-motion machines do not and cannot exist … in half the system, in the material world. But the combination of antipodes, the principle on which a complete (contradictory) system is built, is precisely the only possible Perpetual-Motion Machine: the Absolute. That is, the world owes its indestructibility and eternity to its duality and nothing else. The Law of Antipodes. Like particles, worlds arise only in pairs and disappear in pairs: direct and reverse annihilation of two exact counterpart antipodes. Annihilation alone is the means of converting matter entirely into energy, into radiation, and vice versa. I want to stress that I regard information as a full-fledged physical category, the antipode of the universe's entropy. Its existence does not require an "observer". Cosmic fields of information are an objective reality. Moreover, because waves (massless particles, quanta) can be "their own" antipodes, we can call entropy itself information. The measure of chaos is the same as the measure of order, with the opposite sign. So a negative logarithm, used in place of the positive one used in entropy calculations, is the mathematical principle for calculating negentropy (one of the things you asked about). In general, the principle of antipodes is "easy to use": do not ignore negative and imaginary numbers, which, after all, keep turning up in cosmological calculations. "Algebra is wiser than the natural scientist." And when dealing with cosmological problems, we should put logic first, ahead of experiment. As I have already said, there are no genuine antiparticles in our world. Ours is a world of positive

time. When we obtain antiparticles to examine them, we … extract them from negative time. And the experiment affects its own result: CPT symmetry is violated, the antiparticle changes its properties and, in particular, exhibits positive mass, strictly speaking in violation of the law of conservation. This is no surprise, but it must be understood correctly. It is the “border crossing” itself, a kind of “phase transition”, that changes the parameters, which matter more than the particle’s “name”.

Once more: objects made of antimatter obey the rule of CPT symmetry. One of its requirements is that the time in which such an object exists must run backwards, must be negative. Antimatter lives in negative time (see “the Almighty’s Clock”). Furthermore, only objects moving faster than light can exist in negative time. Remember the classic children’s rhyme “on the subject”: “A rocket went up at noon today, it’s racing far faster than light, and it’ll get there at seven in the morning — yesterday.” Ordinary positive mass, as we know, cannot attain even the speed of light, let alone exceed it. So this is no “ordinary positive mass”. In the antiworld physical quantities change sign.

And finally, another “self-quotation”. I think the following points are worth pondering:

1. The antipode of a particle is an antiparticle.
2. Particle and wave are two names for the same thing.
3. The antipode of a wave is an antiwave.
4. Energy-carrying (massless) particles (waves) are their own antipodes.
5. A quantum of energy is its own antipode.
6. Entropy is a form of energy.
7. The antipode of entropy is called negentropy.
8. Negentropy is synonymous with information.

9. Entropy can also be considered information, since it is “its own” antipode.
10. Entropy is a measure of the chaos, the uncertainty, of a system.
11. Information is a measure of order.
12. In measuring chaos we measure order, but with a minus sign.

I think I’ll stop there. To tell the truth, I’ve given up hope of “persuading my opponents” (the history of science tells us it’s a “lost cause”, but those of us who love paradoxes have a habit of stepping on the same rake again)…

The Time Vector in the Machinery of the Universe

A Casual Lecture for Lazy Rich Kids

If all you're armed with is logic, facts and common sense, you have practically no chance of being understood. So, to save us all time, I'll give you the main points. Any emission of photons happens before it is seen, because the photons take a certain amount of time to reach the light-sensitive receptor cells in the observer's eye. Thus photons go from event to observer in the ordinary positive direction of time, from earlier to later, from the past into the future. Of course, photons that move away from the observer instead of drawing nearer are not registered by the eye at all, unless they meet an object on the way that reflects them and redirects them to the light-sensitive cells of the eye. The later that encounter happens and the farther it is from the observer, the longer the photons take to get back to the recipient. That is, as they recede, the photons find themselves deeper and deeper in the past relative to the observer, so motion away from the recipient is motion backwards in time. Whether the effect is merely a compression of positive time, whether time goes all the way to zero, or whether it actually takes negative values, depends on the speed of the object moving in the streams of light, the potential reflector of those photons.

If the object is travelling below the speed of light, the photons will catch it fairly soon. It will become a source of reflected light, and sooner or later the observer will register it as such. The time registered will slow a little but stay positive. At the speed of light, the negative increment cancels out the entire advance of positive time. Time "goes to zero", stops, and the object displays wave properties, the ability to be in several (many) places at the same instant. All "massless", energy-carrying particles travel at exactly that speed. Some are visible particles and some are "invisibles". Massless particles

can therefore be their own antiparticles. And if they can annihilate under certain conditions, why shouldn't that explain the nature of electricity? Finally, for an object travelling faster than light, time reverses completely. It runs from the future into the past and takes negative values. No cosmic object in negative time is registered visually. Antimatter is permanently in negative time, in accordance with the rule of CPT symmetry. The principle holds for quantum mechanics and for cosmic objects alike. Antiparticles differ from particles in spin and can sometimes be observed, because an observer of quantum particles can be outside the system. In any case, the first thing to understand is that when a traveller changes direction in space, he also changes direction in time.

Unfortunately, time that runs backwards from the future into the past is still a long way from being accepted by all researchers. Yet recognising it would resolve all the outstanding problems in cosmology. In the notorious twin paradox, for example, the round trip is a kind of journey through a loop in time. The difference between the traveller's clock and his stay-at-home twin's, which first appears and grows on the outward leg, diminishes on the way back and disappears altogether when the brothers meet. The logical error that led to the paradox was to ignore the vector of motion and take only the magnitude of the velocity into account. It is hard to imagine a problem that would not be resolved by recognising the existence of invisible antimatter in the world of negative time and faster-than-light speeds (antimatter is most likely the so-called "dark matter" of the universe). The source of the energy of the Big Bang, in particular, becomes clear. It is simply the energy of the latest annihilation of all the matter of the Universe with all its antimatter, in an endless chain of transformations of the antipodal worlds into one another, obeying the Law of Conservation. That is, Friedmann's first model applies, but only as the beginning of a new cycle, rather than the end of everything. The system is self-sufficient and

eternal, like a “particle–antiparticle” pair in the vacuum. Gravity, incidentally, is the natural attraction between antipodes. Are you familiar with “Loop Quantum Gravity”? Okay. That’s all for now. We’ll carry on next Thursday. After the rain.

A Touch of Arrogance

Someone whose opinion I had learned to respect spoke very flatteringly of my “Law of Antipodes”, but added a qualification. This text, he said, was not physics, because physics uses a “different language”. I did not argue. What seemed most important at the time was that a professional, a nuclear physicist, had said just the right things (and, I must say, the things I had been expecting) about my long-suffering “fixed idea”: “Everything you have written is very sound, very sensible, and very witty at the same time. It contradicts nothing and has every right to exist.” I thanked him for his opinion, but was left with the feeling that something had gone unsaid. Now I want to finish with this: the language of the new physics is the one I speak.

The Principle of Causality

Surely to acknowledge that alongside the material universe there is a world made of antimatter, with time running backwards, is itself equivalent to acknowledging that a signal can travel faster than light. Many people mistakenly think the Theory of Relativity imposes a ban on faster-than-light speeds. That is quite wrong. The ban on motion faster than light comes neither from special relativity nor from Maxwell's or Lorentz's mathematical derivations. It comes solely from … the principle of causality. Einstein himself, considering faster-than-light signalling, wrote: "We are compelled to regard as possible a mechanism for transmitting a signal whereby the resulting action precedes its cause." And then: "Although this result contains no logical contradiction, it is nevertheless so contrary to the character of all our experience that the impossibility of the assumption $v > c$ seems sufficiently well founded." Notice that after the word "although", the subject is simply ordinary everyday experience. That once served as the basis for the idea that Earth was flat. It has no direct bearing on science in general or on the Theory of Relativity in particular. That is, if anything, it is this statement by Einstein that conflicts with special relativity. The theory itself, on the contrary, "compels" us to recognise the possibility of faster-than-light speeds.

The belief that a faster-than-light signal supposedly contradicts the Theory of Relativity is so deeply rooted, however, that even L. Wang, who carried out the impressive experiment in which the speed of light was exceeded by a factor of three hundred (!) in a laboratory, believes the experiment … is at odds with relativity! At the same time, the scientist reasonably points out that the claim that speeds above that of light are unattainable apparently applies only to objects whose rest mass is greater than zero. "Light, however, can be represented either as waves, to which the concept of mass does not apply at all, or as photons, whose rest mass, as is well known, is zero," writes

Wang. “So the speed of light in a vacuum is no limit at all.” Incidentally, a photon, strictly speaking, is never “at rest” in the first place.

So let’s sort out what’s what. First of all, we must understand that the principle of causality is violated at and above the speed of light, and its scope is therefore confined to the world of matter. It does not apply to the field, that is, to energy-carrying particles. Upon reaching the speed of light, massless particles can occupy many places at once, in keeping with wave properties, and can serve as their own antiparticles. Still less can it apply to objects made of antimatter, which reside permanently in reversed time. There entropy, a function of time, decreases instead of increasing as in the world of matter. Consequently, processes are reversible, and it is precisely the cause that gets ahead of the effect! So I repeat: the principle of causality is inviolable in the material (visible) universe. But the world of antimatter is just the medium that allows faster-than-light speeds to become a physical reality. In the words of researchers studying antihydrogen, it is “a key to another universe, where time changes direction and runs the opposite way, from the future into the past, everything left becomes right, and everything positive becomes negative”.

Indeed, we need only know that the masses of a particle and an antiparticle can annihilate with each other to understand that there is something radically different about those masses. At least at the moment of annihilation, it is unlikely that they have the same sign. Otherwise they would add up rather than cancel each other. There seems to be a single fundamental entity in nature which changes so much according to its speed that it takes on different signs, from positive through zero to imaginary and negative values (the square of an imaginary value is one possibility), passing into another dimension of spacetime. As inhabitants of a strictly material world, with only a positive “yardstick for taking measurements”, we are apt to

pronounce it “ordinary positive mass” if and when we manage to catch and measure it, violating CPT symmetry in the process. As a result, what is found to have been violated is … the first law of thermodynamics (!). And we like to repeat that an antiparticle’s “ordinary positive mass” is an “established fact”. Yet even the mass of a particle at rest is an elastic concept, since strictly speaking a particle does not exist at rest either. As Einstein wittily remarked, if the fact contradicts the theory, so much the worse for the fact. Then again, anyone can get the nuances of interpretation wrong by yielding to the weight of received opinion. What matters is that the scientific method enables us to perceive the world around us as accurately as possible and to correct our mistakes as new knowledge accumulates. By now even the latest radio observations by astrophysicists give grounds to suspect faster-than-light motion in some cosmic objects, and an impressive number of experiments have been carried out in laboratories around the world to slow light down and speed it up. We can be sure that Einstein, with his remarkable ability to think unconventionally, would never now make such a sweeping remark about the inadmissibility of the faster-than-light speeds his own theory “compels him to talk about”. As for those who are so excessively principled in their concern to keep outdated postulates inviolate, who miss the point of the Theory of Relativity and want to be more Catholic than the Pope… we must admit that they’re a long way from being Einsteins. Unfortunately.

A UFO Has a Wing

What a strange old Whatsisname!
Let me tell you about him all the same:
ears that are enormous,
gills although he lives on land
(mind you, he demands each day
there be water close at hand).
His skin is a deep, rich green,
and he's the smartest you've ever seen!
To our quiet little town
a UFO brought him down.

Space aliens, envoys of advanced extraterrestrial civilisations, and their flying machines, UFOs, surface from time to time in the press and other mass media.

I'm generally reluctant to take up such subjects, simply because, as S. Lem observed, "In science it is not the custom to discuss what we don't know and patch the holes with guesses of our own." A sensible principle, in my view. But … I've suddenly felt like offering a few thoughts. Bear with me.

First, let me say that I'm prepared to agree with R. Ashby: any open dynamic system, if the laws that operate in it remain constant long enough, gives rise to its own form of intelligent life.

The only question, then, is how to delimit the system correctly before looking for intelligent life within it. For instance, we aren't surprised that fingers don't grow on our ears, are we? Yet we regard a hand without fingers as a departure from the norm.

It is worth remembering that on cosmological scales we can't tell for certain whether we are dealing with a complete system (an "organism") or with one of its parts. And we can very easily mistake "ears" for "hands". So I wouldn't be surprised if there were no intelligent extraterrestrial life on Mars, in the solar system, in our galaxy or even in the universe as a whole. Nor

would I be particularly surprised to find it in any of those places.

As far as the solar system is concerned, though, there is evidently no intelligent life apart from life on Earth. We know enough about the solar system to draw that sad conclusion.

But there is also room for cautious optimism. The total number of galaxies in the observable universe is on the order of two hundred million. From the cosmological principle of isotropy, the uniformity of the universe throughout, we can suppose that all structures in its visible part are of the same type, with roughly the same structure and the same age. In other words, the two hundred million galaxies resemble one another closely enough for us to suppose that all of them may harbour life like our own on Earth.

And obviously we should look for it in the spiral arms of those galaxies, where conditions should match those in the Orion Arm of the Milky Way, home to our earthly civilisation. The other side of it is that all these hypothetical technological civilisations, by the very same principle of isotropy, ought to have reached a level of development neither higher nor lower than ours. It is hard to expect a more advanced civilisation in structures of the same type.

In that case, though, we should not expect UFOs from those parts. For the simple reason that we ourselves cannot yet travel to distant galaxies, the principle of isotropy will immediately bar such a journey to any technological civilisation at our level.

In other words, to talk about UFOs we need to find a more advanced extraterrestrial civilisation, and we should try to do so in our own galaxy, the Milky Way. Besides the Sun, it contains another hundred billion or so stars, many of which have acquired planetary systems. Then, when (and if!) the advanced civilisation we seek is found in the Milky Way, we can suppose that such civilisations also exist in distant, alien

galaxies, taking into account that notorious principle of isotropy.

It probably makes sense to look for a civilisation more advanced than ours nearer the galactic centre, where planetary systems could have begun forming a little earlier than the solar system. (I am deliberately saying nothing about certain factors that could hinder the development of civilisations near the galactic nucleus.)

So. It is about thirty thousand light-years from Earth to the centre of the Milky Way. In other words, light takes thirty thousand years to cover such a distance. However far ahead of Earth this hypothetical civilisation may be, the speed of its flying machines will in any case be incomparably lower than that of light. Our “galactic aliens” are made of matter, are they not?

Let me remind you that the escape speed from the solar system (the “third cosmic velocity”) is only … sixteen km/s, while the speed of light is three hundred thousand km/s. Let’s not be stingy, though. Let us allow our highly advanced fellow intelligences such mastery of their energy sources that their craft can travel at … well, let’s say, in our wildest dreams, one tenth the speed of light: thirty thousand kilometres per second (instead of our sixteen!). Such a “flight” would then take … three hundred thousand years. Now tell me, what living organism, protein-based or even silicon–molybdenum-based, can “afford” a journey that long?!

There’s another wrinkle as well. When we travel through space (and when “they” do too!), we are always heading into the past. If a point is thirty thousand light-years from where the cosmonaut started, then on reaching it he will find himself in a time thirty thousand years earlier than the time he set out. Now consider: wasn’t it Cro-Magnons who lived on Earth thirty thousand years ago? Exactly! Our curious travellers will find no

trace of a technological civilisation on Earth, even if they have had enough time and, ahem … other resources (above all their own lifespans, stocks of fuel, and so on and so forth) to “knock about” in space for a whole three hundred thousand years.

So all those UFO encounters of our contemporaries that inspired “contactees” describe are, excuse me, very, very unlikely. Oh, why mince words … they’re utterly unbelievable!

As for other kinds of contact apart from aliens visiting Earth in person, say the transmission of radio signals that travel at the speed of light…

We have only been able to receive radio signals for about ONE HUNDRED years (radio was invented around that time). The fact that no such signals were registered in that period means that there are no technological civilisations at least within a radius of one hundred light-years from Earth. From other civilisations at our level, if they exist farther away, radio signals have naturally not had time to reach us.

Well, then, what is left in the end? Only the unsubstantiated claims of so-called “contactees” about … “telepathic” (mental) contact with representatives of advanced extraterrestrial civilisations.

At this point, surely, “Scheherazade” ought to “cease her permitted discourse”. Because hearing other people’s voices in one’s head is categorically beyond the scope of scientific discussion. Except perhaps among doctors, and then only if the “case” in question is unfortunately a clinical one. Schizophrenia, say, or temporal lobe epilepsy.

Let me say at once that I certainly do not mean that people with schizophrenia are “second-rate”. Not at all. Everyone has probably heard of John Nash, the brilliant mathematician who had schizophrenia. It may well be that his illness was what gave rise to his unique abilities. Nash’s auditory and visual

hallucinations were exceptionally vivid and entirely indistinguishable from reality to him. That is essentially true of hallucinations in general. And incidentally, you do not have to have schizophrenia or temporal lobe epilepsy to experience them. It can be enough, for instance, to have … a congenital or post-traumatic (remember, psychological trauma is trauma too!) increase in the excitability of the temporal lobe of the cerebral cortex. What's more … similar changes in cortical activity can be brought on by far more "ordinary" causes: certain chemical substances, a religious trance, even rock music played excessively loud … in short, ASCs, or "altered states of consciousness", of many different origins.

Scientists have recently established that stimulating the temporal region of the brain with an electric current (an electromagnetic field) causes hallucinations with religious content. Naturally, their "particulars" vary with a person's religion. Christians "communicate" with Christian saints, Muslims with their prophets, and so on and so forth. It may well be that people inclined to believe in aliens accordingly "make contact" with "aliens".

But we can be more than certain that in "pre-Copernican" times, when humanity had no idea there were planets other than Earth, hallucinations about aliens were in principle impossible. Succubi and incubi, mermaids and forest spirits … well, that was one thing. Angels of every rank, demons of every stripe, certainly. But aliens? No, no!

Well, as the esteemed N. Shulgin says, let us "divide the night into the wolf and the moon". There are scientific findings that can be checked through experiment, facts, logic and mathematical calculation. And there are reports that have absolutely nothing to do with science. In a situation like this, what matters is keeping the two straight. In that I wish us all luck. Otherwise we'll end up with an utterly inconceivable jumble that can distort the truth beyond recognition. And when

trying to build an accurate picture of the world, what worse result could there be?

On the bank of a quiet woodland lake sat Gil and Liri, young, slender, emerald-green humanoids, thoroughly dejected. “No, just imagine what she writes! What she writes!!?” Liri kept saying agitatedly, wringing her scaly hands and twitching her large, perfectly shaped ears. “Calm down, darling,” Gil replied sensibly, sighing to the full capacity of his impressive gill slits. “Surely you can see that this member of Homo sapiens, as they call themselves, clearly has low excitability in her temporal lobe. It can’t be helped. We’ll make friends among people who don’t have that congenital defect. Just don’t worry so!” And Gil, trying to soothe his beloved, held his gelatinous lips, stretched into a long tube, to Liri’s lips in a long kiss.

> Their kiss goes on and on…
> Why can’t we sleep till dawn?
> Or have we Earthlings, shame to say,
> forgotten how to kiss that way?
> Is that why we seek them out,
> the clever, kindly ones from far beyond?

The Law of Popularity

Photons take a certain amount of time to reach the observer, so particles are always emitted before the emission is registered. Thus photons go from the “event” to the observer in ordinary positive time, from the past into the future. Motion away from the observer, by contrast, takes place in “reverse” time, from the future into the past. Backwards motion in time can decrease the absolute value of positive numbers or increase that of negative ones, depending on the coordinates of the starting point. Conversely, motion from the event to the observer increases the magnitude of positive values or decreases that of negative ones. This holds for any subluminal observer, wherever they may be. It is evidence both for the isotropy of the universe and for the objective character of the knowledge obtained. Speed plays only a supporting role in the sense that the greater the speed, the more quickly these absolute values of time increase or decrease. If the object under observation is moving below the speed of light and the increment in magnitude is negative, we can register only the compression, the slowing down, of time. At the speed of light (characteristic of energy-carrying particles), however, the negative increment “cancels out” the entire advance of positive time. Faster-than-light objects are in negative time. Rejecting negative time is tantamount to rejecting the theory of relativity. (And that is not permissible even for Einstein himself.) Until physicists acknowledge that negative time exists, come to grips with its nature and recognise the presence of invisible, faster-than-light objects made of antimatter within it, the argument is doomed to remain hopelessly deadlocked.

That’s what happened. A physicist I know said the idea was crazy enough to be true. There’s one snag, the law of popularity: most people understand fools. A minority understand the clever. Only a handful understand geniuses.

I’m a thoroughly singular Cat,
unpopular with physicists,
I’m no Dirac, no Jules Verne,
I write scientific postmodernism.

Schrödinger’s Cat.

